

HUJAK MINING LIMITED COMPANY

FEASIBILITY REPORT AKIM APAPAM GOLD PROJECT



Abuakwa South Municipality

Eastern Region

Ghana

JULY, 2025

O F F I C I A L U S E		Received at: _____	
		On: _____	
		Authorizing officer: _____	
		Report is confirmed complete: _____	
		On: _____	
		Reviewing officer: _____	

Type of application or lease for which the bankable feasibility is submitted:

- ☒ **Application for a Mining Lease**
☐ Application for a Restricted Mining Lease
☐ Existing Mining Lease
☐ Existing Restricted Mining Lease

A. LEASE APPLICANT / HOLDER DETAILS

- Name of Applicant / Lease Holder:** Hujak Mining Limited Company
- Registered Address:** PLT8BLK10, Pentecost Road, Kwadaso - Kumasi
- Postal Address:** P.O Box KA 9403, Accra
Location of lease area: Akyem Apapam
District: Abuakwa South District
- Locality:** Akyem Apapam
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- Name of mine:** Akim Apapam

Table of Contents

List of Figures	vii
List of Tables	viii
EXECUTIVE SUMMARY	ix
1.0 PROJECT INFORMATION.....	1
1.1 Introduction	1
1.2 Project Location and Access	1
1.3 Infrastructure	5
1.4 Physiography, Drainage and Climate.....	5
1.5 Vegetation	6
1.6 Legal and Fiscal Regime.....	6
1.6.1 Mineral Rights in Ghana	6
2.0 GEOLOGY AND RESOURCE.....	10
2.1 Regional Geology and Mineralization	10
2.1.1 Birimian Metavolcanics	12
2.1.2 Birimian Metasediments	12
2.1.3 Birimian Granitoids	13
2.1.4 The Tarkwaian System	13
2.1.5 The Voltaian Basin	14
2.2 Local Geology	14
2.3 Deposit Types.....	16
2.3.1 Lode Gold Deposit.....	16
2.3.2 Alluvial Gold Deposits	17
2.4 Historical Context and Early Mining Activities.....	17
2.5 Modern Exploration and Expansion.....	17
2.6 Depositional Environments of Alluvial Gold	18
2.6.1 Active Stream Gravels and Valley Flats	18
2.6.2 Old (Paleo) Valleys and Terraces	18
2.6.3 Beach Placers	18
3.0 CURRENT EXPLORATION.....	19
3.1 Introduction	19

3.2	Desk Study and Reconnaissance	19
3.3	Ground Geophysics Surveys	19
3.3.1	Resistivity Surveys.....	19
3.3.2	Induced Polarization (IP) and Magnetic Surveys	20
3.4	Alluvial Prospecting.....	21
3.4.1	Characteristics of the Apapam Creeks	21
3.4.2	Phase 1: Exploratory pitting.....	22
3.4.3	Phase 2: Detailed Alluvial Prospecting.....	23
3.5	Resource Estimation.....	31
3.5.1	Overburden Thickness	31
3.5.2	<i>Gravel Thickness</i>	31
3.5.3	<i>Bedrock Lithology</i>	31
3.5.4	<i>Average Gold Grade</i>	32
3.5.5	<i>Paleo-channel Zones</i>	32
3.6	Ore Reserve Estimation.....	34
3.6.1	Proposed Mining Lease.....	38
4.0	MINING.....	39
4.1	Introduction	39
4.2	Mining Method.....	39
4.2.1	Mining Method Parameters.....	40
4.2.2	Ore Deposit Characteristics	40
4.2.3	Strip Mining.....	41
4.2.4	Mining Schedule Analysis	42
4.2.5	Geotechnical considerations - Design Parameters.....	44
4.2.6	Mine Life Determination	50
4.2.7	Mining Layout	50
4.2.8	Water Management in the Pit and Surface Run-Off Management	53
4.2.9	Equipment Requirements.....	54
5.0	MINERAL PROCESSING	59
5.1	Introduction	59
5.2	Processing Test work	59

5.2.1	Gravel Particle Size Distribution	59
5.2.2	Gold Particle Size Distribution	60
5.2.3	Gold Fineness.....	60
5.2.4	Gravel Washability	61
5.2.5	Plant Design	61
5.2.6	Selection of Processing Plant.....	62
5.2.7	Processing Plant Flow Sheet.....	63
6.0	HUMAN RESOURCE REQUIREMENT	67
6.1	Introduction	67
6.2	Staff Roles.....	68
6.3	Local Content	70
6.4	The Legislative Instrument	71
6.4.1	Employment.....	71
6.4.2	The Unskilled Labour	71
6.4.3	Gender Equality	71
6.4.4	Recruitment of Employees.....	72
6.4.5	Training Program for Local Labour.....	72
7.0	INFRASTRUCTURE	73
7.1	Introduction	73
7.2	Social infrastructure	73
7.2.1	Healthcare Facilities.....	73
7.2.2	Educational Facilities.....	73
7.2.3	Water and Sanitation.....	73
7.2.4	Transportation and Roads	74
7.2.5	Electricity and Energy.....	74
7.2.6	Telecommunications	74
7.3	Project-Specific Infrastructure	74
7.3.1	Power Supply	74
7.3.2	Communication System.....	75
7.3.3	Security	75
7.3.4	Water Supply	76

7.3.5	Water Storage Facility	76
7.3.6	Tailings Management System.....	77
7.3.7	Oversized Material.....	77
7.3.8	Tailings	77
7.3.9	Auxiliary Facilities.....	78
8.0	PROJECT COST ESTIMATION	81
8.1	Introduction	81
8.2	Capital Cost.....	81
8.3	Working Capital	81
8.4	Project Cost	81
8.5	Pre-Production Cost	82
8.6	Total Operating Costs	82
8.6.1	Introduction.....	82
8.7	Project Economic and Financial.....	85
8.8	Sensitivity Analysis.....	90
9.0	ENVIRONMENT	91
9.1	Introduction	91
9.2	Baseline Information.....	91
9.3	Environmental Data	92
9.3.1	Air Quality	92
9.3.2	Surface Water.....	92
9.3.3	Flora and Fauna.....	93
9.3.4	Noise and Vibration	93
9.3.5	Soil	94
9.3.6	Groundwater Quality	94
9.3.7	Rehabilitation and Mine Closure	95
9.3.8	Commitment	96
10.0	SOCIO-ECONOMIC BENEFITS	97
10.1	Negative Impact on the Community	97
10.2	Benefits to the Community	98
10.2.1	Job creation	98

10.2.2	Infrastructure development	98
10.2.3	Economic growth	98
10.2.4	Improved education and training.....	98
10.2.5	Enhanced health services	98
10.2.6	Community development programs	98
10.2.7	Corporate Social Responsibility Program.....	99
10.2.8	Community Development	100
10.3	Occupational Health and Safety	101
11.0	SIGNATURE	103
12.0	APPENDICES	104
	Appendix 1: Selected Logging Data Sheet	104
13.0	Organizational Chart Brief.....	108

List of Figures

Figure 1: Approximate location of the Akim Apapam concession.....	2
Figure 2: Akim Apapam Concession Map.....	4
Figure 3: Regional geology of the Akim Apapam project area	11
Figure 4: Geology of the Akim Apapam concession.....	15
Figure 5: Pseudo-section of Measured Apparent Resistivity.....	20
Figure 6: Pseudo-section of Calculated Apparent Resistivity	20
Figure 7: Layout of detailed pitting highlighting mineralized envelopes.....	25
Figure 8: Pitting in progress.....	26
Figure 9: Weight measurent of recovered gold.....	29
Figure 10: Polygonization of target areas	34
Figure 11: Polygonized ore blocks forming delineated reserve pits.	37
Figure 12: Annual Gold production schedule.....	43
Figure 13: Mine Plan Layout	52
Figure 14: Processing Plant Flow Sheet	64
Figure 15: Proposed Organizational Chart	70
Figure 16: Hujak Mining’s Corporate Social Responsibility (CSR) program.....	100

List of Tables

Table 1: Pillar Coordinates of Akim Apapam Concession	3
Table 2: Types of Mineral Rights in Ghana.....	7
Table 3: Summary of Fiscal Regimes in Ghana	9
Table 4: Preliminary Pitting Results	23
Table 5: Gravel Size Distribution	30
Table 6: Projected Alluvial Ore Reserve Estimate	35
Table 7: Proposed Mining Lease Block Coordinates	38
Table 8: Mining Method Selection Template for Hujack Mining	41
Table 9: Projected gold production values in koz.....	43
Table 10: Productivity of Backhoe Excavators	55
Table 11: Volume Estimation of Truck Loaders	56
Table 12: Selected Equipment Summary.....	58
Table 13: Gravel Particle Size Distribution.....	59
Table 14: Distribution of Gold Particle Sizes	60
Table 15: Staff Requirement.....	67
Table 16: Breakdown of Project Cost.....	82
Table 17: Breakdown of total operating cost US\$.....	84
Table 18: Key Project Economic Results	86
Table 19: Project Income Statement in US\$.....	87
Table 20: Base-Case Project Cash flow and Evaluation based on equity funding	88
Table 21: Cash flow Sensitivity Analysis with a 5% decrease in Projected Revenue.....	89
Table 22: Results of Sensitivity Analysis	90
Table 23: Priority Areas for Hujak Mining CSR Activities	101

EXECUTIVE SUMMARY

PROJECT INFORMATION

Hujak Mining Limited Company (Hujak Mining or the company) is a registered Ghanaian company incorporated under the Company's Act of 1963 to undertake large scale mineral exploration and mining.

Establishing a thriving local mining company that prioritizes environmental sustainability and safety while guaranteeing shareholders competitive profits is the company's main goal. As such, the company is dedicated to striking a realistic balance between increasing profits and generating jobs for host communities for mutual development.

To accomplish this objective, Hujak Mining has allocated significant resources to conduct comprehensive alluvial exploration activities across its Akim Apapam concession aimed at assessing and quantifying the alluvial gold potential within the concession.

Consequently, the company has successfully completed successful alluvial prospecting campaign, which identified promising auriferous gravel horizons. These findings have facilitated subsequent resource and reserve estimations, serving as the basis of this feasibility report and as part of the support for the Mining Lease application of the Akim Apapam Concession.

Location and Access

The Akim Apapam Gold Project is situated in the Abuakwa South Municipality of the Eastern Region of Ghana. The 8.4 km² Akim Apapam concession is located on topographic Field Sheet 0601C4. The concession is accessible off the N6 Accra-Kumasi Highway. Apapam can be connected From Suhum through Apedwa.

Infrastructure

The Akim Apapam project area has fairly accessible roads linked to the Accra-Kumasi trunk road, and essential utilities such as electricity, water, and telecommunications are available. Most communities are connected to the national power grid, though occasional outages make standby generators necessary. Water is mainly sourced from boreholes and wells, with some households accessing piped water. The presence of artisanal gold operators and miners has increased housing

demand, with limited lodging facilities available. Mobile network coverage is good across major providers, but internet speed and broadband access vary by location.

Physiography, Drainage and Climate

The concession area features undulating terrain rising between 240m and 300m, with varied rock formations creating both valleys and highlands. It is drained by the Akuasua River and its tributaries, with a dendritic drainage pattern flowing east to west. Located in the west semi-equatorial zone, the area experiences two rainy seasons (May–June and September–October), annual rainfall between 125mm and 175mm, and a dry season from November to February marked by harmattan winds. Temperatures range from 26°C to 30°C, with high year-round humidity between 70% and 80%.

Vegetation

The area lies within the moist semi-deciduous forest. There are few reserves covering about 108.8sq. km. The forest reserves constitute about 15% of the entire surface of the area. Some commercial tree species contained in the forest are *Milicia excelsa* (Odum), *Sterculia rhinopetala* (Wawa), *Terminalia superba* (Ofram), *Khaya ivoriensis* (Africa mahogany), *Antiaris toxicaria* (Kyenkyen) etc.

Mineral Rights in Ghana

License Type	Reconnaissance License (RL)	Prospecting License (PL)	Mining Lease (ML)	Restricted Lease / License	Small Scale Mining
Purpose	Regional exploration not including excavation and drilling	Search for minerals and evaluation	Extraction of Minerals	Building and industrial minerals	Extraction of minerals on small scale basis
Area	Blocks of 0.21 sqkm, not exceeding 5,000 contiguous blocks	Not exceeding 750 contiguous blocks	Not exceeding 300 contiguous blocks	Provisions relating to mineral rights apply	In accordance with the number of blocks prescribed
Maximum Duration	12 Months, Renewable	3 years, renewable, with reduction of area to not less than half	30 years or less, depending on mine life. Renewable	15 years or less, depending on mine life. Renewable	5 years, renewable
<i>Small Scale Mining Licenses are granted to Ghanaians only but foreigners can participate by way of service companies</i>					

REGIONAL GEOLOGY

Ghana lies within the eastern domain of the Man Shield, which occupies the southernmost third of the West Africa craton. It is largely composed of folded and metamorphosed rocks of early Proterozoic age that were intruded by granitoids during the Eburnean orogeny. About two-thirds of Ghana is dominated by Paleoproterozoic Birimian rocks consisting of five evenly spaced volcanic belts trending northeast-southwest. The intervening basins between the volcanic belts are filled by sediments. The supra-crustal rocks are highly deformed. However, the sedimentary rocks are particularly characterized by extensive folding. The lavas are mainly of basaltic composition, though andesitic, dacitic and rhyolitic rocks are also present. Some pattern of facies distribution is shown by the Birimian sedimentary basins from the margins towards the basin centres. The transition zone between the volcanic belts and the sedimentary basins is marked by chemical facies, which has of late been found to be the site of much of gold mineralization in Ghana.

Both the volcanic belts and sedimentary basins are intruded by three types of granitoids differing in age, mineralogy and chemistry, namely:

- Granitoids in the sedimentary basins - the Cape Coast type - are dominated by two-mica granites
- Granitoids associated with the volcanic belts - the Dixcove-type - are dominated by hornblende-bearing granites.
- The late K-rich granitoids (post-Tarkwaian) comprising the Bongo, Tongo and Bansa granitoids

LOCAL GEOLOGY

Extensive knowledge of the geology underlying the Akim Apapam area is derived largely historical mapping and geological works done by a myriad of companies who have worked in the area. The concession sits within the Birim River basin, dominated by detrital sediments; mainly sandstone and conglomerate rock units and the argillitic/pelitic sediments with the presence of graphite. At the far northeastern portion of the concession exists the basaltic flow/subvolcanic rocks and minor interbedded volcanoclastics.

The hornblende-biotite granitoid tend to intrude narrowly outside the southwestern portion of the concession. Typically, the intrusive display foliation (often gneissic). The metamorphic foliation

in the area appears to have a general NE-SW although it is apparently quite variable. Topographic lineaments, remote sensing images, and geophysical data indicate that the area features a number of major faults (Griffis, 1998).

CURRENT EXPLORATION

A structured and systematic alluvial prospecting program has been implemented with the objective of delineating alluvial gold deposits of economic significance within the project concession. The exploration approach integrated a multi-phase methodology encompassing desktop analysis, reconnaissance surveys, geophysical investigations, detailed sampling, mineral resource estimation. Work executed include are summarized below.

Desk Study and Reconnaissance

A comprehensive desk study analyzed regional geological maps, satellite imagery, and historical exploration data to identify prospective alluvial gold zones. Follow-up field reconnaissance verified these findings, assessing geomorphological indicators—including river terraces, paleo-channels, and floodplains—to confirm gold deposition potential. Sedimentological and basin dynamics analysis further refined target areas for detailed exploration.

Electrical Resistivity Survey Summary

Electrical resistivity surveys effectively mapped subsurface lithological variations to identify gold-bearing structures, particularly paleo-channels and gravel beds. Resistivity contrasts distinguished overburden, clay layers, and coarse gravel horizons—key indicators of alluvial gold deposition. High-resistivity zones correlated with gold-bearing gravels, while low-resistivity areas reflected finer sediments.

The data revealed paleo-channel dimensions and depositional environments, improving target accuracy for pitting, trenching, and bulk sampling. This optimized resource evaluation and reduced exploration risks.

Induced Polarization (IP) and Magnetic Surveys

Induced Polarization (IP) and Magnetic surveys were conducted to further refine the geological understanding of the alluvial terrain of the concession. These techniques were used to:

Distinguish sedimentary layers, locate paleo-channels enriched with placer gold deposits, as the presence of heavy minerals and associated sulfides often produce higher chargeability responses and identify variations in bedrock composition.

Exploratory pitting

The initial prospecting phase comprised systematic geological reconnaissance integrated with geophysical assessment. Test pits were strategically positioned across alluvial plains and terraces following a randomized sampling protocol designed for preliminary alluvial grade evaluation.

A total of ten (10) prospecting pits were excavated within selected drainage basins, including historically exploited artisanal workings, to evaluate gravel deposit characteristics. Geographic positioning of test sites was achieved using high precision handheld GPS instrumentation with <3m positional accuracy. Following phase completion, sites indicating positive gravel intersections were prioritized for further evaluation

Detailed Alluvial Prospecting

After the successful completion of initial prospecting identified significant mineralized zones a structured 100m × 100m grid sampling program was implemented. This systematic approach resulted in the excavation of 156 prospecting pits across the targeted zones. Ground condition complications led to the abandonment of 14 pits either through water seepage or total collapse. Therefore, the resulting effective pits that were used in the evaluation were 142.

A distinct lithological contact between the overburden and gravel horizons was consistently observed across multiple test pits. Continuous sampling was conducted at 0.5-meter vertical intervals throughout the gravel horizon, with each interval securely packaged to maintain sample integrity. To optimize gold recovery and assess potential grade dilution, an additional 0.3-meter penetration into the bedrock was implemented, targeting preferential gold accumulation within structural discontinuities and weathering profiles at the gravel-bedrock interface.

ORE RESERVE ESTIMATION

Comprehensive field prospecting has led to the delineation of alluvial plains and isolated creek systems within the concession catchment. Subsequent reserve evaluation indicates an estimated total contained gold reserve of approximately 42,000 ounces. This is hosted within an interpreted

auriferous gravel volume of 2.9 million cubic meters (m³), averaging a gold grade of 0.45 g/m³ and a gravel thickness of 1.5 meters. A summary of the Ore Reserve Estimate is presented in the table below.

Grid	Average Gravel Thickness (m)	Overburden Thickness (m)	Ore Block Area (m ²)	Block Gravel Volume (m ³)	Average Grade (g/m ³)	Assigned Weight (%)	Weighted grade (g/m ³)	Gold Content (g)	Gold Content (oz)
Akim Apapam	1.50	1.3	1,939,500	2,909,250	0.45	100.00	0.45	1,309,162.50	42,091
Total/Avg			1,939,500	2,909,250	0.45			1,309,162.50	42,091

31.103 g = 1 ounce

Life of Mine (Lom) of the Proposed Lease

Based on the delineated reserves, the projected operational Life of Mine (LoM) for the Akim Apapam Gold Project is estimated at approximately 6 years. The proposed Mining Lease application covers the full extent of the original concession area, totaling 8.4 square kilometers, equivalent to approximately 40 cadastral blocks.

Power Supply

The project's power requirements are as follows: 210 kVA from the national grid, 625 kW from an Aksa generator, 275 kW from another Aksa generator, and 20 kW from solar energy.

MINING

The proposed mining framework identifies requisite technical competencies (engineering, geology, environmental management) and non-technical proficiencies (teamwork, safety protocols, leadership) to establish a compliant workforce capable of addressing mining-specific challenges with sustainable solutions. Hujack Mining's management has delineated role-specific task allocations for all operational personnel, providing clear employee guidelines.

In alignment with Ghana's Local Content Regulations (2020), the company prioritizes local workforce integration, encompassing the recruitment and upskilling of both skilled and unskilled labor from Akim Apapam. This initiative mitigates unemployment, stimulates regional economic growth, enhances socio-economic development, and fortifies community engagement. Such a

strategy optimizes operational efficiency while securing the project's social license to operate within its immediate catchment area

MINERAL PROCESSING

Hujack Mining employs advanced, environmentally compliant extraction methods that eliminate prohibited chemicals and avoid primitive processing techniques. The operation begins with mechanical screening to separate oversized debris from gold-bearing fines, followed by size classification to optimize downstream processing. Gravity separation, utilizing sluices, shaking tables, and jigs, exploits gold's density differentials with gangue materials, using water as the primary medium. The resulting concentrate may undergo further refining for enhanced purity.

Test work on representative gravel samples revealed a particle size distribution of 60% +6mm and 40% -6mm, with a sandy-clay matrix. Gold particle distribution analysis showed 70% coarser than 60 mesh and 30% finer, predominantly fine to coarse with sporadic nuggets. Aqua regia digestion and AAS analysis confirmed gold purity of 96% (22.9 carats), with silver as the primary impurity (4%).

Liberation studies confirmed free gold particles fully detached from host minerals, amenable to gravity washing. Pre-concentration is viable, with 70% of gravel in the +6mm fraction, enabling waste removal to upgrade feed grade. The process capitalizes on size disparities to enhance efficiency, integrating washing and screening to retain gold-bearing fines while rejecting barren coarse material. Downstream processing benefits from this optimized feed preparation.

HUMAN RESOURCE REQUIREMENT

The human resource framework for the mining operation is designed to define the core competencies required for successful project execution. It emphasizes a balance between technical expertise—such as mining and processing engineering, geology, and environmental management—and non-technical capabilities, including safety awareness, teamwork, leadership, and adherence to sustainable mining practices. This competency-based approach ensures that the workforce is well-equipped to meet the operational and regulatory demands of the mining industry while addressing project-specific challenges efficiently.

MINE INFRASTRUCTURE

The existing social infrastructure comprises healthcare facilities, educational institutions, water supply systems, transportation networks, and energy services, with roadway conditions assessed as partially adequate during baseline studies. As part of operational development, the company will implement critical infrastructure including:

A dedicated power generation and distribution system, Security installations and monitoring stations, A mineral processing plant with ancillary circuits, Administrative and operational offices, Staff accommodation complexes, Maintenance workshops and equipment storage facilities and Other supporting infrastructure as may be required by operational demands.

ENVIRONMENTAL CONSIDERATIONS

The company has developed a comprehensive Environmental Management Plan (EMP) aimed at ensuring effective mitigation of environmental impacts throughout the operational phase and during mine closure. The mining baseline study for the Akim Apapam Gold Project has identified surface water bodies that may be potentially impacted by mining activities. Notably, the proposed alluvial mining operation will abstain from the use of environmentally hazardous chemicals such as cyanide and mercury. Instead, gold extraction will be conducted using environmentally benign and conventional techniques, including panning, manual sorting, winnowing, and demagnetization.

Specific mitigation measures have been outlined to minimize adverse effects on both the environment and local communities. Post-mining land restoration and environmental rehabilitation strategies have also been incorporated into this feasibility study. The total estimated reclamation liability for the Akim Apapam Gold Project stands at approximately US\$1.33 million.

PROJECT COST ESTIMATE

The total project cost is estimated at US\$6.49 million, covering key components such as the mining equipment fleet, processing plant, power supply infrastructure, vehicles, contingency provisions, and other associated capital items.

Over the estimated life of mine, the project is expected to generate US\$75.76 million in gross revenue and deliver a net operating profit of US\$21.75 million.

The project will contribute an estimated US\$15.50 million to government revenues, comprising US\$3.79 million in royalties and US\$11.71 million in corporate income taxes. In addition to its financial contributions, the project will also create employment opportunities for local youth within and beyond the operational area.

Operational performance metrics indicate strong profitability. The project is expected to generate net cash flows of US\$21.25 million over its life, with a Net Present Value (NPV₁₀) of US\$13.37 million and an Internal Rate of Return (IRR) of 57.30%.

SOCIO-ECONOMIC BENEFITS

The Akim Apapam Gold Project is anticipated to deliver significant positive socio-economic outcomes, primarily through the generation of both direct and indirect employment opportunities. This is expected to contribute meaningfully to reducing the unemployment burden within the host communities and the broader national context, thereby supporting inclusive local economic development.

Corporate Social Responsibility

Hujak Mining's Corporate Social Responsibility (CSR) program goes beyond philanthropy to focus on meaningful social interventions that create shared value for both the company and host communities. Initiatives include support for education, healthcare, and access to potable water. Stakeholder consultations with community leaders and regulators are integral to the project's environmental and socioeconomic assessments.

The CSR program spans the entire mine life and is built on principles of responsible mining, aiming to enhance social well-being, protect the environment, and maintain financial viability. It is structured around four key areas: stakeholder engagement, employee engagement, community development, and environmental stewardship.

1.0 PROJECT INFORMATION

1.1 Introduction

Hujak Mining Limited Company (Hujak Mining or the company) is a registered Ghanaian company incorporated under the Company's Act of 1963 to undertake large scale mineral exploration and mining.

Establishing a thriving local mining company that prioritizes environmental sustainability and safety while guaranteeing shareholders competitive profits is the company's main goal. As such, the company is dedicated to striking a realistic balance between increasing profits and generating jobs for host communities for mutual development.

To accomplish this objective, Hujak Mining has allocated significant resources to conduct comprehensive alluvial exploration activities across its Akim Apapam concession. These efforts are designed to assess and quantify the alluvial gold potential within the concession.

Given the sustained stability of global gold prices and their impact on the mining sector, the company has strategically prioritized the development of its alluvial gold deposits. This initiative aims to generate immediate revenue through alluvial gold production, thereby providing the necessary capital to fund extensive hard rock exploration programs. Such exploration is critical to evaluating and unlocking the concession's underlying hard rock gold potential.

As a result, the company has successfully completed successful alluvial prospecting campaign, which identified promising auriferous gravel horizons. These findings have facilitated subsequent resource and reserve estimations, serving as the basis of this feasibility report and as part of the support for the Mining Lease application of the Akim Apapam Concession.

1.2 Project Location and Access

The Akim Apapam Gold Project is situated in the Abuakwa South Municipality of the Eastern Region of Ghana, with Kyebi serving as the municipal capital (Figure 1). The concession located within Topographic Field Sheet 0601C4 spans approximately 8.4 km². The project is bounded by the following geographic coordinates:

- Longitude: 0° 34' 45" W - 0° 36' 30" W
- Latitude: 6° 5' 30" N - 6° 7' 45"

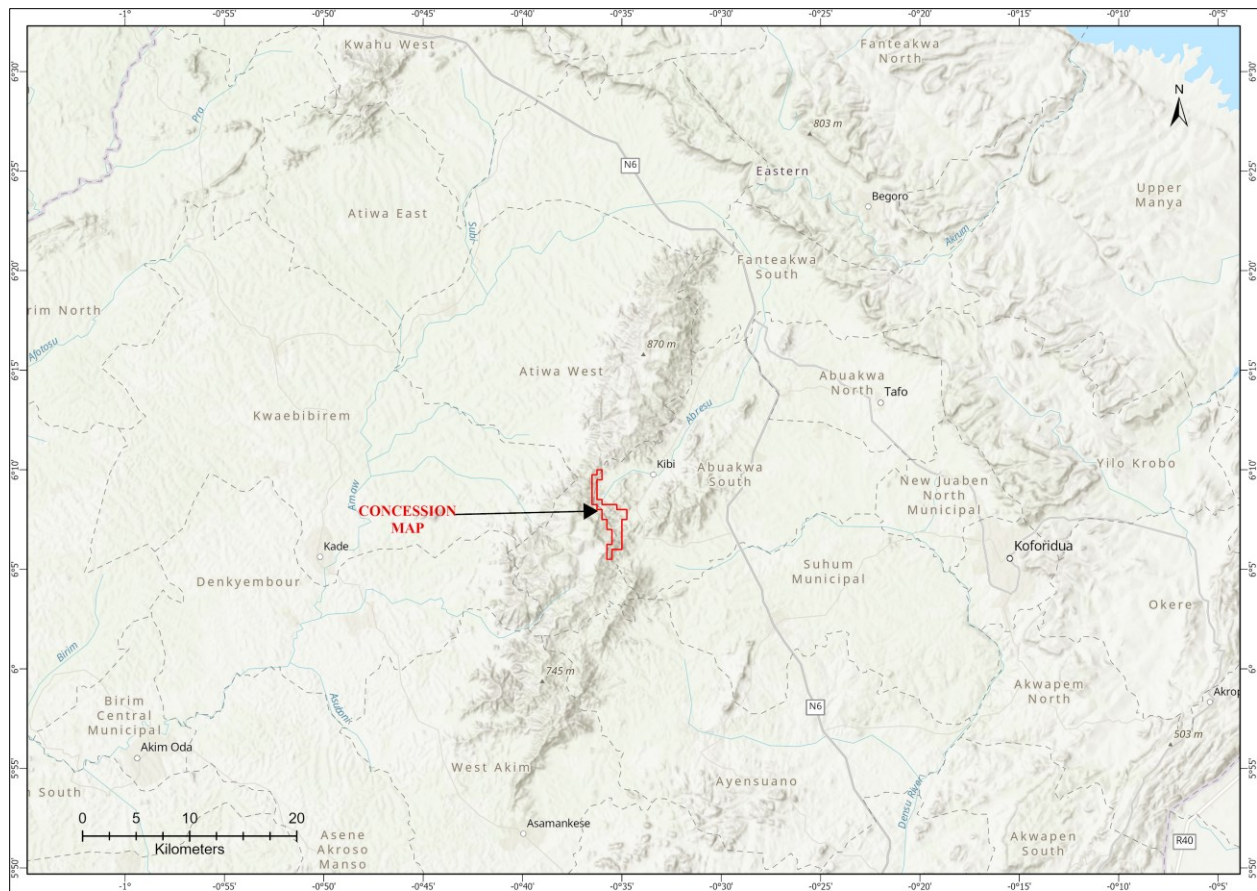


Figure 1: Approximate location of the Akim Apapam concession

The property lies approximately 78 km northwest of Accra, Ghana’s capital. Boundary pillar coordinates are provided in Table 1, with the concession outline illustrated in Figure 2.

The concession is accessible via the N6 Accra-Kumasi Highway. From Suhum, proceed to Apedwa Junction, then turn left onto the Wiren-Odumase-Kyebe road. Upon reaching Kyebe township, a left turn leads directly to the Akim Apapam project site.

Table 1: Pillar Coordinates of Akim Apapam Concession

PILLAR	LONGITUDE	LATITUDE
P1	0° 36' 15"	6° 10' 00"
P2	0° 36' 00"	6° 10' 00"
P3	0° 36' 00"	6° 09' 30"
P4	0° 36' 15"	6° 09' 30"
P5	0° 36' 15"	6° 08' 30"
P6	0° 36' 00"	6° 08' 30"
P7	0° 36' 00"	6° 08' 15"
P8	0° 35' 15"	6° 08' 15"
P9	0° 35' 15"	6° 08' 00"
P10	0° 34' 45"	6° 08' 00"
P11	0° 34' 45"	6° 07' 30"
P12	0° 35' 00"	6° 07' 30"
P13	0° 35' 00"	6° 06' 00"
P14	0° 35' 30"	6° 06' 00"
P15	0° 35' 30"	6° 05' 30"
P16	0° 35' 45"	6° 05' 30"
P17	0° 35' 45"	6° 06' 15"
P18	0° 35' 30"	6° 06' 15"
P19	0° 35' 30"	6° 07' 00"
P20	0° 35' 45"	6° 07' 00"
P21	0° 35' 45"	6° 07' 30"
P22	0° 36' 00"	6° 07' 30"
P23	0° 36' 00"	6° 08' 00"
P24	0° 36' 15"	6° 08' 00"
P25	0° 36' 15"	6° 08' 15"
P26	0° 36' 30"	6° 08' 15"
P27	0° 36' 30"	6° 09' 45"
P28	0° 36' 15"	6° 09' 45"

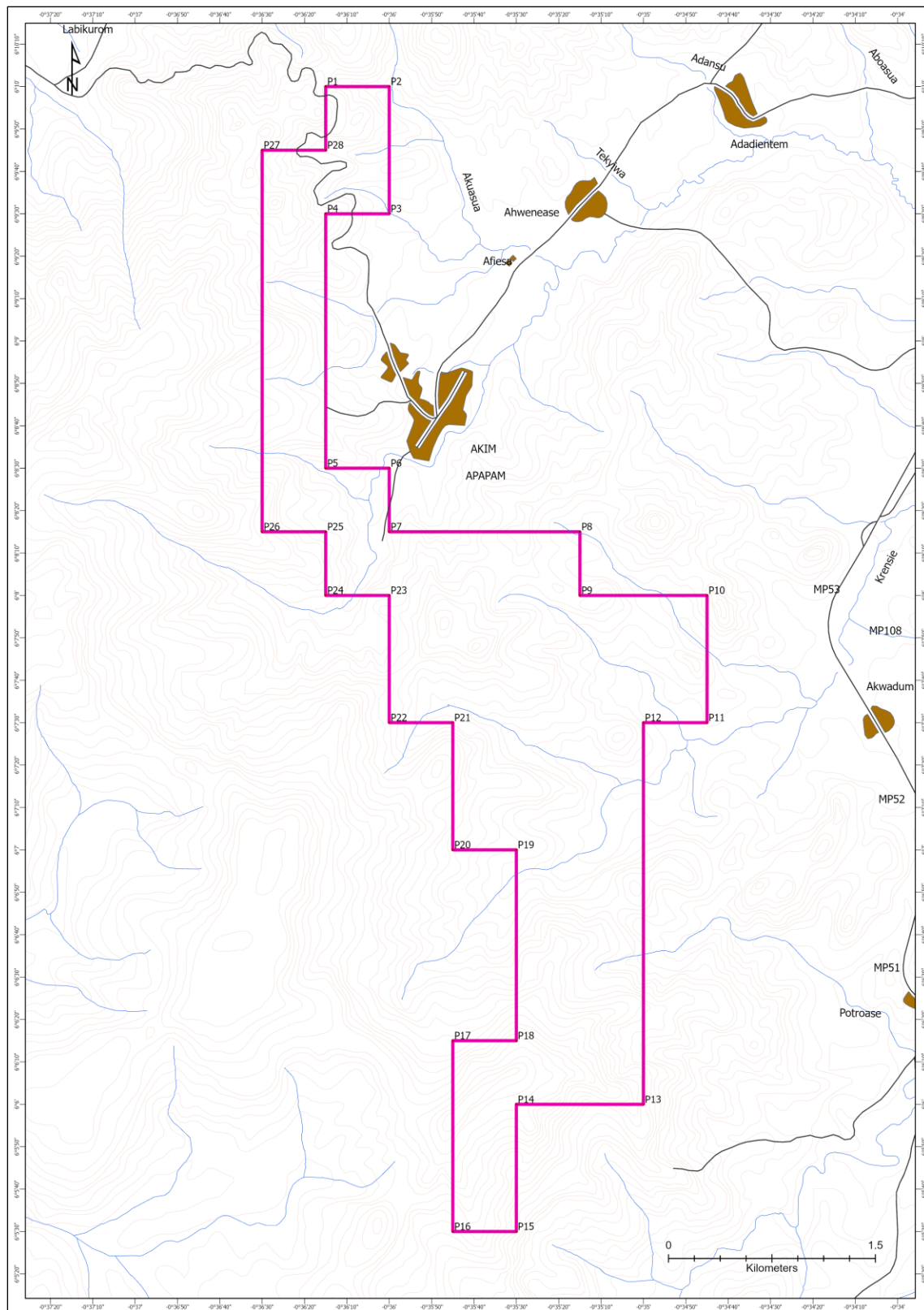


Figure 2: Akim Apapam Concession Map

1.3 Infrastructure

The infrastructure within the project area has a fair distribution of accessible roads, majority of which link the Accra-Kumasi trunk road. Also, utility available in the project area include electricity, water and telecommunication, which are very crucial for development. Most of the communities have been connected to the National Electricity Grid. This has improved the lives of the people.

The Akim Apapam area and its surrounding communities are integrated into the national power grid, providing a stable electricity supply. However, occasional power disruptions may occur, though they are relatively infrequent of which getting a standby generator is importantly necessary. Water supply within the project area is primarily sourced from manually operated boreholes, dug wells, and community-managed mechanized boreholes. Households with the necessary infrastructure also have access to piped water through privately installed transmission lines.

The influx of mine workers and related businesses has led to increased demand for housing. The Akim Apapam township is served by Executive Lodges and guesthouse facility that have the capacity to accommodate fifteen (15) people. Mobile network coverage in the project area is generally good, with all the major telecom providers (Airtel/Tigo, MTN, and Telecel) operating in the area. Internet access is available, though speeds may vary, and reliable broadband services might be limited to certain areas.

1.4 Physiography, Drainage and Climate

The land is generally undulating and rises about 240m to 300m above sea level. There are several different types of rock formation which give the different relief features ranging from flat bottom valleys to steep-sided highlands usually covered with hardpans, bauxite and kaolin. The concession area is drained by Akuasua and its network of tributaries. Several other unknown seasonal streams are found within the other portions of the concession. The pattern is largely dendritic flowing in the east - west direction.

The concession lies in the west semi-equatorial zone which is characterised by two main rainfall seasons occurring in June and October; the first rainy season starts from May to June and the second from September to October. The mean annual rainfall is between 125mm and 175mm. The

dry season are distinct starting between November and late February which usually is characterised by the eastern dry winds also known as harmattan. Temperatures are found to be fairly uniform ranging between 26°C in August and 30°C in March. Relative humidity is generally high throughout the year, ranging between 70% -80% in the dry season and 75% - 80% in the wet season.

1.5 Vegetation

The area lies within the moist semi-deciduous forest. There are few reserves covering about 108.8sq. km. The forest reserves constitute about 15% of the entire surface of the area. Some commercial tree species contained in the forest are *Milicia excelsa* (Odum), *Sterculia rhinopetala* (Wawa), *Terminalia superba* (Ofram), *Khaya ivoriensis* (Africa mahogany), *Antiaris toxicaria* (Kyenkyen) etc.

1.6 Legal and Fiscal Regime

1.6.1 Mineral Rights in Ghana

Between 1986 and 2006, the Mineral and Mining Law 1986, PNDCL.153 was the basic mining legislation in Ghana. While it was regarded as a trailblazer in terms of mining legislation in sub-Saharan Africa, changes in the international mining scene necessitated its revision.

After a protracted review from the early 2000s, the governing legislation for Ghana's minerals and mining sector was updated in 2006 by enacting Act 703. Under the constitution and the mining laws of Ghana, all minerals in Ghana in their natural state are the property of the state and title to them is vested in the President on behalf of and in trust for the people of Ghana, with rights of prospecting, recovery and associated land usage granted by the state under licenses or leases. Three types of tenure are granted as a company progresses through the reconnaissance, exploration, development and production phases: *Reconnaissance License*, *Prospecting License* and *Mining Lease* (Table 2).

In addition, a license is required for the export or disposal of minerals and the government has a pre-emptive right over all minerals produced. Activities such as the diversion of water require

separate licenses or consents. A rental fee is payable to the government in respect of licenses and leases.

Table 2: Types of Mineral Rights in Ghana

License Type	Reconnaissance License (RL)	Prospecting License (PL)	Mining Lease (ML)	Restricted Lease / License	Small Scale Mining
Purpose	Regional exploration not including excavation and drilling	Search for minerals and evaluation	Extraction of Minerals	Building and industrial minerals	Extraction of minerals on small scale basis
Area	Blocks of 0.21 sqkm, not exceeding 5,000 contiguous blocks	Not exceeding 750 contiguous blocks	Not exceeding 300 contiguous blocks	Provisions relating to mineral rights apply	In accordance with the number of blocks prescribed
Maximum Duration	12 Months, Renewable	3 years, renewable, with reduction of area to not less than half	30 years or less, depending on mine life. Renewable	15 years or less, depending on mine life. Renewable	5 years, renewable
<i>Small Scale Mining Licenses are granted to Ghanaians only but foreigners can participate by way of service companies</i>					

1.6.1.1 Reconnaissance License

Reconnaissance License permits the holder to carry out geophysical, geochemical and photo-geological surveys, but not drilling, excavation or subsurface activities. They are granted for a period of up to twelve months and may be renewed, provided the renewal application is made at least three months prior to the expiration of the license and the government determines that the renewal is in the public interest. While a Reconnaissance License is not necessary in order to secure a Prospecting License, it does grant the holder with the exclusive right to obtain a Prospecting License in respect of the minerals being sought within it.

1.6.1.2 Prospecting License

Prospecting License permits the holder to carry out both reconnaissance work and sub - surface excavation such as drilling, trenching and pitting. Prospecting Licenses are granted for a period of up to three years and may be renewed for additional periods of two years, provided the renewal application is made at least three months prior to the expiration of the license and the government determines that the renewal is in the public interest. Upon the renewal of a Prospecting License, the area covered by the initial license may be reduced by one-half of the area covered by the previous license. The license holder is able to select the areas to be eliminated upon renewal. If a

license holder fails to expend any monies that the holder designated in its program for exploration under the Prospecting License, then such non-expended monies will become a debt due to the Republic of Ghana.

1.6.1.3 Mining Lease

A Mining Lease grants rights to take all reasonable measures on or under the surface to mine the mineral to which the mining lease relates, to erect necessary equipment, plant and buildings, to prospect within the mining area and to stack or dump mineral waste in an approved manner. Ghana's mining framework balances investor rights with state oversight, ensuring environmental compliance, revenue sharing (10–30% equity + royalties), and control safeguards. The various components of a mining lease are as summarized below;

1.6.1.3.1 Mining Lease Rights & Requirements

- Grants rights to mine, prospect, erect infrastructure, and manage waste.
- Only prospecting license holders can apply after proving commercial quantities of minerals and submitting an environmentally sound development plan.
- Must apply within 3 months of discovery notice. Non-license holders may apply but approval is discretionary.
- Validity: 30 years (extendable for national interest), renewable for another 30 years if applied 1 year before expiry.
- Requires EPA registration and environmental permit before operations.

1.6.1.3.2 Transfer & Control Restrictions

- No transfer/assignment of mineral rights without Ministerial approval.
- Change of control (e.g., mergers, >20% voting power shifts) requires government consent.
- 5%+ shareholders must be disclosed.

1.6.1.3.3 Government's Role & Interests

- Ministry of Lands & Natural Resources grants/revokes leases, advised by the Minerals Commission. Leases require parliamentary ratification.
- State Equity:

- Automatic 10% free carried interest (via non-assessable shares).
- Optional additional 20% (negotiated or arbitrated).
- Royalties: 3–12% of sales (after expenses), paid quarterly (3% advance), adjusted annually.

1.6.1.3.4 Special Share & Veto Rights

- Government may hold a “special share” (free or negotiated) granting veto power over:
- Changes to company control.
- Liquidation or asset/Mining Lease disposal.
- Control defined as 20%+ voting power or direct influence.

The summary of the fiscal regimes as present in Ghana is shown in Table 3.

Table 3: Summary of Fiscal Regimes in Ghana

Fiscal Instrument	Application Rates
Mineral Royalty	3% - 6%
Corporate Income Tax	35%
Capital Allowance	20% straight line on mining assets for 5 years
Losses carried forward	5 years
Thin Capitalization	3:01
VAT	12.5% (Refundable)
NHIL & GET Fund Levy	2.5% each and tax deductible
Pay As You Earn (PAYE tax)	Graduated rate
Dividend tax	8%
National Fiscal Stabilization Levy (NFSL)	5% of profit before tax
Government Carried Equity	10%
Annual Mineral Right Fees	As prescribed by L.I 2176
Withholding tax on resident entities	Goods 3%, Works 5%, Service 7.5%
Withholding tax on interest payable to non-residents	8%
Withholding tax on royalties, natural resource payments and rents	15%
Ground Rent	As prescribed by L. I. 2357

2.0 GEOLOGY AND RESOURCE

2.1 Regional Geology and Mineralization

Ghana lies within the eastern domain of the Man Shield, which occupies the southernmost third of the West Africa craton. It is largely composed of folded and metamorphosed rocks of early Proterozoic age that were intruded by granitoids during the Eburnean orogeny. About two-thirds of Ghana is dominated by Paleoproterozoic Birimian rocks consisting of five evenly spaced volcanic belts trending northeast-southwest. The intervening basins between the volcanic belts are filled by sediments. The supra-crustal rocks are highly deformed. However, the sedimentary rocks are particularly characterized by extensive folding (Figure 3). The lavas are mainly of basaltic composition, though andesitic, dacitic and rhyolitic rocks are also present. Some pattern of facies distribution is shown by the Birimian sedimentary basins from the margins towards the basin centres. The transition zone between the volcanic belts and the sedimentary basins is marked by chemical facies, which has of late been found to be the site of much of gold mineralization in Ghana.

Previously it was thought that the Birimian sedimentary rocks were older than the volcanic rocks. However, recent studies indicate that the volcanic belts and sedimentary basins are lateral facies equivalents.

Both the volcanic belts and sedimentary basins are intruded by three types of granitoids differing in age, mineralogy and chemistry, namely:

- Granitoids in the sedimentary basins - the Cape Coast type - are dominated by two-mica granites
- Granitoids associated with the volcanic belts - the Dixcove-type - are dominated by hornblende-bearing granites.
- The late K-rich granitoids (post-Tarkwaian) comprising the Bongo, Tongo and Bansa granitoids

Each of the Birimian volcanic belts contains meta-sedimentary rocks - the Tarkwaian Group – that lie unconformably over the Birimian. The Tarkwaian consists entirely of conglomerates, sandstones, phyllites and slates derived from the Birimian country rocks.

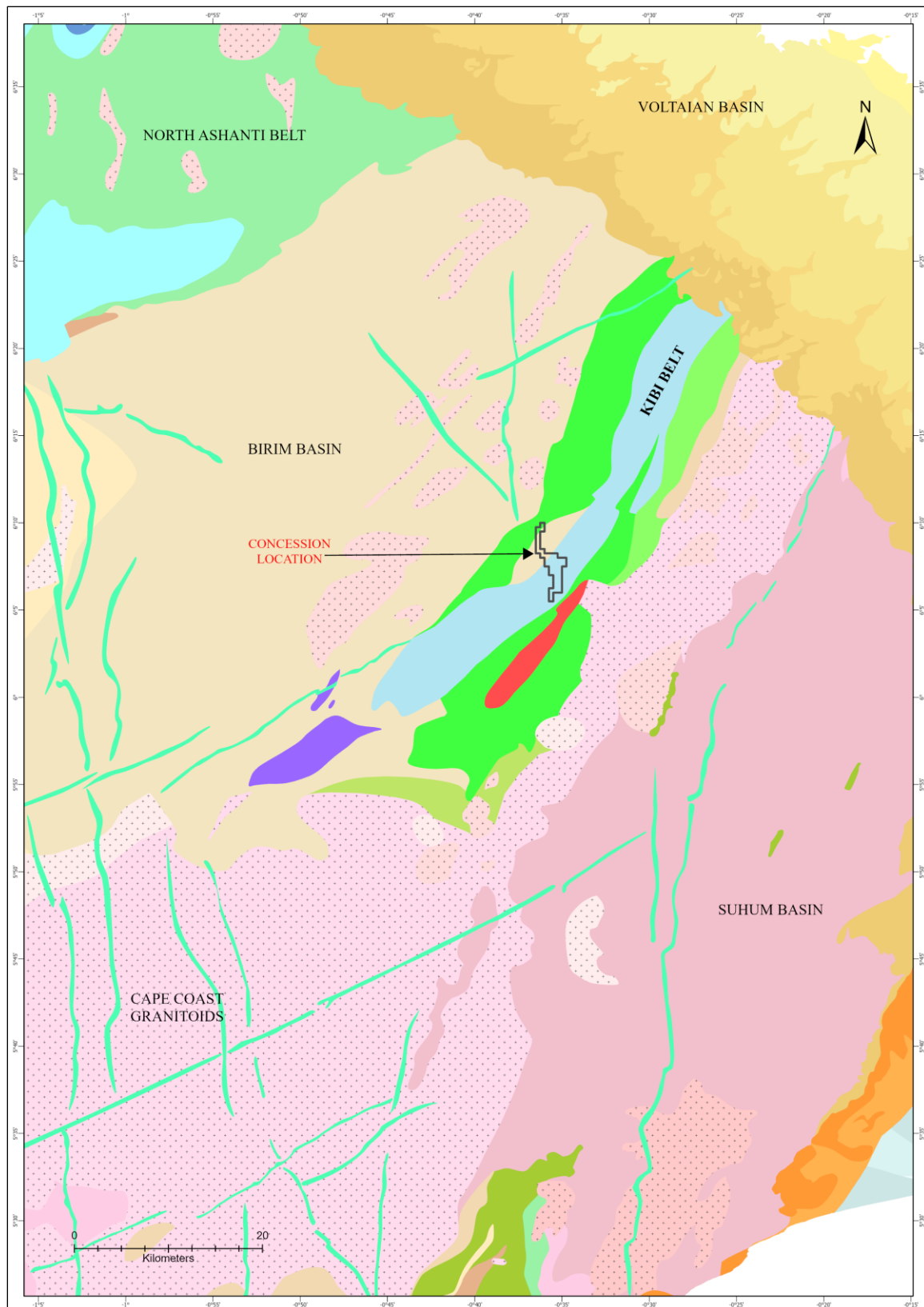


Figure 3: Regional geology of the Akim Apapam project area

The Birimian and Tarkwaian rocks occur mainly in the Eastern, Central and Western Regions, including Ashanti, Brong Ahafo and the western part of the Northern and Upper Regions.

The Dahomeyan System occupies the southeastern corner of Ghana and occurs as four alternate belts of acid and basic gneisses, trending south-southwest to north-northeast direction. A greater part of the System forms a monotonous low-lying plain broken by isolated inselbergs and ridges of mainly ultramafic intrusives and hills forming outliers of Togo rocks. The areas occupied by the basic gneisses are flat (Accra Plains), while the areas occupied by the acid gneisses give rise to gently undulating topography. The Togo structural units are made up of meta-sediments and Buem structural unit dominated by meta-sediments and metavolcanics, support a range of hills lying in the eastern corner of the country. Undeformed, flat-lying Neoproterozoic clastic sedimentary rocks forming the Voltaian System cover the eastern third of the country.

A strip of Paleo-Cretaceous to Tertiary sediments occurs along the coast. These consist of the Accraian Series (in Accra area), the Sekondian Group (Elmina, Komenda and Sekondi areas), Apollonian Formation (Tano and Keta areas), and the Amisian Formation (the Saltpond area).

2.1.1 Birimian Metavolcanics

There are six volcanic belts in the Birimian, namely; the Kibi-Winneba, Ashanti, Sefwi, Bui, Bole-Nangodi and Lawra belts. The belts have a slight keel-shaped outline that is perhaps produced by the associated diapiric, intrusive plutons (Kesse, 1985).

The belts consist mainly of metamorphosed basaltic and andesitic lavas, now hornblende-actinolite schists, calcareous chlorite-schists and amphibolites (the greenstones). Minor intrusions of mafic rocks cut the volcanics. Volcaniclastic sediments occur interbedded within the basaltic flows of all volcanic belts (Leube et al., 1990). Metamorphism in most volcanic rocks is confined to the chlorite zone of the greenschist facies. Amphibolite-facies assemblages occur sporadically but especially along the margins of granitoid bodies.

2.1.2 Birimian Metasediments

Birimian metasedimentary rocks of Ghana are divided into:

- volcaniclastic rocks;
- turbidite-related wackes

- argillitic rocks;
- chemical sediments.

Boundaries between these subdivisions are gradational (Leube et al., 1990). The volcanoclastic sediments comprise chiefly of sand-to silt-sized, partly reworked pyroclastics that is shown by the presence of quartz, idiomorphic plagioclase crystals, chloritised glass fragments, and the absence of heavy minerals (Leube et al., 1990).

2.1.3 Birimian Granitoids

Four main types of granitoids are recognized in the Birimian of Ghana. They include Winneba, Cape Coast, Dixcove and Bongo granitoids (Junner 1935; Kesse, 1985). The latter three have been recently termed - Basinal, Belt and K-rich granitoids. (Leube et al., 1990). The Cape Coast (Basin type) and Dixcove (Belt type) granitoids are widespread in Ghana, the Winneba belt is limited to small areas near Winneba, and the Bongo type crops out in the Bole Navrongo Belt and in the Bansa area (Kesse, 1985).

The (Basin type) granitoids occur only within the Birimian sedimentary basins. Some of them are two mica granites. This group also includes gneisses, and these are especially well developed in the metasedimentary basin. They are typically biotite-bearing. It has been suggested that the Basin granitoids, which appear migmatitic in some localities, might represent an older continental basement on which the Birimian supracrustals were deposited. Based on the degree of foliation, early workers assumed that the Basin type granitoids intruded during regional deformation and that Dixcove granites were emplaced after deformation (Kesse, 1985).

Belt type granitoids are metaluminous and typically dioritic to granodioritic in composition. They intrude Birimian volcanic rocks. They are typically hornblende-bearing and are commonly associated with gold mineralization where they occur as small plutons within the volcanic belts. The granitoids are massive in outcrop, do not have a compositional banding or foliation, and are thus generally considered post-deformation.

2.1.4 The Tarkwaian System

The Tarkwaian is a Proterozoic supracrustal system overlying the Birimian. It consists mainly of shallow water sediments and is present in all the Birimian volcanic belts (Junner, 1935). It consists of coarse clastic sedimentary rocks that include conglomerates, arkoses, sandstones and minor

amounts of shale. The Tarkwaian is usually regarded as the detritus of Birimian rocks that were uplifted and eroded following the Eburnean tecto-thermal event (Eisenlohr and Hirdes, 1992). Economically, the most important unit of the Takwaian is the Banket series which contains economic concentrations of gold in several areas.

2.1.5 The Voltaian Basin

The Voltaian Basin area appears to have been the eastern margin of a large West African cratonic block, which had broken away from the former supercontinent Rodinia (Hoffman, 1999). The Voltaian Basin is a structural basin where sedimentary rocks along the eastern margin were folded during orogenic activity associated with a late Precambrian to early Paleozoic thermal event, the Pan African Thermo-Tectonic Episode (Kennedy, 1964).

Almost one third of Ghana is covered by sediments of the inland Voltaian Basin which covers an area of about 103,600 km²; it consists mainly of flat lying or very gently dipping sediments sitting on a major Precambrian unconformity (Griffis, et al. 2002). This unconformity marks an erosional surface, which apparently covered the entire Man Shield.

2.2 Local Geology

Knowledge of the geology underlying the Akim Apapam area is derived largely from the published works of the Ghana Geological Survey. The concession sits within the Birim River basin, dominated by detrital sediments; mainly sandstone and conglomerate rock units and the argillitic/pelitic sediments with the presence of graphite. At the far northeastern portion of the concession exists the basaltic flow/subvolcanic rocks and minor interbedded volcanoclastics Figure 4.

The hornblende-biotite granitoid tend to intrude narrowly outside the southwestern portion of the concession. Typically, the intrusive display foliation (often gneissic). The metamorphic foliation in the area appears to have a general NE-SW although it is apparently quite variable. Topographic lineaments, remote sensing images, and geophysical data indicate that the area features a number of major faults (Griffis, 1998).

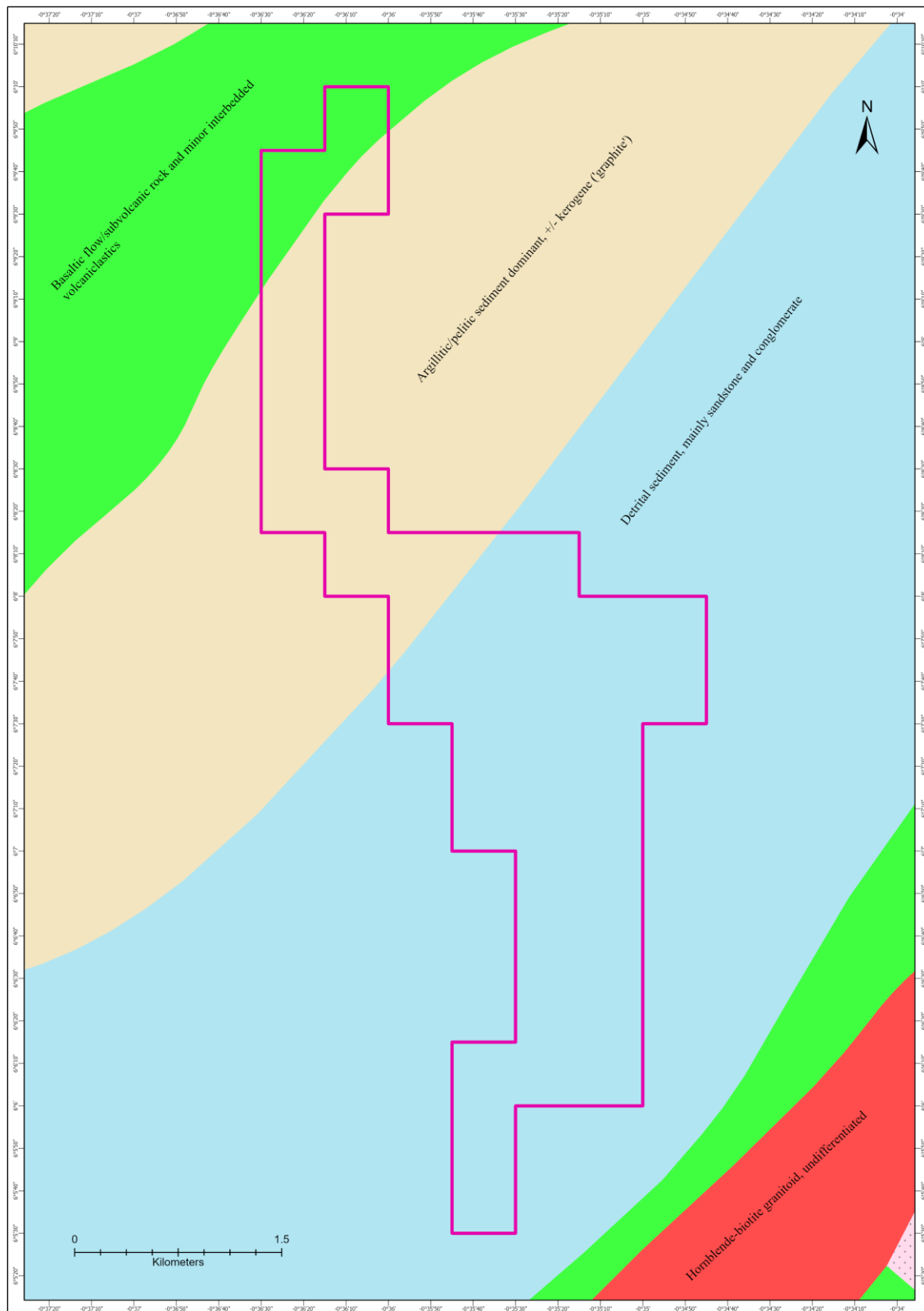


Figure 4: Geology of the Akim Apapam concession

2.3 Deposit Types

2.3.1 Lode Gold Deposit

Primary mineralization within the Akim Apapm Gold Project is considered to be typical of mesothermal vein-style or orogenic-style gold deposits. The discussion below is sourced from Goldfarb et al., (2005), and Groves et al., (1998; 2003).

Orogenic gold deposits are found in diverse metamorphic terranes, ranging from Middle Archean to younger Precambrian ages, with continued formation into the Phanerozoic. These deposits develop in various geological settings, including volcano-plutonic complexes and clastic sedimentary sequences, though they are not restricted to any specific lithology. A recurring feature is their spatial and temporal link to granitoids of varying compositions. Host rocks generally exhibit greenschist-facies metamorphism, though localized amphibolite or granulite facies conditions may occur. Prominent examples in Ghana include Obuasi, Akyem, Prestea, Bogosu, Konongo, Kenyasi, and Bibiani.

Gold mineralization is typically localized near major, deep-seated fault systems. These large-scale structures, extending hundreds of kilometers in length and several kilometers in width, often display intricate deformation histories. Economically viable ore deposits commonly form within secondary shear zones and faults, particularly at structural bends or strike variations along these crustal-scale fault systems. The style of mineralization varies with depth: shallow, brittle environments host stockworks and breccias, while brittle-ductile regimes feature laminated crack-seal veins and sigmoidal vein arrays. Deeper, ductile settings may contain replacement-style or disseminated orebodies.

The mineralization is either disseminated or vein-hosted and typically occurs late in the structural sequence, coinciding with or postdating peak metamorphism. Veins are predominantly composed of quartz, with subordinate carbonate and sulphide minerals. Additional accessory phases may include albite, chlorite, white mica (notably fuchsite in ultramafic rocks), tourmaline, and scheelite. Carbonate minerals such as calcite, dolomite, and ankerite are also present. Sulphides vary by setting, with pyrite, pyrrhotite, chalcopyrite, galena, sphalerite, and arsenopyrite being common. Gold is frequently associated with sulphides but may also occur as native grains. In volcano-plutonic environments, pyrite dominates in greenschist facies, whereas pyrrhotite is more

typical in amphibolite-grade rocks. Sedimentary-hosted mineralization often features arsenopyrite as the primary sulphide.

Detailed mineralogical studies reveal that gold primarily occurs within pyrite, mainly as inclusions. Pyrite crystals, ranging up to 1.5 mm in size, exhibit both euhedral and subhedral morphologies. In oxidized zones, gold is commonly found as inclusions or as liberated grains along the margins of goethite/limonite derived from pyrite alteration.

2.3.2 Alluvial Gold Deposits

Alluvial gold mining in Ghana has a rich history that dates back to the pre-colonial era, with artisanal and traditional miners exploiting gold deposits found in river basins and streams draining the country's gold belts. Over the years, prospecting efforts have expanded across all major gold belts and their intervening basins to locate new deposits, driven by the gradual decline in easily accessible gold reserves.

2.4 Historical Context and Early Mining Activities

Early mining activities were concentrated near major river systems such as the Birim, Pra, Offin, Ankobra, and Tano rivers, where gold was easily accessible in shallow alluvial deposits. These rivers drain Ghana's prolific gold belts—the Ashanti Belt, Sefwi Belt, and Bui Belt - which are known for their rich mineralization. Traditional miners primarily worked along riverbanks and floodplains, using rudimentary tools to extract gold from gravel and sand deposits.

2.5 Modern Exploration and Expansion

In recent years, geoscientists have employed detailed prospecting programs to identify new alluvial gold prospects not just along major rivers but also inland and within tributaries. Advances in geological surveying techniques, including geochemical sampling, geophysical methods, and remote sensing, have improved the accuracy of exploration, leading to discoveries in previously overlooked areas.

2.6 Depositional Environments of Alluvial Gold

2.6.1 Active Stream Gravels and Valley Flats

Gold particles are eroded from primary sources (such as quartz veins) and deposited in riverbeds where water flow slows down, allowing denser gold particles to settle.

2.6.2 Old (Paleo) Valleys and Terraces

Ancient River channels, now uplifted or abandoned, often contain reworked gold deposits that were once part of older drainage systems.

2.6.3 Beach Placers

Along Ghana's coastline, particularly near Elmina and Cape Coast, gold-bearing sands and gravels have been deposited by wave and tidal action.

For the most part, gold is typically fine-grained and widely disseminated, but in some areas particularly where the bedrock consists of soft clay or weathered material concentrations of coarser gold particles can be found. These "pay streaks" are often targeted by miners due to their higher gold content.

Modern alluvial mining has become more efficient and profitable due to:

Improved Equipment: Small-scale miners now use portable dredges, sluice boxes, and mercury-free gold recovery systems, while larger operations utilize heavy machinery like excavators and wash plants.

Improved Processing Techniques: Methods such as gravity separation, flotation, and cyanidation (for larger operations) have increased gold recovery rates.

Regulatory Support: Government initiatives, including the Mineral Development Fund, Community Mining Schemes and the recently formed GoldBod aim to formalize small-scale mining while minimizing environmental damage.

3.0 CURRENT EXPLORATION

3.1 Introduction

A structured and systematic alluvial prospecting program has been implemented with the objective of delineating alluvial gold deposits of economic significance within the project concession. The exploration approach integrated a multi-phase methodology encompassing desktop analysis, reconnaissance surveys, geophysical investigations, detailed sampling, mineral resource estimation, and technical evaluation. The principal activities executed during this phase are summarized below.

3.2 Desk Study and Reconnaissance

An extensive desktop review was undertaken, incorporating the analysis of regional geological maps, high-resolution satellite imagery, and historical exploration datasets. This preliminary phase was aimed at identifying geologically favorable zones for alluvial gold mineralization. Subsequent reconnaissance fieldwork was conducted to validate desktop interpretations, with a focus on evaluating geomorphological indicators such as river terraces, paleo-drainage systems, and floodplain environments - key features associated with alluvial gold deposition. Sedimentological characteristics and basin dynamics were also examined to refine target area delineation for subsequent detailed exploration phases.

3.3 Ground Geophysics Surveys

3.3.1 Resistivity Surveys

Electrical resistivity surveys were employed as a non-invasive geophysical technique to investigate subsurface lithological variations and delineate potential gold-bearing structures. The primary objective was to map buried paleo-channels, gravel beds, and other stratigraphic units associated with alluvial gold mineralization.

The surveys successfully delineated resistivity contrasts between overburden, clay-rich layers, and gravely horizons, thereby enabling the identification of key stratigraphic markers critical to alluvial gold exploration. High-resistivity anomalies corresponded to coarse-grained, consolidated gravel units—typically the principal host of alluvial gold—while low-resistivity zones were indicative of finer-grained clay and silt layers.

The resistivity data provided a reliable framework for interpreting paleofluvial architecture, including channel width, depth, and continuity. This facilitated the identification of paleo-channel networks and depositional environments conducive to gold concentration. As a result, the survey enhanced the precision of target definition for pitting, trenching, and bulk sampling operations, thereby optimizing subsequent excavation and resource evaluation efforts as demonstrated in Figures 5 and 6.

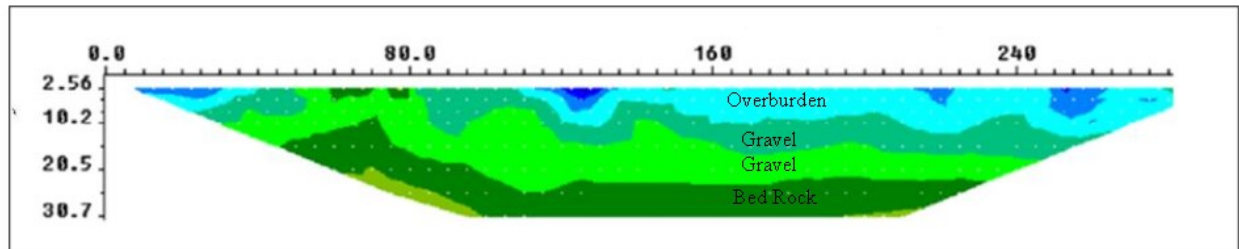


Figure 5: Pseudo-section of Measured Apparent Resistivity

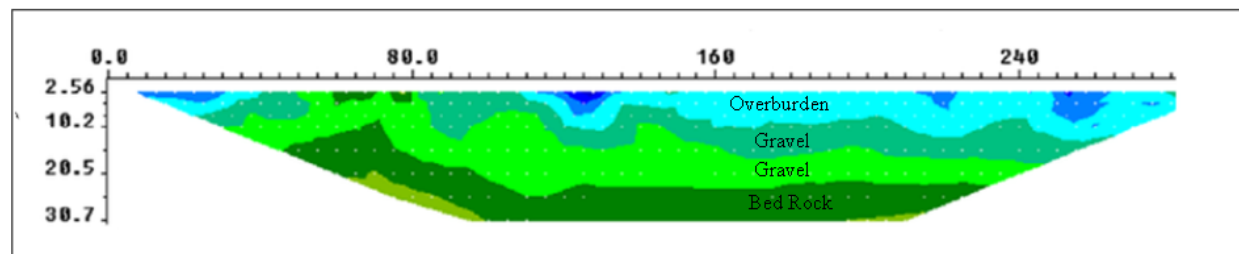


Figure 6: Pseudo-section of Calculated Apparent Resistivity

3.3.2 Induced Polarization (IP) and Magnetic Surveys

Induced Polarization (IP) and Magnetic surveys were conducted to further refine the geological understanding of the alluvial terrain of the concession. These techniques were used to:

Distinguish sedimentary layers by detecting variations in chargeability and conductivity, helping to separate mineralized zones from barren ones.

Locate paleo-channels enriched with placer gold deposits, as the presence of heavy minerals and associated sulfides often produce higher chargeability responses.

Identify variations in bedrock composition, aiding in the delineation of potential gold traps and structural controls.

Ground Penetrating Radar (GPR) was deployed to define gravel thickness and detect subsurface layering, providing critical data for subsequent pitting programmes.

3.4 Alluvial Prospecting

3.4.1 Characteristics of the Apapam Creeks

The concession is drained in the north by a network of tributaries into the Akuasua stream. The midsection and the southern part of the concession are drained by well-defined but unnamed streams.

Exploratory test work has revealed that auriferous (gold-bearing) gravel deposits extend significantly beyond the immediate confines of river catchments, suggesting a broader distribution of alluvial gold than traditionally assumed. This finding has important implications for exploration strategies, resource estimation and mining potential in both active and paleo-fluvial systems.

The Apapam Gold Project exhibits a highly prospective geological setting where tight creek systems are augmented by contributions from alluvial fans and colluvial materials, creating optimal conditions for placer gold concentration. This unique combination of depositional environments significantly enhances both the grade and spatial extent of recoverable mineralization.

The project's tight creek systems serve as efficient natural gold concentrators. These narrow, steep-gradient streams promote effective hydraulic sorting, with bedrock irregularities such as crevices and potholes acting as natural traps for coarse gold particles. The system's effectiveness is further amplified by the presence of alluvial fans, which form at topographic breaks where sediment-laden flows suddenly lose velocity. These fan deposits function like natural sluice boxes, preferentially concentrating heavier gold particles at their bases, often containing coarse gold due to minimal transport from proximal source areas.

Colluvial materials provide a continuous feed of gold into the system. Composed of eroded bedrock and saprolite washed down from adjacent slopes, these deposits introduce freshly liberated gold from nearby primary sources. In the Apapam area's steep topography, this colluvial contribution serves to regularly replenish placer deposits within the creek systems.

3.4.1.1 Sample Location

Sample locations were preselected using government-issued topographic maps, with coordinates extracted and uploaded into a high-precision handheld GPS for accurate field positioning. Each test pit was systematically collared at the surface with a precisely measured dimension of 1m x 1m and excavated through the overburden and gravel layers down to identifiable bedrock using local equipment, including pickaxes, shovels, and chisels (sorsor).

In instances where the predetermined location was deemed unsuitable for excavation, an alternative site within the same vicinity was selected. Once a sample location was established in the field, the 1m x 1m pit was excavated until the gravel horizon was intercepted. The siting and excavation of test pits adhered to standard industry protocols commonly employed in Ghana for alluvial gold exploration.

Field geologists recorded detailed site descriptions along with the geographic coordinates of each pit for documentation and analysis. This methodology ensured precision in sampling, adaptability in site selection, and compliance with Ghanaian mining exploration protocols, maintaining reproducibility and traceability for feasibility studies and regulatory submissions.

3.4.2 Phase 1: Exploratory pitting

The initial prospecting phase comprised systematic geological reconnaissance integrated with geophysical assessment. Test pits were strategically positioned across alluvial plains and terraces following a randomized sampling protocol designed for preliminary alluvial grade evaluation. This stage facilitated quantitative and qualitative assessment of gravel deposits, enabling targeted selection of high-potential zones for subsequent detailed investigation.

A total of ten (10) prospecting pits were excavated within selected drainage basins, including historically exploited artisanal workings, to evaluate gravel deposit characteristics. Geographic positioning of test sites was achieved using high precision handheld GPS instrumentation with <3m positional accuracy. Following phase completion, sites indicating positive gravel intersections were prioritized for further evaluation.

Preliminary metallurgical assessment involved panning of composite gravel samples extracted from pit sections. Details of the results from the initial pitting are presented in Table 4. This quantitative data informed subsequent resource delineation strategies and deposit modeling.

Table 4: Preliminary Pitting Results

SITE	OBT (m)	GRT (m)	Grade (gm ³)	Sample Medium	Nugget count
KP1	1.3	1.6	0.29	Test Pit	
KP2	0.5	0.5	0.27	Test Pit	2
KP3	1.4	0.8	0.95	Test Pit	
KP4	1.8	1.2	0.25	Test Pit	5
KP5	1.1	2.05	0.21	Test Pit	
KP6	1.4	0.9	0.80	Test Pit	12
KP7	1.7	0.7	0.86	Test Pit	
KP8	1.5	1.7	0.57	Test Pit	1
KP9	0.8	0.8	0.67	Test Pit	2
KP10	1.5	1.9	0.49	Test Pit	6

3.4.3 Phase 2: Detailed Alluvial Prospecting

3.4.3.1 Gridding and Surveying

Following the identification of prospective zones during the preliminary prospecting phase, a systematic and detailed pitting program was undertaken to further evaluate the spatial continuity and distribution of gold-bearing gravel horizons. This phase employed a standardized grid-based approach, designed by the geological team and executed in the field under the supervision of qualified field technicians.

The grid layout was established across delineated drainage systems, specifically along valley flats and plains interpreted to host alluvial mineralization. A regularized 100m x 100m grid interval was adopted to ensure consistent spatial coverage and data comparability across the target areas.

Grid lines were accurately demarcated in the field using handheld GPS units and flagging, with pit collar positions precisely located at grid intersections. This structured approach enabled systematic excavation, sampling, and logging of test pits, providing high-resolution data on gravel thickness,

lithological variability, and gold distribution. The detailed grid-based methodology significantly enhanced the understanding of mineralization continuity, facilitated volumetric resource estimation, and informed subsequent resource modeling and economic evaluation efforts.

3.4.3.2 Pitting

After the successful completion of initial prospecting identified significant mineralized zones a structured 100m × 100m grid sampling program was implemented. This systematic approach resulted in the excavation of 156 prospecting pits across the targeted zones. Ground condition complications led to the abandonment of 14 pits either through water seepage or total collapse. Therefore, the resulting effective pits that were used in the evaluation were 142.

The exploration grid configuration and mineralized envelope boundaries is presented in Figure 7 showing detailed pit locations and resource outline. This high-density sampling matrix ensured robust alluvial evaluation of the Akim Apapam Gold Project.

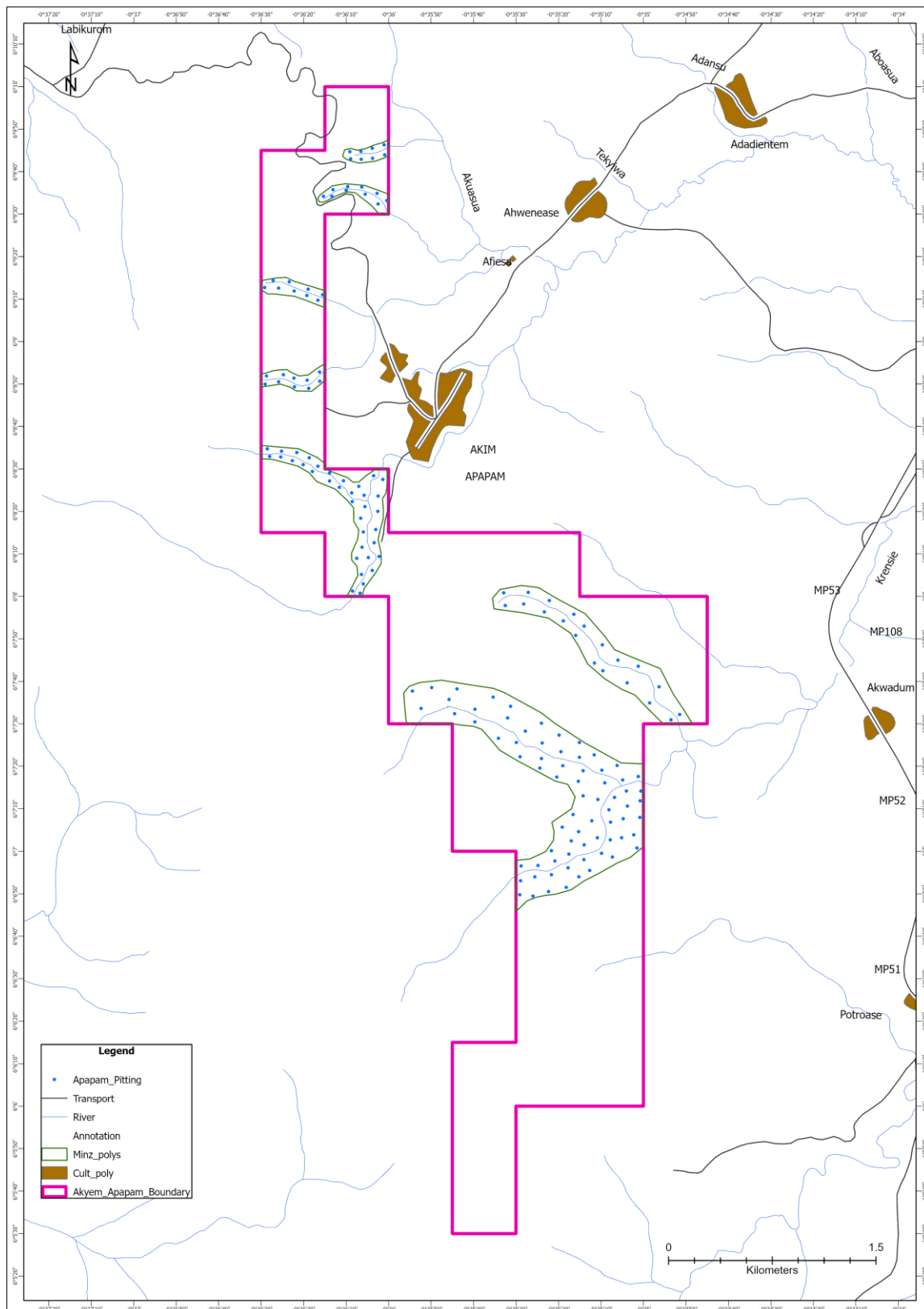


Figure 7: Layout of detailed pitting highlighting mineralized envelopes

Pre-determined sampling coordinates were accurately located in the field using high-precision handheld GPS units to ensure spatial accuracy. Upon confirmation of each sample point, excavation was initiated and carried out in accordance with standardized industry practices widely adopted in Ghana for alluvial gold investigations.

Each test pit was systematically excavated through the overburden and auriferous gravel layers until the competent bedrock surface was encountered. Manual excavation was undertaken using locally available traditional tools, including pickaxes, shovels, and chisels (*sorsor*), ensuring effective penetration through variable sedimentary layers. A representative pit under excavation is illustrated in Figure 8.



Figure 8: Pitting in progress

A distinct lithological contact between the overburden and gravel horizons was consistently observed across multiple test pits. The gravel pay zone was systematically extracted in its entirety and properly contained for downstream processing. Continuous sampling was conducted at 0.5-

meter vertical intervals throughout the gravel horizon, with each interval securely packaged to maintain sample integrity.

To optimize gold recovery and assess potential grade dilution, an additional 0.3-meter penetration into the bedrock was implemented, targeting preferential gold accumulation within structural discontinuities and weathering profiles at the gravel-bedrock interface. This methodology ensured comprehensive material characterization while accounting for potential metallurgical losses during extraction.

The field geologist overseeing the pitting operations conducted systematic logging of each excavation, documenting both subsurface characteristics and immediate environmental conditions.

3.4.3.3 Gravel Processing and Treatment

Sluicing represents the predominant technique for gravel treatment in global alluvial mining operations, utilizing cost-effective sluice box constructions fabricated from readily available materials such as wooden planks or metal sheeting. The standard sluice configuration consists of a wooden trough measuring 0.3 meters in width, 3 meters in length, and 0.04 meters in depth, structurally reinforced with lateral wooden battens to contain feed material during processing. The interior substrate incorporates specialized carpet linings optimized for the retention of gold particles across all morphological classifications, including nuggets, granular grains, fine particles, and flake formations.

Feed material quantification employs a $0.5\text{m} \times 0.5\text{m} \times 0.4\text{m}$ (0.1m^3 capacity) wooden measuring box, with cumulative processing volumes calculated through systematic box-count methodology. This volumetric measurement protocol facilitates accurate reconciliation between processed gravel quantities and in-situ bank volumes. Processing involves individual sample batches washed under controlled hydraulic conditions, followed by manual panning of concentrates using standardized swirling techniques to isolate valuable constituents, ensuring representative grade determination for each sampled interval.

The washing regime adapts to material characteristics, employing high-pressure water jets for clay-dominated matrices to achieve effective disaggregation and carpet scrubbing, while sandy

substrates undergo low-pressure washing to prevent gold particle displacement. This differential washing approach optimizes recovery efficiency while maintaining the integrity of gold-bearing concentrates throughout the beneficiation process.

3.4.3.4 Extraction Processes

The pan-derived concentrate underwent systematic refinement through sequential beneficiation stages. Initial processing involved manual sorting followed by desiccation and gravimetric measurement. Given the predominance of medium-to-fine gold particulates in the concentrate matrix, mercury amalgamation was employed to optimize fine gold recovery. This metallurgical process facilitated the formation of a mercury-gold amalgam, with subsequent retorting to volatilize the mercury and isolate the precious metal fraction.

The recovered gold was quantified using a precision electronic balance capable of 0.01-gram resolution, ensuring accurate metallurgical accounting. This rigorous protocol enabled precise determination of gold content within each concentrate sample.

3.4.3.5 Weight Measurement

Gold is typically weighed in troy ounces, with 1 troy ounce equivalent to 31.1034768 grams. However, its purity is commonly measured in 'carats'. Carats denote the degree of purity of gold when alloyed with other metals. Pure gold, with no other metals added, is rated at 24 carats.

A standardized weight of 60 grams was carefully positioned on the scale to verify the machine's calibration and ensure its proper functioning before measuring the actual gold grains. This calibration process was conducted indoors, with all air conditioning units and fans deactivated to prevent external factors from affecting the weight measurements. Gold grains that did not record any value on the scale were assigned a below detection limit value of (<0.03 grams) and a zero (0) value to pits with no gravel recovery. Figure 9 shows the weight measurement of recovered gold undertaken using high precision scale.

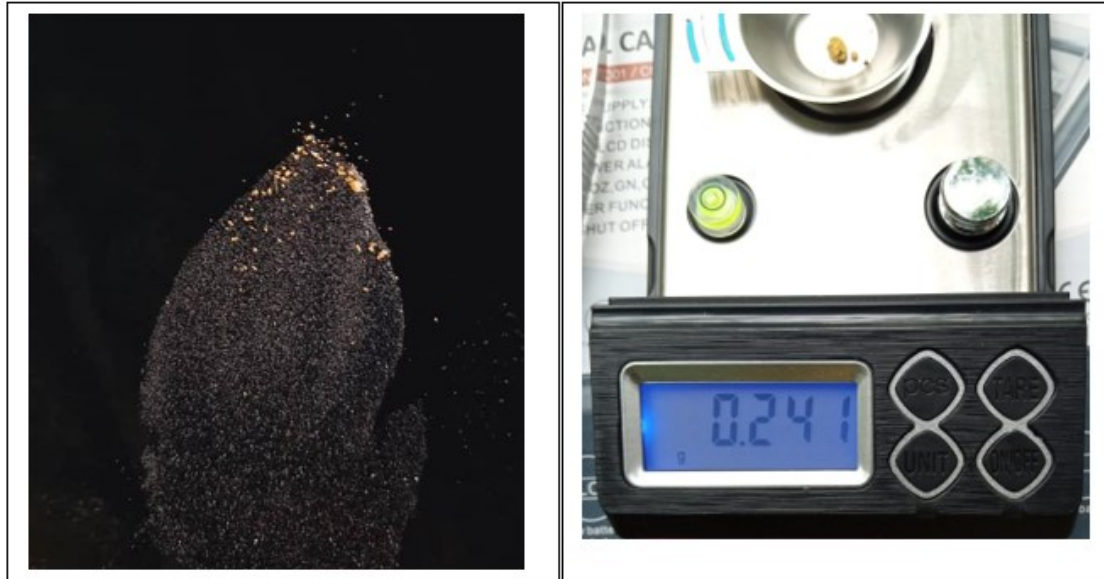


Figure 9: Weight measurement of recovered gold

3.4.3.6 Grade Calculations

The economic valuation of alluvial gold deposits relies on quantitative grade assessment, defined as the concentration of metallic constituents within the ore matrix. Grade quality correlates directly with gold particle density per unit volume, expressed in grams per cubic meter (g/m^3). This metallurgical parameter was derived through gravimetric analysis, combining precise mass measurements obtained from calibrated electronic scales with volumetric data collected during field sampling using standardized 0.1 cubic meter containment units.

The grade (ρ) was calculated through the fundamental density equation $\rho = m/V$, where m represents the measured mass in grams and V denotes the sampled volume in cubic meters. This calculation methodology is illustrated in Figure 12, demonstrating the direct relationship between mass and volume in determining ore grade. The resulting values provide critical data for resource estimation, economic evaluation, and metallurgical recovery planning.

The weighted average method was applied to different gravel volumes from the same pit that had all their gold grains registering weights on the scale. This was done by multiplying its relative proportion or percentage by the volume in the sequence and adding those sums to obtain a single grade for an intercept.

3.4.3.7 Particle Size Distribution

The alluvial gravels displayed poor granulometric sorting, exhibiting a polymodal particle size distribution ranging from clay (<0.002 mm) to cobble/boulder fractions. Stratigraphic analysis revealed a matrix-supported fabric, with fine-grained sediments constituting the dominant interstitial material. Particle size classification followed standardized geotechnical parameters:

- Gravel: >2 mm
- Sand: 0.05–2 mm
- Silt: 0.002–0.05 mm
- Clay: <0.002 mm

Processing optimization employed size-selective separation at 6 mm, with the undersize fraction (-6 mm) beneficiated through:

1. Primary concentration: Inclined sluice with riffled Astroturf lining
2. Secondary recovery: Recirculation of sluice tailings
3. Quality control: Manual panning verification of +6 mm tailings

The -6 mm fraction underwent tertiary concentration via traditional Calabash panning. The gravel particle size distribution is shown in Table 5.

Table 5: Gravel Size Distribution

Scream size (mm)	% Fractions
Clay	27
- 6	33
+ 6	18
+ 15	7
+ 45	5
+ 60	6
+ 100	4

3.4.3.8 Volume Calculations

The resource estimation involved calculating the volume of auriferous gravel by multiplying the average gravel thickness with the surface area of the pay zone, and determining the gold content as the product of this volume and the average gravel grade. A polygonal method was used to define

each sample's area of influence, adjusting for edge samples using natural boundaries. The gravel thickness within each polygon was incorporated to refine volume estimates.

This method, suitable for deposits with simple geometry and moderate grade variability, produced a global estimate through a weighted linear combination of sample data. Mineralized zones were classified according to JORC standards: areas with sample spacing between 50–100 meters were categorized as Measured or Indicated, while more widely spaced or less certain areas were classified as Inferred.

3.5 Resource Estimation

The following critical parameters were systematically evaluated to establish comprehensive resource definition for the alluvial deposit:

3.5.1 Overburden Thickness

Measurement of unconsolidated superficial material overlying the productive gravel horizon. This parameter determines stripping ratios and influences extraction economics by quantifying the volume of non-economic material requiring removal to access mineralized zones.

3.5.2 Gravel Thickness

Assessment of the productive alluvial layer thickness containing economic mineralization. This measurement is essential for calculating reserve tonnages and establishing the vertical extent of mineable material within each excavation unit.

3.5.3 Bedrock Lithology

Characterization of the underlying geological substrate controlling gravel deposition patterns and gold concentration mechanisms. Bedrock composition influences hydraulic conditions during placer formation and determines the morphology of auriferous zones.

3.5.4 Average Gold Grade

Quantitative determination of gold content per unit volume of processed gravel material. This parameter establishes the economic viability of extraction operations and forms the basis for resource classification and mine planning calculations.

3.5.5 Paleo-channel Zones

Identification and delineation of ancient drainage systems exhibiting enhanced gold grades and optimal gravel characteristics. These buried channel systems represent primary targets for detailed exploration due to their elevated economic potential and concentrated mineralization patterns.

The alluvial gold resource was calculated using a grid pitting method, focusing on areas with consistent auriferous gravels. A lower cutoff grade of 0.20 g/m³ was applied, implying only pit sections meeting or exceeding this threshold were included in the resource estimation. This method not only identified economically viable gold-bearing material but also delineated distinct zones within the concession. These zones aligned with the area's drainage patterns, providing further understanding into the depositional environment and the spatial distribution of gold within the alluvial and colluvial system.

For the Akim Apapam alluvial gold estimation, the polygonal method was employed. This approach uses spatial data to create polygons around each sample point, ensuring that every location within a polygon is closer to its assigned point than to any other.

The polygonal method (area of influence method) was employed for resource estimation of the Apapam alluvial gold deposit. This geostatistical technique assigned areas of influence to the prospecting pits based on geometric proximity principles, with each sample point representing the nearest and most representative data for its designated polygon.

Voronoi tessellation construction was carried out using algorithms within ArcGIS Pro environment. Perpendicular bisectors were constructed between adjacent sample points. The tessellation process defined individual polygons where each encompasses the area closest to its respective pit location.

The Voronoi tessellation generated distinct polygons corresponding to the alluvial prospecting pit locations. Polygon boundaries were refined to exclude non-mineralized areas and conform to the geological limits of the alluvial deposit.

The polygon method involves assigning different impact polygon to each sample. The areas of these polygons are then used as weights to remove clustering. The boundaries of the polygon are determined by the perpendicular bisector of a line segment. This bisector is a line where points are equidistant from either end of the line segment. The perpendicular bisectors between a sample and its neighbors establish the boundaries of the polygon of influence.

The result of polygonised zones with the various areas of influence is as shown in Figure 10.

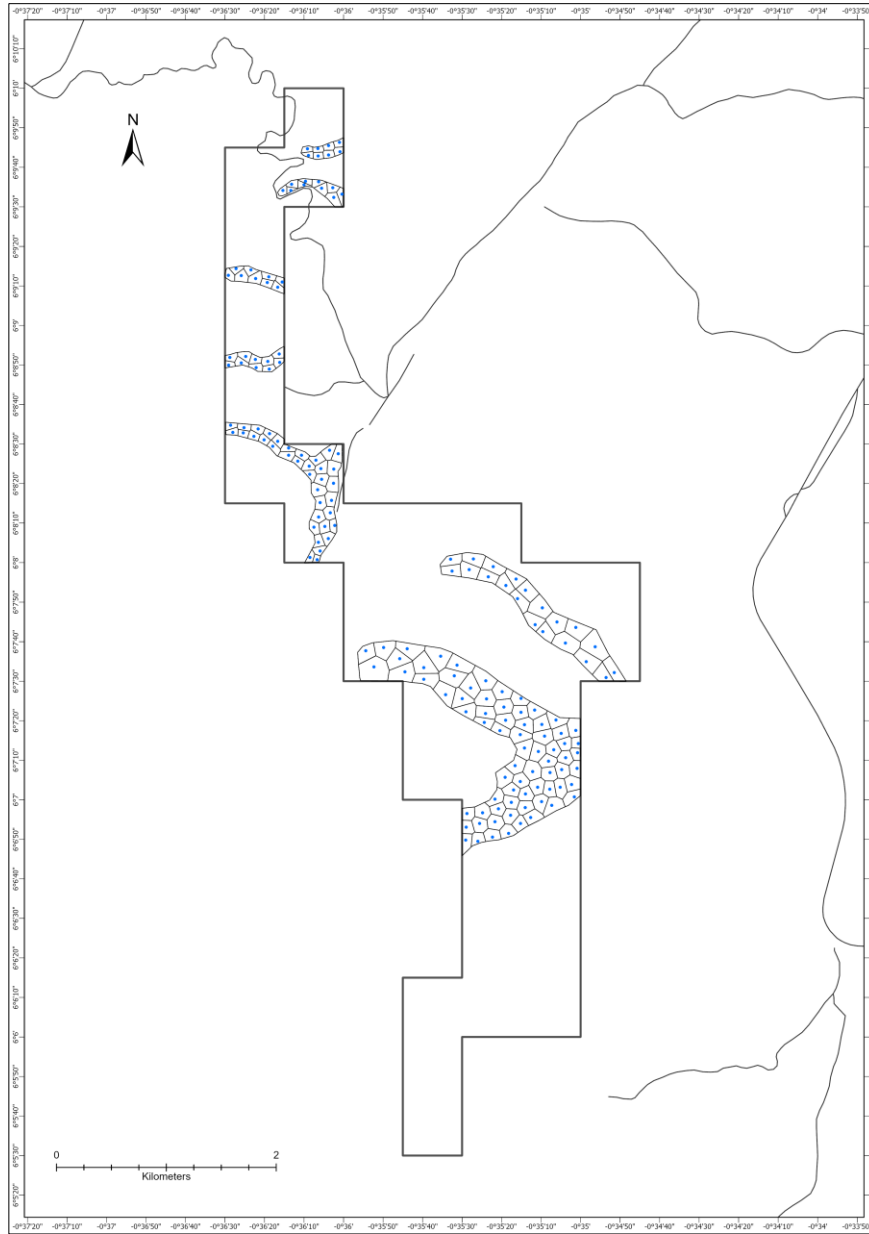


Figure 10: Polygonization of target areas

3.6 Ore Reserve Estimation

The advanced pitting program maintained inter-pit spacing of less than 100 meters throughout the majority of the exploration area. This systematic grid density enabled the classification of resources into either Indicated or Measured categories, reflecting the high confidence levels achievable through close-spaced sampling. The spatial distribution of sample points provided sufficient interpolation control to support elevated resource confidence classifications in accordance with acceptable reporting standards.

The conversion of resources to reserves was restricted exclusively to samples falling within the Measured and Indicated resource categories, ensuring adherence to industry standards for reserve classification. This conservative approach excluded all Inferred resources from reserve calculations, maintaining the requisite confidence levels for mining feasibility assessment and operational planning.

A total of 135 test pits met the stringent criteria for inclusion in reserve calculations, representing the most reliable and well-constrained portion of the total resource database. Reserve delineation was based on the implementation of a conservative cutoff grade of 0.20 g/m³ applied to the total pit section composite grade. This economic threshold established the fundamental criterion for distinguishing between economically pay-load and sub-economic (waste) material within the deposit.

Pit sections demonstrating total composite grades equal to or exceeding 0.20 g/m³ were classified as pay deposit material. This classification formed the basis for economic evaluation and mine planning calculations. The projected alluvial ore reserve estimate is shown in Table 6.

Resource Classification

- **Resource Category:** Measured or Indicated
- **Grade Threshold:** Total pit section grade $\geq$ 0.20 g/m³
- **Spatial Control:** Inter-pit spacing $\leq$ 100 meters

Reserve Classification

- Proven and Probable Reserves were derived from Measured Resources with demonstrated economic viability

Table 6: Projected Alluvial Ore Reserve Estimate

Grid	Average Gravel Thickness (m)	Overburden Thickness (m)	Ore Block Area (m ²)	Block Gravel Volume (m ³)	Average Grade (g/m ³)	Assigned Weight (%)	Weighted grade (g/m ³)	Gold Content (g)	Gold Content (oz)
Akim Apapam	1.50	1.3	1,939,500	2,909,250	0.45	100.00	0.45	1,309,162.50	42,091
Total/Avg			1,939,500	2,909,250	0.45			1,309,162.50	42,091

31.103 g = 1 ounce

The projected reserve estimate form the technical foundation for the current feasibility study, demonstrating the deposit's potential for economic extraction under the proposed mining lease. The estimated parameters indicate a viable alluvial gold deposit that confirms a substantial gold reserve with consistent grade distribution across the deposit.

The Summary presentation is as shown below;

- **Gold Content:** 42,000 ounces (42 koz Au)
- **Reserve Volume:** 2.91 million cubic meters
- **Average Grade:** 0.45 g/m³
- **Average Gravel Thickness:** 1.5 meters

The ore reserve polygonised blocks are as shown in Figure 11.

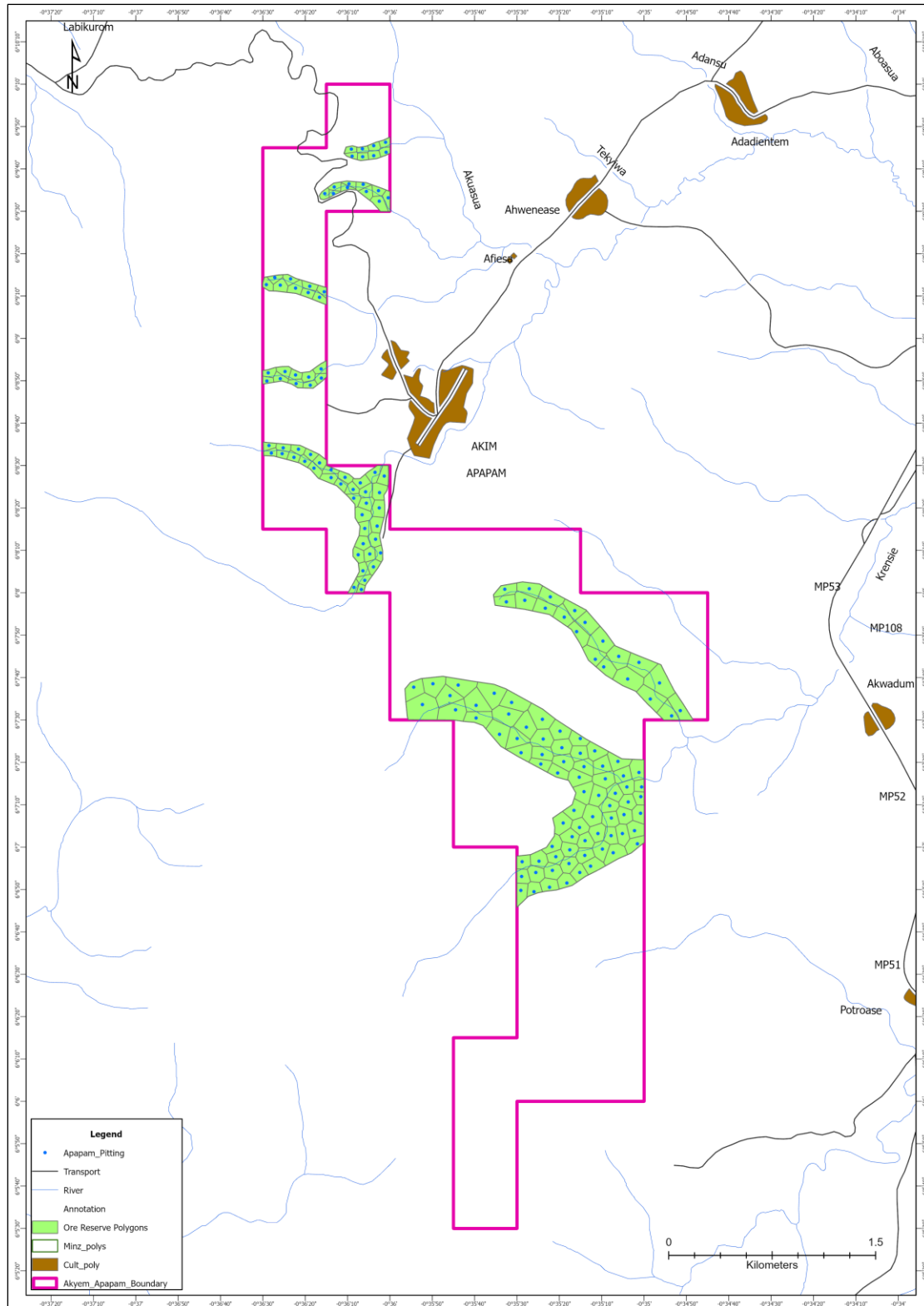


Figure 11: Polygonized ore blocks forming delineated reserve pits.

3.6.1 Proposed Mining Lease

The company intends to retain the entirety of the concession bounded by the pillar coordinates given in Table 7. The area of the proposed lease is given as 8.4 km² which translates into approximately 40 cadastral blocks.

Table 7: Proposed Mining Lease Block Coordinates

PILLAR	LONGITUDE	LATITUDE
P1	0° 36' 15"	6° 10' 00"
P2	0° 36' 00"	6° 10' 00"
P3	0° 36' 00"	6° 09' 30"
P4	0° 36' 15"	6° 09' 30"
P5	0° 36' 15"	6° 08' 30"
P6	0° 36' 00"	6° 08' 30"
P7	0° 36' 00"	6° 08' 15"
P8	0° 35' 15"	6° 08' 15"
P9	0° 35' 15"	6° 08' 00"
P10	0° 34' 45"	6° 08' 00"
P11	0° 34' 45"	6° 07' 30"
P12	0° 35' 00"	6° 07' 30"
P13	0° 35' 00"	6° 06' 00"
P14	0° 35' 30"	6° 06' 00"
P15	0° 35' 30"	6° 05' 30"
P16	0° 35' 45"	6° 05' 30"
P17	0° 35' 45"	6° 06' 15"
P18	0° 35' 30"	6° 06' 15"
P19	0° 35' 30"	6° 07' 00"
P20	0° 35' 45"	6° 07' 00"
P21	0° 35' 45"	6° 07' 30"
P22	0° 36' 00"	6° 07' 30"
P23	0° 36' 00"	6° 08' 00"
P24	0° 36' 15"	6° 08' 00"
P25	0° 36' 15"	6° 08' 15"
P26	0° 36' 30"	6° 08' 15"
P27	0° 36' 30"	6° 09' 45"
P28	0° 36' 15"	6° 09' 45"

4.0 MINING

4.1 Introduction

The alluvial prospect within the Akim Apapam concession has an established total reserve of 2,909,250 m³ at an average grade and gravel thickness of 0.45 g/m³ and 1.5 m respectively. With an estimated overburden volume of 2,531,048 m³ (volume of waste material) to be mined, an estimated overburden-to-gravel stripping ratio of 0.87:1 has been projected, which implies that 0.87 m³ of overburden must be removed for every 1 m³ of auriferous gravel extracted.

Mining activities will utilize heavy-duty machinery, including excavators, dump trucks, backhoes, dozers, wheel loaders, graders, and water tankers. Well-organized mining systems will be implemented to ensure optimal recovery, minimize costs, and maximize revenue, thereby sustaining the project throughout its mine life.

Existing access routes developed during the exploration program will provide initial access to the mining site.

This section outlines the mining method to be adopted on the project.

The section provides details of the parameters used in the selection of the.

- mining method,
- safe geotechnical conditions,
- mining schedule,
- equipment,
- personnel requirements, and
- water management.

4.2 Mining Method

The mining department's main objective is to provide ore to meet the requirement of the processing requirement (80 m³/h). The selected method will therefore ensure optimum mining and hauling of the ore to the plant. The mining method selected will have to ensure optimum recovery of ore, concurrent reclamation and minimum damage to the environment. Some alluvial mining methods are discussed below.

The mining department is mandated to ensure smooth mining operation, the supply of ore to the processing plant and to have adequate gravels ore at the stockpile. Hujack Mining will secure efficient trucks with good fuel consumption to reduce transportation costs.

The mining operations conducted in the Kwaebibirem district utilize a haul and transport system, which optimizes ore recovery and ensures efficient use of mining equipment. Trucking ore contributes to maintaining the cleanliness of the open pit area and enhances machinery maneuverability. While the transportation of ore to the stockpile incurs costs, it offers significant benefits, such as reducing theft and enabling controlled washing processes. Furthermore, it ensures maximum security surveillance within the mine site yard, as opposed to operations conducted in remote bush areas.

Additionally, haulage trucks will transport slime and tailings on their return trips to the open pit area, where they will be used for backfilling the pits.

4.2.1 Mining Method Parameters

The company is considering the following parameters in selecting an appropriate mining method for the Akim Apapam Gold Project:

- I. It will be technically and economically feasible
- II. It shall have negligible effect on the immediate environment
- III. The equipment will be adaptable in the terrain and suitable to the weather conditions
- IV. The method will be adjustable to the ground conditions present in the different mining blocks
- V. The method should lessen ore dilution from overburden and barren bedrock
- VI. The system should uphold high level of safety for workers and residents in the locality of the operations.

4.2.2 Ore Deposit Characteristics

The alluvial ore deposit on the concession has the following characteristics:

- I. The overburden is generally barren, with the grades increasing with proximity to the gravel layer on the alluvial plains, but within the terrace environment the grades are not depth dependent.

- II. The boundary between the overburden and the ore is distinct.
- III. The average overburden thickness is 1.3 m thick.
- IV. The gold bearing gravel layer is averagely 1.5 m
- V. The ore matrix is clayey within the eluvial regions and silty sand on the alluvial plains.
- VI. The average stripping ratio is of 0.87:1.
- VII. The bedrock recorded less mineralization and barren at depth as well

4.2.3 Strip Mining

Strip mining, which involves the removal of the topsoil or rock (overburden) above the ore layer to extract the exposed mineral, can be completed in rectangular blocks in plain view called pits or strips by the local operators, and this exposes the gold bearing gravels to stripping extraction. According to mining experts, a typical alluvial deposit with a vertical cross section made up of vegetation, topsoil (0.2 m), average overburden thickness of 4 meters, gold-bearing gravels (1.2 m) and the underlying bedrock (0.3 m) respectively can be mined using both manual and mechanized mining methods (Table 8).

Construction of haul roads will be carried out by the use a dozer in conjunction with a grader. Clearing the land in preparation for mining will be done using a dozer. Striping will be done using backhoe, excavators assisted by the dozer when necessary and applicable. Loading and hauling of ore will be carried out using excavators and dump trucks. Mining sequence will be carried out to ensure a continuous reclamation for the restoration of the farmlands for immediate cropping season.

Table 8: Mining Method Selection Template for Hujack Mining

S/N	Type of Deposit	Proposed Mining Method
1	Alluvial / elluvial / colluvial deposits	Strip mining
2	Weathered gold loads, phyllitic, quartzitic and lateritic materials	Terrace mining
3	Auriferous lode bearing gravel deposits in hilly terrains or flat plains in lateritic cover with depth >5 m < 20 m	Adit mining (ancient pits done during the colonial era)

4.2.4 Mining Schedule Analysis

4.2.4.1 Mining Schedule

The company has established a total proven and probable alluvial reserve of 2,910,000 m³ with a weighted average grade of 0.45 g/m³ and an average stripping ratio of 0.87:1. The ore will be processed through a gravity concentration plant with a designed capacity of 80 m³ per hour. The primary objective of the mining operations is to systematically extract and transport ore to maintain continuous plant feed while establishing adequate stockpile reserves to prevent processing interruptions.

4.2.4.2 Operational Parameters

The mining department will implement a three-shift operational schedule, with each shift operating for 8 hours to supply ore at a rate of 80 m³/h to the treatment facility. The crew will operate on a five-day weekly schedule, with the remaining two days allocated for planned maintenance, equipment servicing, operational planning and resting. The processing plant will undergo scheduled maintenance shutdowns for sluice box cleaning operations, including carpet removal and gold recovery procedures, requiring approximately 20 minutes per cleaning cycle.

4.2.4.3 Production Scheduling

The mining schedule has been optimized to account for initial site preparation, infrastructure development and operational ramp-up requirements. Production output is anticipated to be constrained during the first year due to mine development activities, equipment mobilization, crop compensation issues and regulatory compliance procedures. Steady-state production is projected to commence in Year 2, with peak production rates achieved during Years 3-5 when operational efficiency reaches optimal levels.

4.2.4.4 Projected Gold Production

Table 10 presents the annual projected gold production values over the 6-year mine life. The production schedule accounts for operational constraints, seasonal variations and equipment availability factors. Initial production of 4 koz is projected for Year 1, ramping up to peak

production of 9 koz during Years 4-5, before declining to 6 koz in the final year due to reserve depletion and mine closure preparation.

Table 9: Projected gold production values in koz

Year	Gold (koz)	Gravel Volume (m ³)	Cumulative Production (koz)
1	4	277,071	4
2	6	415,606	10
3	8	554,141	18
4	9	623,358	27
5	9	623,358	36
6	6	415,606	42
Total	42	2,909,250	

Figure 12 shows the annual gravel volumes processed (bar chart) and the corresponding gold production (line plot) over the 6-year mine life.

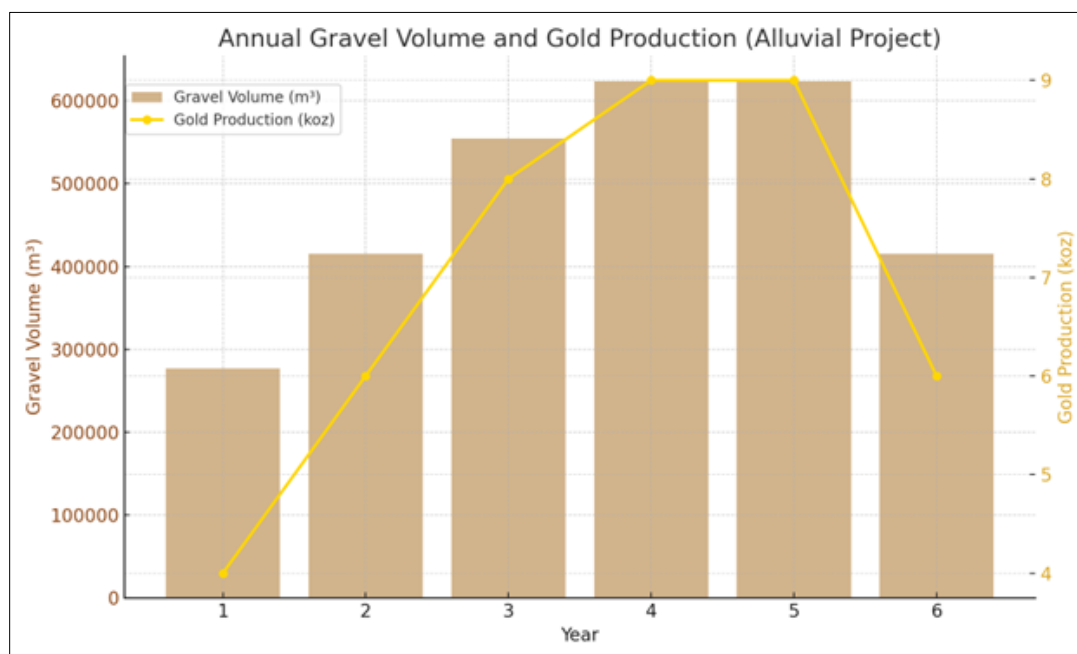


Figure 12: Annual Gold production schedule

4.2.5 Geotechnical considerations - Design Parameters

4.2.5.1 The project backgrounds

The Akim Apapam Gold Project holds potential for both alluvial and hard rock gold exploitation, though recent exploration efforts have prioritized the development of alluvial deposits within the concession. Alluvial operations are advantageous due to their minimal geotechnical complexities and operational challenges.

To maintain safety and mitigate pit-wall failures, mining activities will be strategically confined to terrace terrains during rainy seasons and flood plains during dry seasons. Proactive mine planning has already addressed potential hazards, with waterlogged areas identified and demarcated during comprehensive pitting and site inspection programs.

4.2.5.2 The site geology

The lease area's geology primarily consists of laterite, clay, sand, and silt. The gravel deposits terminate at a parent material layer identified as saprolites, which form the bedrock exposures. Pit excavations revealed that the underlying bedrock likely originates from metasedimentary and metavolcanic units present across the property. The gravel payload itself is composed of heterogeneous mixtures of pebbles, cobbles and boulders.

4.2.5.3 Hydrogeology

The proposed mine pits are predominantly situated on hill slopes and plateaus, areas with minimal to negligible groundwater recharge. During the rainy season, surface runoff will be directed into adjacent streams. To maintain pit stability and prevent cave-ins, the hydro department will implement a dewatering system to manage water accumulation within the active mining pits.

4.2.5.4 Geotechnical Design

Detailed surface topographic data obtained from prospecting activities has been used to generate a preliminary Digital Terrain Model (DTM). This initial model will be enhanced through drone surveys to achieve higher precision in topographic mapping.

Initial open pit configurations are designed with an overall slope angle of 40°, which will be subsequently refined based on geotechnical assessments. Given the lateritic to clay overburden

composition at the Hujack Mining Project, final batter angles are expected to range between 35°-45°. These angles may require adjustment with increasing depth and variations in ground competency.

In unconsolidated or weathered overburden zones, bench heights will be maintained at 2.0-5.0 meters. Hard rock sections may accommodate significantly higher benches due to their greater structural integrity. A series of 2.0-5.0-meter benches will be implemented immediately above the gravel bed interface to minimize dilution and facilitate efficient loading and haulage of both waste material and ore from the pit.

- The shallower bench heights in overburden zones address stability concerns in weaker materials
- The transitional bench design at the gravel contact optimizes material handling while controlling grade dilution
- Slope angle flexibility accommodates variable ground conditions encountered with depth

4.2.5.4.1 Road Designs

The Akim Apapam project's haul roads will be designed considering construction, maintenance, geometry, materials, and truck specifications. Key parameters include road width, gradient, camber, and curvature radius to ensure safe and efficient operations. Durable materials and proper drainage will be incorporated to maintain road integrity under heavy vehicle traffic. Regular maintenance will preserve optimal conditions throughout the mine life.

The design will balance operational needs with safety requirements, including adequate signage and visibility. Final specifications will be determined through site-specific engineering assessments prior to construction.

4.2.5.4.2 Road Width

The company will ensure that the width of haul roads on both straight and curved sections are adequate to permit safe vehicle maneuverability and maintain road continuity. It is also within the plan of the company to have road design width that will not be too narrow to cause trucks to have their tyres run on the berms when side-passing other vehicles. The road widths will be 2 to 3 times the widest heavy hauler, this will ensure that future heavy trucks fit into the haulage system.

4.2.5.4.3 Gradient

For safety and draining reasons, long steep gradients such as 30 m will include a padding truncation point which will serve as break point for vehicles to readjust stability and maneuvering.

4.2.5.4.4 Camber

Camber design will be incorporated in areas with gullies and creeks which will provide cross slopes that will raise the middle of the road surface to drain off the water to avoid pollution of the streams. Management will also install road signs to help truck drivers reduce speed limits when approaching camber and hazardous areas. The cross slopes should be approximately 1:30 to ensure proper drainage of the road.

4.2.5.4.5 Radius of Curvature

The company will ensure that curves and switchback design will improve the trucks performance since poorly designed curves will increase the haulage cost and create queuing. Hujack Mining engineers will pay close attention to those rugged terrains to ensure seamless transportation.

4.2.5.4.6 Unit Procedures in Mining

The mining operations will follow a systematic sequence beginning with site preparation, which involves clearing vegetation and surface debris while separately removing and storing topsoil for rehabilitation purposes. This is followed by overburden removal, where waste material is systematically stripped and stockpiled to expose the pay gravel horizons.

Once the ore body is accessible, excavation and loading of mineralized gravels commence, with material being loaded into haul trucks for transport. The ore is then delivered to the processing plant feed stockpile, while waste material is transferred to designated dump sites.

This structured approach ensures efficient material handling while maintaining clear separation between waste and ore streams. Each stage adheres to strict safety and operational protocols, with continuous progress monitoring to optimize recovery and minimize dilution. The integrated workflow is designed to maximize productivity while ensuring environmental compliance throughout the mining cycle.

4.2.5.4.7 Ground and Surface Clearing

To prepare a mine block for mining, the removal of trees and other vegetation is necessary. Chainsaw machines will be used to push down trees, and any timber that can be utilized for civil works within the mine will be collected. A D9 dozer will be used for clearing the site effectively.

4.2.5.4.8 Pre-Stripping (Removal of Topsoil)

The operation will implement a structured topsoil management protocol focusing on the preservation of the biologically rich A-horizon (0-30 cm). This organic surface layer will be carefully stripped using low-impact equipment and stored separately to maintain its structural and biological integrity. Stockpiles will be strategically located within 200m of active work areas, with heights limited to 2.5m to prevent compaction and oxygen depletion. Dedicated storage zones will incorporate drainage systems and remain isolated from operational traffic.

Handling will occur under dry conditions (<12% moisture) to preserve soil quality. Stockpiles exceeding 90-day storage will be covered with geotextile fabric and monitored for microbial activity. During rehabilitation, laser-guided equipment will ensure precise 30cm topsoil redistribution, followed by immediate stabilization measures. This systematic approach optimizes soil conservation while supporting efficient mine reclamation.

The plan integrates operational requirements with ecological restoration objectives, ensuring the preserved topsoil maintains its rehabilitation potential throughout the project lifecycle.

4.2.5.4.9 Removal of Overburden

Overburden refers to the non-economic material overlying the mineralized zone that must be removed to access the ore body. The operation will employ an optimized stripping approach where overburden is directly stockpiled in adjacent, pre-planned areas rather than transported off-site. This methodology minimizes haulage costs while providing readily available material for subsequent rehabilitation activities.

The excavation process will utilize a backhoe excavator as the primary stripping unit, selected for its precision in selective material removal. A bulldozer will provide supplementary support for

stockpile formation and grade control. This equipment combination ensures efficient overburden handling while maintaining a minimal operational footprint.

Key operational advantages include reduced transportation requirements, immediate availability of backfill material for reclamation, and minimized disturbance to surrounding areas. The approach demonstrates an effective balance between operational efficiency and environmental management, supporting sustainable mining practices throughout the project lifecycle.

The methodology aligns with industry best practices for overburden handling while optimizing both economic and ecological outcomes. Regular monitoring will ensure stockpile stability and maintain material suitability for future rehabilitation use.

4.2.5.4.10 Excavation and Loading of Ore

The removal of the overburden exposes the gold-bearing gravel (ore) for excavation and loading. In areas where the ore is compact, considerable amount of force will be required for excavating. Loose ore (gravel) will not require much force to excavate. The choice of appropriate equipment is therefore essential. Excavators will be the best choice, since they are able to navigate their way in the pit and can easily hold up collapsing walls more firmly than any other machine. The excavators can also load the trucks easily and be able to compress the ore with their buckets to prevent ore from falling off on the haul route during trucking.

Excavation of the gravel will be undertaken with extreme care to avoid dilution. Strict supervision and constant grade control (sampling and panning) will be undertaken to ensure that under/over mining is kept to a minimum, especially in areas with sparse or abundant gravel bands. Higher gold values can be expected within the basal gravel horizon within the river valleys; for this reason, it will be necessary to include a minimum amount of bedrock with the bottom gravels (about 30cm). It will not be prudent to discard the topmost gravel layer on the pretext that grades increase at depth without sampling the different layers to ascertain them qualitatively. Detail grade control during the mining operation will determine the proper removal of the top gravel layer and will also aid in categorizing the ore by low, medium and high grade for proper blending during treatment.

An Excavator has been selected as the most suitable for the operations due to its versatility and ability to cope with adverse ground conditions. It also has a comparatively short cycle time and could be utilized for general duties around the mining site. The machine selected which conforms to these requirements is the CAT 330 or equivalent fitted with a 1.4m³ (nominally heaped) bucket.

4.2.5.4.11 Transportation of Ore to the Processing Plant

Gold-bearing gravels will be transported to the processing plant for treatment. Haul distances will range from a few hundred meters to a few kilometers. It is planned to use heavy – duty CAT or Volvo articulated trucks for this operation. Haulage trucks will be used on the return trip to transport tailings from the treatment plant to be deposited in the mined-out areas backfilling. All-wheel drive articulated trucks are recommended for haulage operation. Four (4) trucks will be required for continual operations. Considering breakdown/maintenance, etc., an additional truck will be included as a back-up. Total haulage truck requirement is therefore five (5).

4.2.5.4.12 Transportation of Washed Gravel (Tailings) for Backfilling

The processing plant will generate tailings consisting of washed boulders, cobbles, pebbles, sand, and slimes, which will be continuously transported to mined-out areas for backfilling. This rehabilitation process will occur simultaneously with active mining operations. The same haul trucks used for ore transport will carry the washed gravel tailings on their return trips to the mining face, optimizing equipment utilization through a closed-loop material handling system.

Dried slimes will be separately collected from tailings ponds and repurposed for site reclamation programs. This operation will require specialized loading equipment, specifically a CAT 320 excavator or equivalent, to be stationed at the tailings storage facility. The selected excavator, featuring a long boom and medium-weight design, will perform multiple functions including loading trucks with backfill material and maintaining tailings dams through regular desilting operations.

This integrated approach to tailings management enables progressive rehabilitation during active mining while ensuring efficient use of equipment and resources. The system provides both operational and environmental benefits by combining material handling with concurrent reclamation activities.

4.2.6 Mine Life Determination

Ore mining gross rate has been projected to be 100 m³/h inclusive of 20 m³/h contingency stockpile provision to accommodate operational disruptions such as mechanical downtime and adverse weather conditions. Consequently, the effective ore feed rate to the gravity treatment plant is estimated at 80 m³/h.

The processing plant facility is designed to operate on a three-shift system, with each shift lasting eight (8) hours, across five (5) working days per week. The remaining two days of the week are allocated for routine plant maintenance, operational housekeeping and observance of local traditional practices (*otherwise known as taboo days*). On this operational schedule, the plant will achieve a total of 240 processing days annually.

The projected mine life has been calculated based on the aforementioned operational assumptions and parameters, as detailed below:

Key Assumptions;

Plant availability – 0.85

Plant Utilization – 0.90

$$\begin{aligned}\text{Plant Daily Throughput} &= (80 \times 24 \times 0.85 \times 0.90) \text{ m}^3 \\ &= 1,468.8 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Plant Annual Throughput} &= (1,468.8 \times 240) \text{ m}^3 \\ &= 352,512 \text{ m}^3\end{aligned}$$

$$\begin{aligned}\text{Life of Mine, } L_m &= \text{Reserve/Plant Annual Throughput} \\ &= (1,939,500 \text{ m}^3 / 352,512 \text{ m}^3) \\ &= 5.5 \text{ years} \\ &\sim \mathbf{6 \text{ years}}\end{aligned}$$

4.2.7 Mining Layout

The mine plan has been developed through systematic pitting and sampling programs that successfully delineated alluvial gold deposits within the Apapam concession floodplains and its tributaries. The comprehensive pit network provided critical grade distribution data across the concession, enabling optimized placement of key infrastructure including the processing plant.

Mining operations will progress parallel to but countercurrent with natural water flows while maintaining a 100-meter riparian buffer zone in compliance with regulatory requirements. The deposit will be systematically divided into mining blocks, strips, and bites to facilitate phased extraction.

The extraction sequence begins with careful topsoil removal and stockpiling for future reclamation purposes. Overburden from initial cuts will be strategically placed on protective streamside pillars, preserving existing vegetative buffers. The ore extraction process utilizes a bite-mining approach, with the first bite deepened to create a water collection sump. Access ramps with a maximum 10% grade ensure efficient haul truck mobility throughout operations.

The mining lease application for the Akim Apapam Gold project seeks to cover the entire 8.4 square kilometers of the concession, ensuring exclusive rights and access for comprehensive operational activities, mine development, mining and future exploration.

This integrated plan combines detailed resource characterization with sustainable mining practices, incorporating environmental controls such as maintained riparian buffers, progressive rehabilitation and systematic water management.

The technical design demonstrates a balanced approach to resource recovery, operational efficiency and environmental stewardship throughout the project lifecycle, ensuring compliance while optimizing gold recovery from the alluvial deposits.

The mine layout plan is shown in Figure 13.

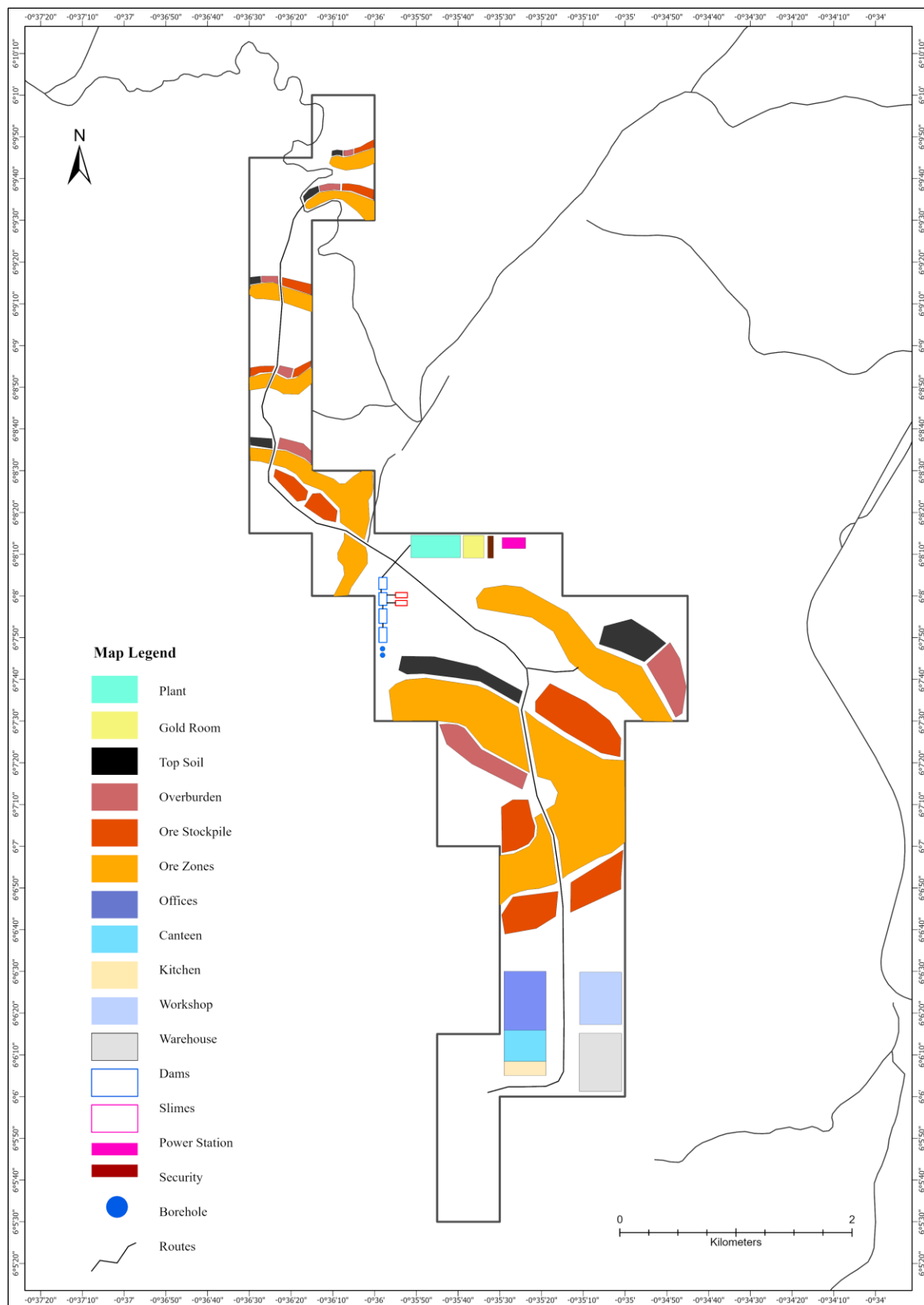


Figure 13: Mine Plan Layout

4.2.8 Water Management in the Pit and Surface Run-Off Management

The pit hydrological system is primarily influenced by two water sources: groundwater infiltration through the pit walls and floor, and surface runoff during precipitation events. To mitigate groundwater ingress, a strategic backfilling protocol will be implemented using oversize tailings from processing operations and overburden material from adjacent mining cuts to immediately backfill worked-out areas, creating a hydraulic barrier.

During periods of intense precipitation or when operating in high-permeability zones, an engineered drainage system will be employed. This system features:

- ✓ A purpose-designed sump at the pit's nadir point, continuously deepened to maintain hydraulic gradient control
- ✓ High-capacity pumping systems (minimum 100 L/min diesel pumps) to transfer collected water to sedimentation basins
- ✓ Perimeter berms constructed from compacted overburden material to divert surface runoff

The concession's tropical climate, characterized by bimodal rainfall patterns (March-July and September-November) with annual precipitation averaging 2000mm, necessitates robust water management protocols. The dewatering system incorporates multiple engineered controls:

- Surface drainage networks surrounding active mining areas
- Plant-site perimeter drainage channels
- Angled diversion berms on adjacent slopes to control erosional processes

The integrated dewatering strategy provides multiple operational and environmental benefits:

- Enhanced geotechnical stability of pit slopes
- Optimized mining conditions through reduced water interference
- Protection of local groundwater resources
- Maximization of operational efficiency
- Preservation of aquatic ecosystems and microbial communities

This comprehensive approach to water management addresses both operational requirements and environmental stewardship objectives, ensuring compliance with best practices for alluvial mining

in tropical environments. The system design allows for adaptive management in response to real-time hydrological monitoring data.

4.2.8.1 Pump head Selection

Water ingress in open pits primarily occurs through intersection of gravel aquifers or excavation in marshy zones, with dewatering conducted using 3-inch or larger capacity pumps. During rainy seasons, operations will prioritize lateritic terrains and drier areas to minimize gold losses associated with high-volume pumping requirements.

The Akim Apapam project's geotechnical characteristics - featuring thin overburden (1.3m average) and a low stripping ratio (0.87:1) - result in short pit lifespans typically under 12 hours per excavation cycle. This operational reality makes the frequent relocation of heavy dewatering equipment between pits both logistically impractical and economically inefficient.

To address these constraints, the operation will employ lightweight, portable pumping systems and implement strategic pit sequencing to maintain water management efficiency while preserving project economics. This approach optimizes resource recovery while mitigating the hydrological challenges inherent to alluvial mining in tropical environments.

Key Parameters:

- Maximum pump capacity: 100 L/min
- Target dewatering rate: <0.5m accumulation per 12-hour shift
- Operational cutoff: Cease mining in pits requiring >3 hours continuous pumping

The system balances effective water control with the practical realities of rapid pit turnover in alluvial gold operations.

4.2.9 Equipment Requirements

4.2.9.1 Mining Excavator

The Akim Apapam Gold Project will utilize a fleet of hydraulic excavators and backhoe loaders, supported by bulldozers, for overburden removal and ore excavation operations. The equipment configuration has been designed to exceed the processing plant's 80 m³/h feed capacity, with a target excavation rate of 100 m³/h to maintain operational continuity and account for potential mining delays.

The excavation system incorporates a dedicated pre-stripping unit consisting of one excavator-dozer combination to maintain adequate ore exposure, supplemented by three production excavators for primary ore extraction. This configuration ensures continuous availability of sufficient ore reserves to support eight hours of uninterrupted plant operation.

Equipment selection was based on rigorous analysis of multiple technical factors including bucket capacity requirements, hydraulic system performance under medium digging conditions (as detailed in Table 10), and the specific material handling characteristics of alluvial deposits. The system design incorporates necessary buffers to accommodate typical operational variables while maintaining production efficiency.

Key operational considerations include the maintenance of a minimum 25% production buffer, systematic overburden management protocols, and optimization of equipment utilization rates. The selected configuration provides robust production capacity while ensuring consistent feed quality to the gravity treatment plant, accounting for the unique challenges of alluvial mining operations.

The technical specifications of the excavation system reflect a balance between production requirements, equipment capabilities, and operational contingencies, ensuring efficient mineral extraction while maintaining necessary production buffers. This approach has been validated through analysis of similar alluvial mining operations and reflects industry best practices for placer deposit exploitation.

Table 10: Productivity of Backhoe Excavators

Bucket Size (m³)	Cycle Time T (sec)	Production Rate Q (m³/h)
0.9	17	79.9
1	17	87.8
1.2	18	98.7
1.4	19	107.7
1.6	19	121.8
1.7	20	129.9
2.3	21	162.8
2.8	22	186.7
3.7	24	227.2
4.6	26	262.5
9.1	27	205.5

Deducing from Table 11, the required production rate of 100 m³/h, an excavator with a bucket size of 1.4 m³ will be appropriate to take care of spillage and underloading. A CAT 330B excavator with 1.4 m³ bucket capacity or equivalent will be used.

4.2.9.2 Tailings Handling Excavator

The project will employ a long-reach excavator such as a CAT 320 or Daewoo 220 for tailings dam desilting and loading oversize waste material for reclamation. The selected excavator features an extended boom for effective sediment removal while maintaining an optimal weight distribution to prevent ground subsidence in saturated conditions. This configuration ensures efficient tailings management while supporting ongoing rehabilitation efforts.

The equipment's balanced design provides sufficient reach and capacity for both desilting operations and material handling, with adequate ground clearance to operate effectively in soft tailings environments. The system maintains operational efficiency while meeting the project's environmental management objectives.

4.2.9.3 Trucks

The Akim Apapam Gold Project will use trucks for hauling of ore to the treatment plant and waste from the treatment plant for backfilling. The size of the trucks to be selected for a particular job depends on the size of the loader. The proposed mining excavator for this project will have a bucket size of 1.4 m³. The number of passes and estimated volume of a 1.4 m³ excavator operating under optimum conditions are shown in Table 11.

Table 11: Volume Estimation of Truck Loaders

No. of Passes	Volume (m³)
4	4.56
5	5.7
6	6.83
7	8.4
8	10.33

As shown in Table 12, 7 m³ CAT articulated trucks for 5 passes of the excavator per truckload will be a good choice.

Cycle time studies of similar trucks working under similar conditions in a similar mine in the country have an average cycle time of about 10 minutes. An average cycle time of 10 minutes is therefore used to determine the number of trucks required to meet the production target.

Truck output formula is given as:

$$P = 60 \times C \times E / T$$

Where P = Truck Output (m³/h)

C= Truck Capacity (7 m³)

E= Truck Efficiency

T = Truck Total Cycle Time (10 mins)

Assuming an Efficiency of 85%

$$P = 60 \times 7 \times 0.85 / 10 = 36 \text{ m}^3 / \text{h}$$
$$\sim 36 \text{ m}^3 / \text{h}$$

Number of trucks = required production / truck output

$$100 \text{ m}^3/\text{h} / 36 \text{ m}^3/\text{h} = 2.78.$$

Approximately = 3

Therefore, the base operational requirement calls for three (3) trucks to maintain continuous operations. To account for scheduled maintenance and avoid operational redundancy, one (1) additional truck will be incorporated into the fleet. Therefore, the project will require a total of four (4) trucks to maintain production efficiency while accommodating necessary maintenance cycles.

A summary of selected equipment required for the project is shown in Table 12.

Table 12: Selected Equipment Summary

Equipment	Quantity
Major Mining and Processing Equipment	1
80 m ³ Gold Trommel Wash Plant	1
CAT 330B Excavator or equivalent	1
CAT 966F Front End Loader	1
CAT 320 Excavator or equivalent	2
7m ³ Trucks	4
B. Supporting Equipment	
CAT D6 Bulldozer	1
Tractor Loader Backhoe (TLB)	1
Utility Truck	2
Dewatering Pump	2
Water Cat (50t)	1
CAT 14G Grader	1
Pick Up (4 x 4)	4
Nissan Bus	2
Solar Energy provision (20 kwt)	Complete setup
Aksa Genset 625kva - Diesel Engine	1
Aksa Genset 275kv-Diesel Engine	1

5.0 MINERAL PROCESSING

5.1 Introduction

Hujack Mining utilizes advanced mining techniques that exclude prohibited chemicals in treatment processes while avoiding environmentally detrimental primitive gold extraction methods. The process begins with passing excavated gravel through screening systems to segregate oversized pebbles, cobbles, and debris from finer particulate matter. The screened fines, containing gold-bearing material, undergo size classification to optimize efficiency in downstream processing stages.

Given gold's higher density relative to associated gangue materials, gravity separation serves as the primary recovery mechanism. This method leverages density differentials through sluices, shaking tables, and jigs to concentrate gold particles. Water acts as the principal medium in the gravity separation process, selected specifically for treating the alluvial ore in this project. The resulting gold concentrate may then undergo additional refining to achieve higher purity. The subsequent section details the optimal processing equipment, ore characteristics, plant layout and relevant technical specifications.

5.2 Processing Test work

5.2.1 Gravel Particle Size Distribution

Representative gravel samples collected from the concession area underwent drying, manual sorting, and sieving analysis to determine particle size distribution. As presented in Table 13, the granulometric analysis indicates that approximately 60% of the gravel fraction consists of particles exceeding 6mm (+6mm), while the remaining 40% falls below 6mm (-6mm). The matrix analysis revealed a sandy clay composition, with the finer fraction exhibiting characteristic clay-sand intermixing. This size distribution provides critical input for optimizing the screening and classification stages of the processing circuit.

Table 13: Gravel Particle Size Distribution

Screen Size (mm)	% Fraction
100	25
50	15
6	20
-6	30
Clay/Silt	10

5.2.2 Gold Particle Size Distribution

The distribution of gold particle sizes by fractions on the Akim Apapam project is shown in the Table 14. The breakdown shows a distribution of 70.0% plus 60 mesh and 30% minus 60mesh. The gold on the concession is primarily fine to coarse, with erratic nugget occurrence in some of the pits.

Table 14: Distribution of Gold Particle Sizes

Appearance	Sieve Mesh Size	Gold Particles (%)
Coarse	16	12.0
Medium	30	18.0
Fine	62	38.0
Very Fine	150	22.0
Flour Gold	-150	10.0

5.2.3 Gold Fineness

Gold particles from the concession were subjected to aqua regia digestion followed by atomic absorption spectroscopy (AAS) analysis, yielding a gold purity of 96% (equivalent to 22.90 carats). Silver was identified as the primary impurity, constituting approximately 4.0% of the composition. Additional mineralogical testing would be required to quantify other trace elements present in the final saleable concentrate.

5.2.3.1 Gold Liberation Characteristics

Since over 70% of the gravel is imbedded in the sandy-silt media, liberation of the gold is through the normal gravity washing. Gold particles on the Akim Apapam concession are fully liberated from the constituent minerals (free gold).

5.2.3.2 Pre-Concentration Amenability of Ore

The gravel deposits within the concession demonstrate suitability for pre-concentration, with particle size analysis indicating that 70% of the material falls within the +6mm fraction. The pre-concentration process will be integrated into the washing and screening operations, where the systematic removal of oversized particles (exceeding typical gold particle dimensions) will effectively upgrade the remaining material.

This size-based separation enhances the feed grade to downstream processing units by eliminating barren coarse fractions while retaining gold-bearing material in the finer fraction. The process

leverages the natural size disparity between gold particles and waste gravel to improve overall processing efficiency.

5.2.4 Gravel Washability

The gravel washability indicates the ease with which gravel could be washed. Generally, the greater the clay content the more difficult it is to wash. The washability varies from free wash, normal wash to difficult washing. Where there was a sandy matrix with very little clay, it was free wash. A sandy clay matrix was normal, while a plastic clay matrix and the terrace deposit were difficult washing.

To ensure effective washing of the gravel prior to the sluicing process, high volume of spray water applied counter flow to the screen surface will be applied. The residence time in the Trommel will ensure full liberation of particles for the subsequent gravity separation process. Clay by visual inspection in the field showed that the clay content of the gravel varied from 10% in the alluvial deposits to 30% within the terraces.

5.2.5 Plant Design

The design of the alluvial gold treatment plant must take many factors into consideration to achieve the best recovery of gold. Among the factors taken into consideration are:

- Characteristics of ore to be processed
- Size distribution of ore and gold particles. Fractions of the gravel reporting to the various sections of the plant.
- Volume of ore/gravel to be processed per day to recover gold to maintain economic viability for the project.
- The effective maximum and minimum sizes of the liberated gold particles and their distribution in the ore.
- Optimum operating conditions of the gravity separating equipment
- Infrastructure requirement
- Robustness of the final design regarding operating conditions.

5.2.6 Selection of Processing Plant

The gravel and gold within the Akim Apapam concession exhibit distinct characteristics that inform the selected processing methodology. The deposit contains free-washing gravel alongside normal-washing gravel with clay matrix components requiring soaking pretreatment. Gold particle sizes present a distribution ranging from fine to coarse grains, necessitating a robust recovery system.

Considering these material characteristics, the processing circuit will employ sluice box concentration followed by secondary treatment using a 20-inch Knelson Concentrator. The gravel will undergo initial washing and screening prior to sluice box processing. Sluice boxes remain the preferred gravity separation device in alluvial operations due to their operational simplicity and effectiveness in recovering gold particles ranging from 25mm to 100 microns. Recovery efficiencies vary significantly based on operational parameters, with modern commercial operations achieving 80-100% recovery compared to sub-50% recovery in artisanal setups.

The GOLD AR 28 sluice box design has been used on similar projects with similar gravel in the country. The GOLD AR 28 sluice box is therefore selected. The GOLD AR 28 has the following advantages:

- Fast waste flush of sands and gravels.
- Very high retention of all heavier minerals
- Variable flow adjustment for fast and slow waterways.
- Water stop feature for clean up or a quick check of current gold levels in the sluice.
- Coarse gold and / or gold particles are trapped in the lower flow zones.
- Fine gold recovery is increased by changing flow rate inside the sluice

The GOLD AR 28 operates on a continuous capture principle, utilizing a progressively decreasing hydraulic pressure gradient along its length. This design facilitates differential settling of gold particles according to size and density, with the entire sluice length functioning as an integrated capture system. The selected configuration provides optimal recovery performance while accommodating the specific material characteristics of the Akim Apapam concession.

5.2.7 Processing Plant Flow Sheet

The Hujack Mining plant will operate at a feed rate of 80 m³/hour, with material sourced from the ROM pad via Front-End Loader (FEL) or directly from haul trucks dumping into the feed hopper. The optimal feed rate will be finalized during plant commissioning and test runs. Pressurized water from a high-capacity monitor will fluidize the gravel in the feed hopper, initiating material breakdown and facilitating flow into the grizzly section. A dedicated clay blaster, positioned at the grizzly edge, will disintegrate clay-rich ore components.

The grizzly, featuring a 50mm aperture, will separate +50mm oversize material (stones and boulders), which will be conveyed to a designated stockpile. Stringent washing protocols will be implemented to minimize gold losses in the oversize fraction. The -50mm underflow will report to the scrubbing trommel, followed by the screen trommel. A nugget trap installed at the trommel inlet will capture coarse gold particles.

The screen trommel will classify material at 6mm, with the +6mm fraction conveyed to an oversize gravel stockpile for subsequent use in pit backfilling, access road construction, and dam rehabilitation. The -6mm fraction will gravitate to a slurry tank, then flow through an 18" pipeline to three parallel sluice box units. Flow distribution will be achieved via three 14" branch pipes equipped with control valves, allowing individual sluice isolation during cleaning cycles (conducted twice per shift at 4-hour intervals).

Slurry tailings will report to a three-stage settling pond system. The processing flow is detailed in Figure 14.

To achieve this, certain key performance indicators are set and monitored closely during plant operations. These key parameters include.

- Feed rate optimization: feed rate is monitored closely and ensured that it's within set limit of 80 m³/hour.
- Slurry density and flow on sluice boards
- Water quality and pressure at the plant
- Checking the state of trommel screen frequently and ensuring that it's not blocked by near screen size stones.

- Continuous taking tailings samples to monitor and minimize gold losses
- Constantly checking the conditions of carpets and rifles on sluice boxes and changing them when necessary
- Determining plant efficiency (recovery rate, tailings grade etc.

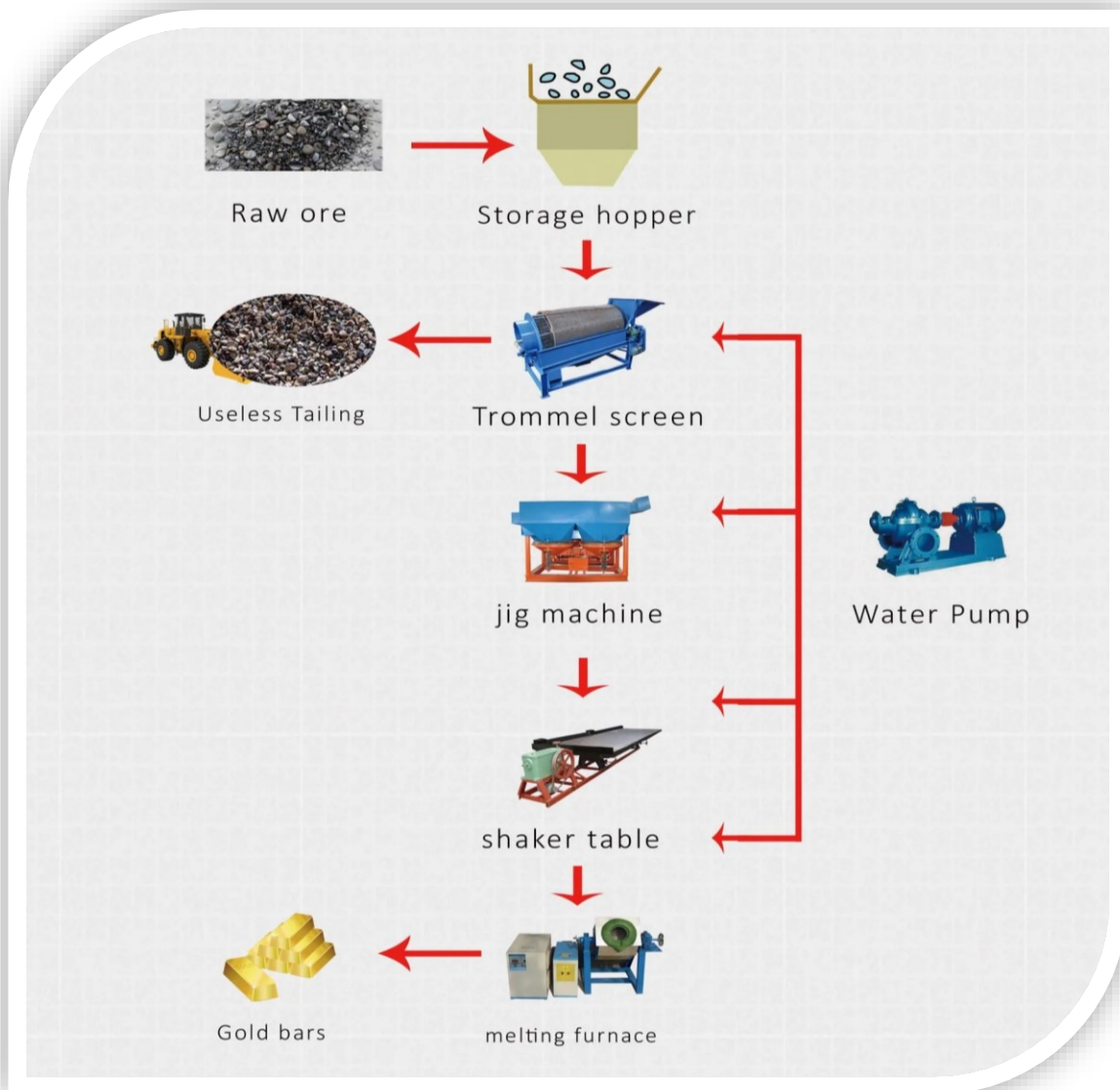


Figure 14: Processing Plant Flow Sheet

5.2.7.1 Process Water

The treatment plant will utilize a closed-loop water system primarily sourced from decanted tailings settling ponds, with supplemental supply drawn from an off-channel reservoir fed by the Akuasua stream and its tributaries. Initial water supply will be delivered through a 12-inch diesel-powered centrifugal pump system from the reservoir, transitioning to predominant use of recycled process water from the settling ponds during normal operations.

The system design accounts for seasonal water losses, with approximately 20% evaporative loss anticipated during dry periods and reduced losses during wet seasons, accounting for both pond surface evaporation and incidental process spillage. Water balance will be maintained through regulated makeup supply from the Akuasua stream when required.

Tailings management infrastructure will consist of engineered impoundments constructed from initial pit excavations, serving dual functions as containment for slurry tailings and water collection basins. These lined ponds will accumulate process water, pit dewatering discharge, and site runoff, enabling maximum water recovery and reuse while minimizing environmental discharge. The treatment plant will incorporate water recycling systems to maintain near-zero liquid discharge operation, with all containment structures designed to prevent groundwater infiltration and meet environmental compliance standards.

5.2.7.2 Water Conservation

Water conservation represents a critical operational consideration in alluvial mining, particularly during periods of seasonal aridity. The operation will implement a closed-circuit water recycling system to maximize hydrological efficiency and minimize freshwater extraction. This system will integrate sedimentation tanks equipped with particle separation technology to remove suspended solids, ensuring water quality meets reuse specifications.

Preemptive desilting of retention dams will be conducted prior to dry seasonal onset to optimize reservoir capacity and maintain process water availability. Erosion control measures will include the establishment of vegetative stabilization using native grass species, which will enhance soil structure integrity and reduce sediment mobilization. These combined strategies ensure compliance with sustainable water management practices while maintaining operational productivity during variable climatic conditions.

The system design prioritizes water recirculation efficiency, with continuous monitoring to balance process requirements with environmental stewardship objectives. Containment infrastructure will incorporate impermeable liners and overflow controls to prevent uncontrolled discharge, aligning with best practices for responsible resource extraction.

6.0 HUMAN RESOURCE REQUIREMENT

6.1 Introduction

The human resource requirement aims to identify and define the core competencies essential for success in the intended mining operations. These competencies may include technical skills such as mining engineering, processing engineering, geology, and environmental management, as well as non-technical skills like safety consciousness, teamwork, leadership, and sustainable mining practices (Table 15). By outlining these competencies, the framework ensures that the industry's workforce possesses the necessary expertise to address industry-specific challenges effectively

Table 15: Staff Requirement

Designation	Number
Managing Director	1
Mines Manager	1
Deputy Mines Manager	1
Mining Engineer	1
Geologist	2
Geological Technician	4
Processing Manager	1
Deputy Processing Manager	1
Engineering Manager	1
Environmental Health & Safety Officer	1
Mining Labour	6
Plant Labour/Processing Staff	8
Pit Foremen	2
Machine Operators	8
Machine Training Officers	2
Auto Mechanic & Electrician	4
Finance & Administration	2
Administration Officer	1
Administrative Assistants	2
Accounts Officer	1
Community Relations Officer	1
Cook	2
Cleaner/Casual	3
Security Officers and Chief Security	8
Total Labour	64

6.2 Staff Roles

Hujak Mining Limited Company will have all their employees' roles spelt out to them before they are employed at the mine site as shown in the organizational chart in Figure 15.

Managing Director: The Managing Director will oversee overall company operations, sets strategic direction, ensures profitability, and maintains compliance with regulations.

Mines Manager/Deputy Manager: Supervises mining operations, ensures efficiency, safety, and regulatory compliance in extracting resources.

Mining Engineer: Designs and oversees mine construction, extraction methods, and equipment usage, responsible for creating access routes. Will be responsible for the volume of ore transported to stockpile.

Finance & Administration: Manages budgeting, accounting, and financial reporting; oversees administrative operations and resource allocation.

Administration Officer: Will handle the day-to-day managerial tasks including correspondence, scheduling, and maintaining company records and updating the finance office about workers payroll.

Processing Manager/Deputy Manager: will be part of the team that will undertake the construction of the processing plant, ensuring management of the plant, efficient extraction of gold, overseeing volume of ore processed daily, weekly and monthly, shall report on gold recovery on daily, weekly and monthly basis.

Geologist: the geological team will be responsible for the resource and reserve definition; they will continue to prove the ore body and move it from inferred to indicated as mining progresses. The geologist shall be in charge of grade control and mine reconciliation.

Mine Surveyor: shall be involved in measurements of the farmlands before compensations are paid to farmers, shall ensure accurate planning and updating mine out and reclamation activities for bonds to the EPA.

Geological Technician: Supports geologists by collecting and analyzing samples, maintaining equipment, and recording data.

Environmental Health & Safety Officer: Ensures compliance with environmental regulations and promotes workplace safety standards.

Community Relations Officer: Engages with local communities, addresses concerns, and builds positive relationships to foster cooperation.

Administrative Assistants: Provide office support, manage schedules, and handle communication within departments.

Engineering Manager: Supervises maintenance and repair of machinery and infrastructure, ensuring operational efficiency. He shall be responsible for the routine services of machinery. Electrical work and civil work shall be in his purview.

Accounts Officers: Manage financial transactions, payroll, and other accounting tasks to ensure financial accuracy.

Pit Foreman: Supervises pit operations, ensuring safety and adherence to work schedules. He shall enforce the right area to open the pit based on the geological data and the right land to enter.

Machine Operator: Operates heavy equipment, such as excavators and bulldozers used in mining activities. Shall report changes in ground conditions or workflow to pit foreman.

Machine Training Officers: Train workers on the safe and effective use of mining machinery and tools.

Auto Mechanic & Electrician: Maintain and repair vehicles and electrical systems to keep operations running smoothly.

Chief Security: he shall ensure the roster of the security force, monitoring the CCTV cameras to have full control of the operations, especially at the plant site.

Security Officers: Protect company assets, ensure the safety of personnel, and manage access control

Cook: Prepares meals for onsite staff, often in remote mining locations

Cleaner/Casual: Maintains cleanliness in offices, accommodation, and common areas.

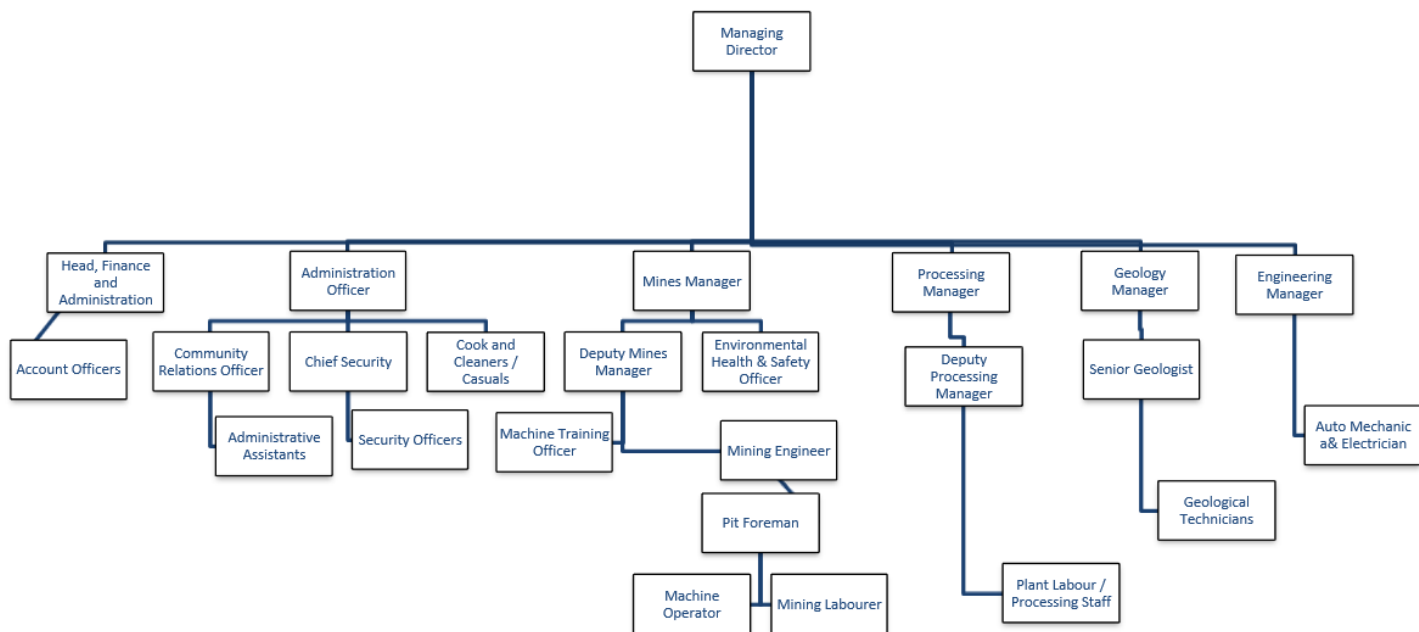


Figure 15: Proposed Organizational Chart

6.3 Local Content

Hujak Mining is committed to the promise of engaging local content for employment within the Akim Apapam and its catchment communities. Local content has been a subject of public debate in Ghana between the mining companies, the youth and their community rulers. But the company is poised to accomplish its local content mandate by following the due procedures stipulated in the legislative instrument.

In line with the local content and local participation regulation of 2020, Hujak Mining has a well-crafted program tailored towards building the local economy in its primary catchment area. This will address unemployment challenges and build capacities to achieve the requirements of the local content and local participation regulation 2020 to enhance socio-economic development of the country.

The skilled and unskilled labour within the operational areas of the company will benefit from the training programs, which will in turn render the company with profitable and efficient delivery of its core mission. The local employment strategy is to aid Company in the process of gaining and

maintaining a social license to operate as well as building an interpersonal relationship with the local dwellers.

6.4 The Legislative Instrument

The focus of developing a Local Content framework for Ghana's mining industry is to create local employment and to promote growth of domestic manufacturing. These two aspects, (Human Resource and Goods & Services) of Ghana's local content framework are supported by the Minerals and Mining Act, 2006 (Act 703) and L.I. 2173, 2012.

The company is committed to the Minerals and Mining Regulations for local Content and Local Participation. The company has outlined plans for recruiting and training potential staff, effective measures have also been put in place in order not to flout this mandatory local content obligations. The Commission will be updated with regards to the dynamics of the company's modus operandi on employment-related issues.

6.4.1 Employment

Hujak Mining is committed to promoting labor satisfaction within the areas it will be operating when this proposed lease is granted. The skilled, unskilled, and gender equalization on labor issues are of utmost importance in this local content plan for the company.

6.4.2 The Unskilled Labour

The Unskilled labour shall be made up of people who do not have any artisanal skills, but whose job prescription falls into the category of; cleaning, manpower, shoveling, assisting experts to undertake tasks. They may not require any curriculum vitae submission but by recommendation from opinion leaders with good conduct and no criminal records.

6.4.3 Gender Equality

The company will ensure that the ratio of men to women gap would not be below the reasonable gender parity range. Hujak Mining is targeting a 70% to 30% ratio of men to women within its first three (3) years of mining, the ratio shall be leveled up from the fourth year of the company's operations through to the end of the mine life.

6.4.4 Recruitment of Employees

The company seeks to recruit, place, promote, compensate, and retain employees based on their qualifications for the work to be performed. This may include experience, merit, and other work-related criteria without regard to gender, race, nationality, ethnicity, social and indigenous origin, religion or belief, disability, age or sexual orientation, or any form of discrimination protected by applicable law.

The company shall employ anybody who is qualified without respect to their political affiliation, ethnicity and or social standing, etc.

The mode of recruitment will be via the following job posting avenues:

- Job boards in communities
- Recruitment firms
- Candidate referrals
- Networking through other companies
- The community information centers

6.4.5 Training Program for Local Labour

The company's policy is to keep the team small and flexible, thus minimizing costs. To achieve this, staff would need to be multi-skilled and adaptable. Additional capacity will be needed to cover for absence due to leave or sickness and although additional capacity is planned particularly where shift work is an important element of the work, it is intended that staff will learn to cover each other's roles. In the beginning, there will be considerable input from the project manager and the consultant. As the project develops and employees gain confidence and knowledge, more operational capabilities will be assimilated by the workforce. Accountancy, financial, and legal services will be outsourced to professional advisers and their respective staff in Accra.

The company shall employ over 90% of unskilled job positions from the host communities. In terms of skilled labor, the host communities shall be given priority before staff outside the catchment areas are considered. Two (2) training officers shall be recruited to oversee the skill enhancement of newly recruited machine/truck operators and to organize periodic refresher courses for the company's employees. This will help minimize work-related accidents and incidents at the mine premises.

7.0 INFRASTRUCTURE

7.1 Introduction

Infrastructure has been categorized into two main sections, namely Social Infrastructure and Project-Specific Infrastructure.

Social infrastructure here includes health facilities, education, water and sanitation, transportation, and energy services, which contribute significantly to improving the living conditions of residents. Project-Specific infrastructure will focus on the proposed infrastructure to be implemented to meet the needs of the Akim Apapam Gold Project, such as power supply, telecommunication system, security post, water supply, tailings management systems and auxiliary facilities.

The Company will collaborate with other specialists, including quantity surveyors and electrical engineers, to provide full service to obtain the optimal solution for a functional and effective mine infrastructure.

7.2 Social infrastructure

7.2.1 Healthcare Facilities

The area's healthcare system is administered mainly by the Kyebi Government Hospital, which serves as the primary medical facility. Additionally, there are Community-Based Health Planning and Services (CHPS) compounds and smaller clinics scattered across the area, particularly in rural communities. However, healthcare services face challenges such as inadequate staffing, limited access to specialized care, and a shortage of medical equipment.

7.2.2 Educational Facilities

There are primary and secondary education, with schools like Kibi Senior High Technical School providing secondary education. Vocational and technical schools are also present, though expansion and improvement are needed. Primary education is available in most communities.

7.2.3 Water and Sanitation

Water supply systems include pipe-borne water, boreholes, and wells but access to clean and reliable water remains a problem in rural areas because of the indiscriminate illegal mining within open water sources such as rivers and stream leading to rampant water pollution and chemical

contamination. Poor sanitation practices, especially open defecation still occur in some communities due to inadequate sanitation infrastructure, though efforts through Community-Led Total Sanitation (CLTS) programs are helping to improve conditions.

7.2.4 Transportation and Roads

Between 2019 and 2024, most roads within the district's communities have been tarred, significantly improving transportation. Notably, the road connecting Kibi to Akim Apapam is now fully paved, leading to reduced travel time.

7.2.5 Electricity and Energy

Electricity is available in many parts of the district, with the national grid extending to most towns and villages. However, cottages and hamlets still lack power supply as result of their isolation from the grid lines.

7.2.6 Telecommunications

Mobile network coverage is generally available, provided by networks like MTN, Telecel (formerly Vodafone) and AirtelTigo. However, remote areas may suffer from poor reception. Internet access within the project is very good and weak into the northernmost portions of the concession. Generally, mobile communication within the Akim Apapam Gold Project Area is fairly good.

7.3 Project-Specific Infrastructure

7.3.1 Power Supply

The power supply to the mine site will be made up of the energy mix; the national grid, renewables (solar) and the diesel engine generator. Power from the national grid would be used to run the processing plant. This will ensure uninterrupted power supply to the vital parts of the proposed mine operations. The backup generator would come on board during power failures and voltage drop on the national grid to help maintain and to reduce equipment damage.

The genset will be linked directly to a Motor Control Circuit (MCC) Room to operate all the plant machinery. The control room will house all the circuit breakers and fuse boxes for the plant machinery. The freshwater feed pump will be diesel-powered and separated from the plant energy source.

Solar panels would power offices, accommodation, and auxiliary facilities with low energy consumption gadgets. This will help the balance of power. The project's power requirements are as follows: 210 kVA from the national grid, 625 kW from an Aksa generator, 275 kW from another Aksa generator, and 20 kW from solar energy.

7.3.2 Communication System

Hujak Mining will provide a two-way radio system for the mining, plant, and security departments, as well as other essential departments. General communication among staff will be conducted via cellular phones, with Vodafone, MTN, and Airtel-Tigo networks available in the area. Management, supervisors, and other key personnel will be provided with airtime for cellular phones.

Additionally, communication and coordination will be facilitated through departmental WhatsApp groups, company email, and other social media platforms such as Zoom and Microsoft Teams. These channels will be used for announcements, meetings, operational updates, and safety alerts to ensure quick and effective responses.

7.3.3 Security

The security post has been incorporated into the site layout and will be staffed by ex-military personnel equipped with Hujak Mining licensed firearms to deter armed robberies. Additionally, civilian security personnel will be stationed at the post to conduct daily searches of staff members, mitigating attempts to smuggle gold and other company assets. The security team will utilize metal detectors to perform thorough inspections of bags and concealed items, including gold and electronic devices such as phones. Monitoring towers will be strategically positioned on elevated terrain to provide comprehensive visibility over the site layout, enabling the detection of potential threats. A siren system will also be implemented to promptly alert staff members in case of imminent danger.

A comprehensive security system and protocol will be implemented to ensure the safe extraction, storage and transportation of gold. Gold concentrates will be transported to the gold room under the direct supervision of security personnel and management. To prevent theft, Closed-Circuit Television (CCTV) cameras will be installed in high-risk areas, including the gold room, fuel

depot, plant site and other sensitive locations, enabling continuous monitoring and deterrence of unauthorized activities.

7.3.4 Water Supply

The water supply for the project will be a combination of boreholes and water abstraction from open streams and rivers. Bore-holes shall be provided on-site for uninterrupted clean water supply to the whole site for domestic usage, the proposed pump station shall be approved by the EPA to ensure the necessary steps are taken to prevent the destruction of the aquatic habitat during full operations.

The company will also draw significant volumes of water from surface water bodies for mineral processing, dust suppression and other operational purposes. A Water Abstraction Permit will be secured from the Water Resources Commission to authorize the use of these water resources.

The operational water supply will be sourced from recycled water collected from the dams, while raw water will be drawn from the Akuasua Stream, which drains the concession. Potable water for households will be provided through boreholes, and drinkable water will come from bottling water companies.

7.3.5 Water Storage Facility

The water storage facility will serve as a reservoir for holding the water pumped from the mechanized boreholes, ensuring a steady and reliable supply for mining operations. The facility will be designed to accommodate fluctuations in water demand and supply, as well as to provide a buffer in case of equipment downtime or maintenance activities.

The storage facility, consisting of high-capacity tanks, pumps, piping networks, and monitoring systems, will enable the company to meet its operational water needs while maintaining flexibility in water use and protecting the environment from potential over-extraction or contamination. By implementing proper water storage infrastructure, the company can support sustainable operations and comply with Ghana's water management regulations.

7.3.6 Tailings Management System

The Akim Apapam Project processing plant will produce two primary waste streams: oversize material discharged from the grizzly and screen trommel, and fine tailings generated from the sluice boxes.

7.3.7 Oversized Material

Oversize material from the grizzly and screen trommel will be discharged onto conveyors and stacked. This material will be repurposed for reclaiming mined-out pits and constructing haul roads.

7.3.8 Tailings

Meanwhile, the tailings, comprising finer waste material from the sluicing process, will be collected and managed in accordance with environmental and operational protocols to ensure proper containment and minimize environmental impact.

The settling pond system operates as follows:

First Pond: Coarse particles settle in the first pond, with coarse sand and pebbles accumulating near the sluice discharge. These materials will be regularly excavated for use in reclamation.

Second Pond: Water, along with finer suspended particles flows from the first pond into the second pond via a weir. Here, further settling of suspended particles occurs.

Third Pond: Clean water flows from the second pond to the third pond through another weir. The third pond is connected to the fourth pond.

Fourth Pond: Clean water is received from the third pond through another weir. The third pond is connected to the fresh water pond also known as the fourth pond. Water from the freshwater pond will be pumped back to the treatment plant for reuse through a pipeline.

Reclamation and Rehabilitation

After backfilled pits have settled, they will be capped with the topsoil removed at the start of mining operations to restore the mined-out areas to their original condition. An afforestation program will then be initiated to further rehabilitate the site.

7.3.9 Auxiliary Facilities

Other supplementary facilities shall be constructed to support the mining operations.

7.3.9.1 Oil Depot

The oil depots shall consist of impermeable concrete walls within which the fuel and lubricant tanks are stored to curb the seepage of oil into the ground. Runoff from these depots shall be channeled through oil water separators before discharge into the environment.

7.3.9.2 Water for Utilities

Mechanized boreholes will be provided at the camp, of which permission will be obtained from the Water Resource Commission (WRC).

7.3.9.3 Site Clinic

An infirmary shall be set up on-site within the office block to take care of daily health care needs such as minor injuries, and daily vital checks for workers with high blood pressure and those prone to diabetes. Emergency units with a stand-by vehicle designated as an ambulance shall always be ready to deliver patients when the need arises. Health professionals will run the Clinic facility on part-time basis. A fully stocked pharmacy shall be provided on-site (and first aid kits) at all operational areas to deal with minor site injuries. Severe injuries shall be taken to the government hospitals by the ambulance or mine transport for emergency medical treatment.

7.3.9.4 Workshop

The plant, equipment workshop and ancillary structures shall be located appropriately with connected access to achieve a reasonable haulage time. This will help with the smooth and faster delivery of gravels to the stockpile/ROM pad to optimize production.

The workshop will have both exit and entry points to avoid traffic and congestion. It will also be well-ventilated to allow free flow of air in and out to allay fire outbreaks during the dry seasons. The workshop will be partitioned and well demarcated to separate the working areas from the storage of the machine parts and working tools. Grinding tools and moving machines will be installed at vantage points to help curb injuries during operations. Oxyacetylene cylinders will be stored well with sealed caps on when not in use to avoid leakage into the environment. Welding

workshops will be provided with enough ventilation vents to conduct noxious gases away. Goggles shall be provided to artisans to protect their eyes as well.

The workshop shall be compartmentalized to fit all the departments within the company's operations. This may range from the mechanical, plant and processing, electrical, light duty and heavy-duty units. This will aid in easy location when certain expertise is needed to solve problems and to relay other jobs to other units to complement the task.

7.3.9.5 Project Camp

Project Camp will be constructed which will comprise of a mine office, residential buildings, a kitchen, and a gold room. The kitchen will be located adjacent to the office. The gold room will house all the equipment required for the final concentration ("clean up") of the high-grade gold concentrates from the sluice boxes. The gold room will be a fully secured building. All of the entrances will be double locked with access requiring two sets of keys held by the security and gold room supervisors.

7.3.9.6 Accommodation

The company will provide accommodation for senior mine management staff at the mine camp. This accommodation will be made up of a concrete bunkhouse for 10 senior staff members separated into individual units within the building. Shipping containers shall be converted to decent accommodation for 20 junior staff members. All other mining staff will be expected to find accommodation in one of the adjacent local communities.

Sewer and toilet facilities will be discharged into a buried septic tank located at the camp. This tank will be the receptacle for both sewage and grey water from the bunkhouses and the kitchen. It shall be designed for the expected sewage volume of the camp during operational hours. Used water from the gold room for final clean-up will be diverted to a separate tailings channel for settling and treatment before being released into the local environment.

The gold room shall house a 20-inch Knelson Concentrator, a standalone water source, a smelting furnace and equipment necessary for bar casting. Power to these facilities shall be provided through an onsite generator with the capacity to ensure the smooth running of the facilities within the site and residential Camp.

7.3.9.7 Warehouse and Stores

A well-managed warehouse shall be run by the procurement unit, stocked with spare parts of machinery, personal protective equipment, consumables, and sundries shall be distributed based on requisitions placed by the various departmental heads. This shall be tracked within the procurement database for a continuous supply to avoid shortages on site.

7.3.9.8 Car parks

Car parks shall be designated for Managers, visitors, and departmental unit managers to allow easy access and evacuation of staff during emergency.

7.3.9.9 Wash Bays

Trucks, excavators, and field machinery shall be weekly/daily washed off grease so leakages and hose-pipe leakages could be monitored to prevent seeps into the environment.

7.3.9.10 Access Routes

The company plans to create access routes connecting the mine blocks to key facilities, including the processing plant, offices, accommodation areas, mess hall, workshops, and fuel depot. Approximately 1 kilometer of routes is proposed for construction, utilizing gravel and laterite for paving.

8.0 PROJECT COST ESTIMATION

8.1 Introduction

The Akim Apapam Golf Project cost estimation is based on general budget pricing from suppliers, consultants, contractors and other similar sized projects. Smaller equipment and facilities component costs were factored in based on norms for the kind of facility being considered to be procured and constructed, and where possible, adjusted to reflect local conditions.

The costs estimates are based on recent quotations for new and old mining equipment, and materials, and a fixed price lump sum proposal for alluvial process plant construction. Other capital costs were derived from quotations and estimates based on operational experience in Ghana. The company intends to finance the project with an amount of US\$6.49 million. The objective of the cost estimates is to provide comprehensive and realistic data that will serve as a basis to evaluate the economic and financial viability of the gold project.

Construction and installation labour rates are based on contractor costs in Ghana and the West African region for similar type of work. Where costs were either not available or irrelevant, costs from other similar projects in the sub region were used.

8.2 Capital Cost

Capital cost is estimated to be US\$5.99 million, and it is made up of the cost of mining processing plant and machinery, motor vehicles, computers and accessories, office equipment and pre-production cost.

8.3 Working Capital

Start up or working capital is the capital required to cover all the operational expenses prior to cash inflows from initial gold sales. In the case of a typical alluvial gold project, 2-month of operating cashflow is often considered adequate to achieve full ramp up production. A total of US\$500,000 is expected to cover the day-to-day operating expenses.

8.4 Project Cost

The cost estimates are based on recent quotations for new and old mining equipment, and materials, and a fixed price lump sum proposal for alluvial process plant construction. Table 16 presents an overview of the development cost.

Table 16: Breakdown of Project Cost

Project Item	Quantity	Unit Cost US\$	Total Cost US\$
Treatment Plant	1	726,000	726,000
CAT 330B Excavator	1	457,380	457,380
CAT 966F Front End Loader	1	328,394	328,394
Dump Trucks	4	317,020	1,268,080
CAT D6 Bulldozer	1	381,150	381,150
Tractor Loader Backhoe	1	477,950	477,950
Dewatering Pump	2	55,176	110,352
Generator	2	59,532	119,064
Utility Truck	2	96,800	193,600
CAT 14G Grader	1	111,925	111,925
Pick Up (4 X 4)	2	55,055	110,110
Nissan Bus	2	151,250	302,500
Station Wagon	4	175,450	701,800
Gold Room	1	154,000	154,000
Office Equipment	1	78,650	78,650
Office furniture & fittings	1	90,750	90,750
Pre-production Expenditure			375,000
Start up capital			500,000
Total Project Cost			6,486,705

8.5 Pre-Production Cost

Pre-production cost is associated cost related to exploration activities and has been capitalized line with section 28 of the Minerals and Mining Act, 2006, (Act 706).

8.6 Total Operating Costs

8.6.1 Introduction

Operating cost of the project is made of all expenses to be incurred in the mining, processing, and administration of the mining operations. Mining cost relates to the costs allocated to the actual mining activities. The activities include stripping of the earth's surface to excavating and the transportation of gold bearing material.

Total operating cost comprises the cost of mining, processing and administration cost incidental to getting the ore to dore' stage. It also includes provisions for depreciation and reclamation.

Total operating cost of US\$38.51 million is estimated to be incurred during the six (6) year life of mine. This is made up of US\$8.42 million labour cost for all operations including mining, processing and admin related activities, US\$8.09 million consumables cost, US\$10.98 million utilities cost, US\$0.84 million rent cost and US\$2.29 million for other costs, direct and indirect. The other costs include US\$5.99 million and US\$1.33 million provisions for depreciation and reclamation respectively.

In the first year it is estimated that US\$5.90 million will be incurred as total operating cost. In the third of operation US\$6.81 million, fifth year US\$8.15 million and US\$4.08 million in the final year of operation. The composition of the total operating cost is shown in Table 17.

Table 17: Breakdown of total operating cost US\$

Operating Cost	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Total
Labour	(1,460,681)	(1,313,984)	(1,398,865)	(1,585,112)	(1,626,519)	(1,035,235)	(8,420,396)
Consumables	(1,020,493)	(1,177,969)	(1,413,563)	(1,696,276)	(2,035,531)	(746,727)	(8,090,560)
Utilities	(1,445,288)	(1,589,816)	(1,987,271)	(2,185,998)	(2,404,597)	(1,365,766)	(10,978,736)
Rent	(120,000)	(132,000)	(145,200)	(145,200)	(150,000)	(150,000)	(842,400)
Other direct cost	(313,573)	(282,081)	(300,303)	(340,285)	(349,175)	(370,400)	(1,955,817)
Depreciation	(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	-	(5,986,705)
Reclamation provision	(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(1,332,583)
Community Devt	(75,000)	(82,500)	(90,750)	(99,825)	(100,008)	(120,788)	(568,871)
Indirect Cost	(43,786)	(48,165)	(52,981)	(58,279)	(64,107)	(70,518)	(337,837)
Total Operating Cost	(5,898,260)	(6,045,953)	(6,808,372)	(7,530,413)	(8,149,375)	(4,081,532)	(38,513,905)

8.7 Project Economic and Financial

The Gold Project has prepared a financial model for the purpose of evaluating the economics of the project. The model is presented as an equity model assuming 100% equity financing. No allowance has been made in the model for the effects and levels of debts financing required. The results of the economic analysis represent forward-looking information that is subject to several known and unknown risks, uncertainties and other factors that may cause actual results to differ materially from those presented within this report.

The model projects the revenue, operating costs, royalties and taxes payable on annual basis throughout the life of the project.

The project has been evaluated using a discounted cash flow (DCF) analysis. Cash inflows consist of annual revenue projections. Cash outflows consist of capital expenditures, operating costs, taxes and royalties. These are subtracted from the inflows to arrive at the annual cash flow projections. To reflect the time value of money, annual net cash flow (NCF) projections are discounted back to the project valuation date using selected discount rates. The discount rate appropriate to a specific project depends on many factors, including the type of commodity and the level of project risks (e.g. market risk, technical risk and political risk). The discounted, present values of the cash flows are summed to arrive at the Project's net present value (NPV). The internal rate of return (IRR) was also calculated. The IRR is defined as the discount rate that results in an NPV equal to zero.

The following assumptions were considered in estimating the economics result in Table 18:

- a. Average gold price, US\$1,800/oz
- b. 5% royalty gross mineral revenue
- c. Corporate tax at 35% of profit
- d. 10% Discount factor

Table 18: Key Project Economic Results

Parameter	
Production Au(ounces)	42,091
Assumed Gold Price/ounce (US\$)	1,800
Revenue (US\$ M)	75.76
Total Operating Cost (US\$ M)	38.51
Operating Profit (US\$M)	21.75
Royalties (US\$M)	3.79
Corporate Tax (US\$M)	11.71
Net Present Value (US\$M)	13.37
Internal Rate of Return (%)	57.30

The operational efficiency of the projects indicates strong profitability. The project is estimated to generate approximately US\$75.76 million of gross revenue over the six (6) year life of mine. The Net Present Value is estimated at US\$13.37 million with an Internal Rate of Return (IRR) of 57.30%. The project income statement and cash flow and evaluation on equity funding are presented in Tables 19, 20 and 21.

Table 19: Project Income Statement in US\$

Year	0	1	2	3	4	5	6	TOTAL
Gold (Ounces)		4,209	5,612	9,120	9,821	10,523	2,806	42,091
Gold Prices (US\$)		1,800	1,800	1,800	1,800	1,800	1,800	
Gross Revenue US\$		7,576,380	10,101,834	16,415,496	17,678,214	18,940,950	5,050,926	75,763,800
Royalties US\$	5%	(378,819)	(505,092)	(820,775)	(883,911)	(947,048)	(252,546)	(3,788,190)
Net Revenue US\$		7,197,561	9,596,742	15,594,721	16,794,303	17,993,903	4,798,380	71,975,610
Labour		(1,460,681)	(1,313,984)	(1,398,865)	(1,585,112)	(1,626,519)	(1,035,235)	(8,420,396)
Consumables		(1,020,493)	(1,177,969)	(1,413,563)	(1,696,276)	(2,035,531)	(746,727)	(8,090,560)
Utilities		(1,445,288)	(1,589,816)	(1,987,271)	(2,185,998)	(2,404,597)	(1,365,766)	(10,978,736)
Rent		(120,000)	(132,000)	(145,200)	(145,200)	(150,000)	(150,000)	(842,400)
Other direct cost		(313,573)	(282,081)	(300,303)	(340,285)	(349,175)	(370,400)	(1,955,817)
Depreciation		(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	-	(5,986,705)
Reclamation provision		(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(1,332,583)
Community Devt		(75,000)	(82,500)	(90,750)	(99,825)	(100,008)	(120,788)	(568,871)
Indirect Cost		(43,786)	(48,165)	(52,981)	(58,279)	(64,107)	(70,518)	(337,837)
Total Operating Cost		(5,898,260)	(6,045,953)	(6,808,372)	(7,530,413)	(8,149,375)	(4,081,532)	(38,513,905)
Operating Income		2,837,526	5,100,892	10,349,519	10,841,432	11,428,080	1,130,252	41,687,701
Tax payable	35%	(993,134)	(1,785,312)	(3,622,332)	(3,794,501)	(3,999,828)	(395,588)	(14,590,696)
Operating Profit		1,844,392	3,315,580	6,727,187	7,046,931	7,428,252	734,664	27,097,006

Table 20: Base-Case Project Cash flow and Evaluation based on equity funding

Year	0	1	2	3	4	5	6	TOTAL
Gold (Ounces)		4,209	5,612	9,120	9,821	10,523	2,806	42,091
Gold Prices (US\$)		1,800	1,800	1,800	1,800	1,800	1,800	
Gross Revenue US\$		7,576,380	10,101,834	16,415,496	17,678,214	18,940,950	5,050,926	75,763,800
Royalties US\$	5%	(378,819)	(505,092)	(820,775)	(883,911)	(947,048)	(252,546)	(3,788,190)
Net Revenue US\$		7,197,561	9,596,742	15,594,721	16,794,303	17,993,903	4,798,380	71,975,610
Labour		(1,460,681)	(1,313,984)	(1,398,865)	(1,585,112)	(1,626,519)	(1,035,235)	(8,420,396)
Consumables		(1,020,493)	(1,177,969)	(1,413,563)	(1,696,276)	(2,035,531)	(746,727)	(8,090,560)
Utilities		(1,445,288)	(1,589,816)	(1,987,271)	(2,185,998)	(2,404,597)	(1,365,766)	(10,978,736)
Rent		(120,000)	(132,000)	(145,200)	(145,200)	(150,000)	(150,000)	(842,400)
Other direct cost		(313,573)	(282,081)	(300,303)	(340,285)	(349,175)	(370,400)	(1,955,817)
Depreciation		(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	-	(5,986,705)
Reclamation provision		(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(1,332,583)
Community Devt		(75,000)	(82,500)	(90,750)	(99,825)	(100,008)	(120,788)	(568,871)
Indirect Cost		(43,786)	(48,165)	(52,981)	(58,279)	(64,107)	(70,518)	(337,837)
Total Operating Cost		(5,898,260)	(6,045,953)	(6,808,372)	(7,530,413)	(8,149,375)	(4,081,532)	(38,513,905)
Operating Income		1,299,301	3,550,789	8,786,350	9,263,890	9,844,527	716,848	33,461,705
Tax payable	35%	(454,756)	(1,242,776)	(3,075,222)	(3,242,361)	(3,445,585)	(250,897)	(11,711,597)
Operating Profit		844,546	2,308,013	5,711,127	6,021,528	6,398,943	465,951	21,750,108
Add:								-
Depreciation		1,197,341	1,197,341	1,197,341	1,197,341	1,197,341	-	5,986,705
Capex	(5,986,705)							
Working Capital	(500,000)							
Net Cash Flow	(6,486,705)	2,041,887	3,505,354	6,908,468	7,218,869	7,596,284	465,951	21,250,108
Discount Factor 10%	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209	0.5645	
Present Value of NCF	(6,486,705)	1,856,261	2,896,987	5,190,434	4,930,585	4,716,695	263,017	13,367,274
Cost Escalator	5%	1.0500	1.1025	1.1576	1.2155	1.2763	1.3401	

Net Present Value: US\$13.37 million

Internal Rate of Return: 57.30%

Table 21: Cash flow Sensitivity Analysis with a 5% decrease in Projected Revenue

Year	0	1	2	3	4	5	6	TOTAL
Gross Revenue		7,197,561	9,596,742	15,594,721	16,794,303	17,993,903	4,798,380	71,975,610
Royalties US\$	5%	(359,878)	(479,837)	(779,736)	(839,715)	(899,695)	(239,919)	(3,598,781)
Net Revenue US\$		6,837,683	9,116,905	14,814,985	15,954,588	17,094,207	4,558,461	68,376,830
Labour		(1,460,681)	(1,313,984)	(1,398,865)	(1,585,112)	(1,626,519)	(1,035,235)	(8,420,396)
Consumables		(1,020,493)	(1,177,969)	(1,413,563)	(1,696,276)	(2,035,531)	(746,727)	(8,090,560)
Utilities		(1,445,288)	(1,589,816)	(1,987,271)	(2,185,998)	(2,404,597)	(1,365,766)	(10,978,736)
Rent		(120,000)	(132,000)	(145,200)	(145,200)	(150,000)	(150,000)	(842,400)
Other direct cost		(313,573)	(282,081)	(300,303)	(340,285)	(349,175)	(370,400)	(1,955,817)
Depreciation		(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	(1,197,341)	-	(5,986,705)
Reclamation provision		(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(222,097)	(1,332,583)
Community Devt		(75,000)	(82,500)	(90,750)	(99,825)	(100,008)	(120,788)	(568,871)
Indirect Cost		(43,786)	(48,165)	(52,981)	(58,279)	(64,107)	(70,518)	(337,837)
Total Operating Cost		(5,898,260)	(6,045,953)	(6,808,372)	(7,530,413)	(8,149,375)	(4,081,532)	(38,513,905)
Operating Income		939,423	3,070,952	8,006,614	8,424,175	8,944,832	476,929	29,862,925
Tax payable	35%	(328,798)	(1,074,833)	(2,802,315)	(2,948,461)	(3,130,691)	(166,925)	(10,452,024)
Operating Profit		610,625	1,996,119	5,204,299	5,475,714	5,814,141	310,004	19,410,901
Add:								
Depreciation		1,197,341	1,197,341	1,197,341	1,197,341	1,197,341	-	5,986,705
Capex	(5,986,705)	-	-	-	-	-	-	-
Working capital	(500,000)							-
Net Cash Flow	(6,486,705)	1,807,966	3,193,460	6,401,640	6,673,055	7,011,482	310,004	18,910,901
Discount Factor 10%	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209	0.5645	
Present Value of NCF	(6,486,705)	1,643,606	2,639,223	4,809,647	4,557,786	4,353,579	174,989	11,692,124
Cost Escalator	5%	1.0500	1.1025	1.1576	1.2155	1.2763	1.3401	

Net Present Value:US\$12.46 million

Internal Rate of Return: 54.40%

8.8 Sensitivity Analysis

A sensitivity analysis was conducted by varying two key inputs that significantly influence the project's profitability: gold prices (revenue) and operating costs. The results indicate that the project's return is most sensitive to fluctuations in these two factors. However, even under adverse conditions, the analysis confirms that the company's profitability remains resilient. In contrast, capital expenditures have the least impact on project returns, as the majority of costs are attributed to long-term operational expenses rather than upfront construction capital.

To test the project's robustness under conservative scenarios:

- A 5% reduction in revenue resulted in a post-tax NPV (at a 10% discount rate) of US\$11.69 million and an Internal Rate of Return (IRR) of 52.40%.
- A 5% increase in operating costs yielded a post-tax NPV10 of US\$12.46 million and an IRR of 54.40%.

The sensitivity analysis results is shown in Table 22.

Table 22: Results of Sensitivity Analysis

Scenario	NPV _{10%} (US\$ M)	Post-Tax IRR (%)
Base Case	13.37	57.30
- 5% Rev	11.69	52.40
+5% Opex	12.46	54.40

9.0 ENVIRONMENT

9.1 Introduction

Since 1989, an Environmental Impact Assessment (EIA) has been a mandatory requirement for developments, projects, or undertakings in Ghana. In June 1995, the Ghana Environmental Protection Agency (EPA) introduced new procedures for EIAs, incorporating gradual phases based on the nature, complexity, and location of the undertaking (Ghana Environmental Impact Assessment Procedures, 1995). Between 1995 and 1999, the EPA undertook a review and revision of these procedures. In June 1999, the revised procedures were officially adopted and passed by Parliament as Legislative Instrument 1652, known as the Environmental Assessment Regulations (L.I.1652). These procedures necessitate the submission of an Environmental Impact Statement (EIS) to the EPA for review and approval, ultimately leading to the issuance of an Environmental Permit. This permit is crucial for a project to proceed on environmental grounds.

Environmental Impact Assessment (EIA) means an evaluation of a project to determine its impact on the environment and human health and to set out the required environmental monitoring and management procedure and plans that are required to mitigate pollution, land degradation as well as negative impact on other land users and socio-economic hardships of affected communities.

Environmental Management Plan means a plan that mitigates pollution, land degradation and negative impact on other land users and socio-economic hardships of affected communities during and after the implementation of the project. The baseline study for the alluvial mining project on the Akim Apapam concession was focused on the water bodies within the project area that are likely to be affected during the operations. Chemical spillages are not anticipated to occur as harmful chemicals will not be employed for gold extraction.

9.2 Baseline Information

The concession area has generally low to moderate undulating topography with elevations in the range of 150-210 meters above sea level (ASL), with the lowest elevation about 120m ASL along the river valleys. The concession area sits within the Birim valley with all streams draining the concession flowing into it. Hujak Mining plans to mine responsibly and would not attempt to pollute any river body when this lease application is granted.

Comprehensive data was gathered via desk study during the prospecting activities it embarked on, historical data, Governmental agencies, academic papers and any reliable source that may be of help.

- **Environmental Data:** Details about the natural environment, such as the quality of air, water, and soil; the presence of wildlife and plant species; climate conditions; and the landscape and geology of the area.
- **Socio-Economic Data:** Information about the local communities, including demographics, health, employment, cultural heritage, land use, and economic activities. This helps in understanding the potential social and economic impacts of a project.
- **Corporate social responsibility (CSR)** is a self-regulating business model that helps a company be socially accountable to itself, its stakeholders, and the public as a whole.

9.3 Environmental Data

9.3.1 Air Quality

The Akim Apapam area is considerably rural and has relatively good background on air quality. Before mining, air quality impacts were mainly due to smoke from slash-and-burn agriculture, charcoal production, and fires caused naturally or by human activities such as hunting. Background dust levels were primarily influenced by vehicular traffic on the mostly unpaved roads in the area and the dust-laden Harmattan winds that occur annually from December to February.

The company will make available water bouncers to water down the dust both the communities and the mine site.

9.3.2 Surface Water

The processing plant site will be in the hilly areas where water shall be drawn from several streams draining the property. Generally, surface water will be diverted around areas of disturbance to avoid impacting water quality. This is because surface water flows in the sub-basins are impacted due to the presence of the tailing's storage facility and water storage facility.

9.3.3 Flora and Fauna

The natural vegetation of the Akim Apapam area is part of the broad Moist Semi-deciduous Zone. Vegetation in the Project Area is composed of a mixture of natural plant communities in early stages of ecological succession, crops and plantations, patches of second-growth forest, and riparian communities along rivers and streams. The natural vegetation in the Project Area has been extensively fragmented because of agricultural activities, fire, removal of timber and, consequently, has little resemblance to the native forest communities once typical of the region.

The Fauna diversity within the Project Area is slightly high but varies depending on the vegetative habitat. There are mammals, amphibians, reptiles, different species of butterflies, and birds forming the fauna within the catchment area, Mammals such as the antelope and grasscutters are poached daily. Studies have shown different kinds of plant species that have been threatened by illegal timber lumbering.

9.3.3.1 Mitigation

Company staff shall not be allowed poaching, falling of trees, and killing of poisonous reptiles. The concession does not sit near any forest cover. It therefore offers staff the opportunity to steer clear of practices of poaching and killing of dangerous reptiles.

9.3.4 Noise and Vibration

The activities of Hujak Mining shall be comprised of conventional mining using excavators, and haul trucks, together with support and service equipment such as dozers, graders, trommels, power generating plants, etc. are heavy and noisy, which may impact of the overall noise and vibration in the area.

9.3.4.1 Mitigation

To mitigate noise pollution, the company shall maintain its entire fleet of vehicles in good order using a planned maintenance schedule. The vehicles will be fitted with appropriate noise muffler systems and primarily exhaust silencers. Ear protectors shall be given daily to workers and the safety coordinators will insist on the wearing of the protectors in case the allowable noise pollution is being exceeded.

9.3.5 Soil

Soil conservation is an integral part of environmental protection and as such the soil will be safely kept for later environmental restoration purposes.

The key methods for soil conservation that will be used by Hujak Mining may include:

- I. removal and stockpiling of soil from impacted areas. This material is later utilized in the site revegetation program.
- II. minimizing the area of land required for project infrastructure, so that the impact on land users will be minimized.
- III. progressive revegetation of disturbed land to minimize the amount of time that bare soil is exposed to the erosive effects of rain impact and runoff.
- IV. the burial and encapsulation of all potentially contaminative materials to isolate them from the final soil surface.
- V. where practicable, earthwork activities in sensitive areas will be undertaken during dry periods; and
- VI. the use of standard erosion control measures, such as interception of drains, contour planting, silt fences, establishment of groundcover species, optimal drainage construction, use of silt ponds, erosion matting, etc.

9.3.6 Groundwater Quality

Potential contamination of groundwater through oil and grease spills from machinery and running plants will be curbed through constant servicing and patching of leaks. Monitoring of the groundwater will be conducted periodically to track the water quality.

Groundwater within the communities is drinkable and clean, according to a periodic check that is done by the adjoining mining companies such as Elite Minerals, BSD Mining etc.

9.3.6.1 Solid and Liquid Waste Management

Solid waste bins will be provided and categorized into different ways of dumping: plastic, organic, metallic, and rag into classified bins. The plastics will be given out to the recycling companies, whilst the organic waste generated from the cuisine will be used by the environmental department for producing compost to aid in soil reclamation and revegetation. The remaining solid waste will be subcontracted to the commercial waste management companies to be disposed off appropriately according to the sanitation laws of the country.

Public toilets and urinary facilities will be provided on-site to curb open defecation which can lead to waterborne diseases such as cholera. Disposal of grease, oil, and liquid waste contaminants shall be done appropriately following the Health, Safety and Environment protocols.

9.3.6.2 Restoration

Trucks that will be carrying ore to the treatment plant will carry washed gravel (oversize) or tailings on their return trip to the mining face. The processed tailings: made up of washed boulders, cobbles, pebbles, and sand will be continuously conveyed to the mined-out areas for backfilling. This operation will be carried out concurrently with the mining of ore. The entire mine out pits will be backfilled with tailings and overburden, and subsequently planting trees to set the land back to its original state.

9.3.6.3 Impact Identification

Project-related air quality impacts include dust generation attributable to mining activities (mainly hauling) and emissions from the mining equipment fleet. Point source emissions include emissions from the trash/medical waste incinerator and emissions from the gold room processing facility. Management of air quality impacts is primarily focused on dust control. Measures will include extensive road watering, applying rock running courses to road surfaces, and speed limit controls to minimize dust. An air quality monitoring program shall be in place to evaluate respirable dust (PM10) and dust accumulation. Source testing of the point source emissions will be conducted periodically.

9.3.7 Rehabilitation and Mine Closure

The EPA requires a Reclamation Bond to be posted as part of any mine permitting process. The bond is required to provide financial surety against non-compliance under the approved Closure and Reclamation Plan and is required within six months after the start of operations. The bond looks at the closure liability associated with the current mine plan and projected two years out to the future. The Estimated reclamation liability for the Akim Apapam Gold Project is approximately US\$ 1,333,000.

Hujak Mining is committed to rehabilitation and Mine Closure reports to ensure that the environment is restored to its primary state. The waterbodies, the air, the trees, cash crops, infrastructure sited on arable lands, dams, tailings dams, ponds, and sundries shall all be restored

and the report submitted to the Mine Inspectorate division not later than sixty (60) days before abandonment or closure. Monthly mined-out maps, of partially backfilled, reclaimed, and revegetated areas shall be prepared regularly to ascertain the land status for them to be restored during operations.

This will help the company to zero in on areas during the mine closure and folding plans actions. Hujak Mining will ensure that total mine-out figures and maps are well documented. This aids in ongoing budget preparations for the environment department to embark on seamless backfilling and revegetation programs during the mine life. Consultation with landowners will help define a map-out plan as to how their respective lands shall be put to use. This will inform the environmental unit as to how to put the end-use of their land before commencing the restoration.

The site infrastructures including dams, ponds, offices, routes, bridges, culverts, workshops, and the proposed processing plant; shall have their decommissioning programs roll out to the Inspectorate Division before the closure or abandonment in compliance with the Mining Act. Hujak Mining shall comply with all the Minerals Commission regulations concerning the closure of the processing plants, removal of hazardous objects, and closure of ancillary facilities if the Lease is granted.

9.3.8 Commitment

The company is committed to following these restrictions during operations:

- 100 metres buffer zone from Roads, Forest Reserves, Cultural Heritage Sites, Shrines, Cemeteries, streams, railway lines and 300 metres from pylons
- The company will not embark on any exploitation before the mining right is granted
- No use of mercury and cyanide or any other chemical on site
- Ore stockpiles, slimes dams or settling ponds may not be constructed within the river bank or the highest flood-line of any watercourse. All slimes dams and settling ponds shall be lined as approved by the authorities.

10.0 SOCIO-ECONOMIC BENEFITS

The impending gold project shall have both negative and positive impacts on the community and as well as the country's growth. This shall be as the result of the resources being used judiciously to the benefit of the community as well as the nation.

10.1 Negative Impact on the Community

- Environmental degradation: mining activities can lead to deforestation, soil erosion, and loss of biodiversity, affecting local ecosystems and wildlife habitats.
- water pollution: gold mining can contaminate local water sources with hazardous chemicals, as well as sediments, leading to water quality issues and affecting aquatic life.
- air pollution: dust and emissions from mining operations, including heavy machinery and processing plants, can degrade air quality and cause respiratory problems for residents.
- noise pollution: the use of heavy equipment, tipping trucks unloading materials could create loud bangs by slamming their tailgates and other mining activities can create significant noise, disrupting the peace and tranquility of local communities.
- displacement of residents: mining projects may require land, leading to the displacement of local communities or indigenous people and loss of homes and livelihoods.
- health risks: exposure to toxic chemicals and pollutants from mining can pose significant health risks to workers and nearby residents, including respiratory illnesses, skin diseases, and long-term health effects.
- economic disruption: while mining can bring jobs and development, it may also disrupt traditional economies and livelihoods, such as agriculture and fishing, creating economic instability.
- social conflicts: the presence of a mine can lead to social tensions and conflicts within communities, particularly if there is a perceived unfair distribution of benefits or if mining activities negatively impact communities.

10.2 Benefits to the Community

10.2.1 Job creation

The project is expected to provide employment opportunities for residents, boosting the local economy and reducing unemployment.

10.2.2 Infrastructure development

Projects often require the development of infrastructure such as roads, schools, hospitals, and housing, which can benefit the entire community.

10.2.3 Economic growth

Local businesses can benefit from increased demand for goods and services due to the influx of workers and the operational needs of the project.

10.2.4 Improved education and training

Projects may offer educational programs and vocational training to the local workforce, enhancing skills and increasing employability.

10.2.5 Enhanced health services

Increased investment in local health services, including the construction of clinics and hospitals and improved access to healthcare, can result from project activities.

10.2.6 Community development programs

Many projects include community development initiatives, such as funding for local schools, water supply projects, or community centers, enhancing the quality of life for residents.

10.2.6.1 Increased local revenue

Local governments may receive additional revenue from taxes and royalties, which can be reinvested into community development projects.

10.2.6.2 Benefits to the Nation

The Akim Apapam Gold Project will contribute to national GDP growth by increasing production, exports, and attracting foreign investment. It will generate significant foreign exchange earnings through the export of gold, which will help stabilize the national economy.

10.2.7 Corporate Social Responsibility Program

In addition to doing its core business of mining, Hujak Mining is committed to demonstrating social responsibility towards the communities that host our projects. The scope of our citizenship moves beyond traditional philanthropy to include engagements and critical social interventions that will create shared value for us and the local communities. These will include the provision of educational and health facilities, potable water supply, etc.

Consultation with stakeholder communities, traditional and elected leaders, and regulators forms an integral component of the project's environmental and socioeconomic impact assessment process. Stakeholder consultation follows a typical pattern reflecting the stages of evolution of project knowledge and permitting.

In addition to compliance with regulatory requirements, Hujak Mining will design and commit to a robust Corporate Social Responsibility (CSR) program, which cascades across the project's mine life to mine closure. The company's core belief is to undertake responsible mining which is premised on innovation through demonstrating responsibility for enhancing social well-being, protecting the environment, whilst maintaining the financial viability of its operations.

The CSR program comprises of four key activities, namely, Stakeholder engagement; Employee Engagement; Community development; and Environmental stewardship. Each of these activities is mutually reinforced and will be planned and implemented in a way that they extend credible development opportunities for local communities. Figure 16 summarizes the CSR program, reflecting the company's dedication to responsible mining and community development.



Figure 16: Hujak Mining’s Corporate Social Responsibility (CSR) program

10.2.8 Community Development

Hujak Mining focuses on creating shared value with local communities through successful mining operations. As part of this, the company will work with the District Assembly to prioritize and invest in areas of Education and Skills Development, Health infrastructure, Local economic development, as well as Water and Sanitation facilities. In particular, the skills development module will be aligned with the labor requirements of the project to ensure that youth are prioritized for employment throughout the project life. Added to the operational requirements, the initiatives under each of the prioritized modules above must align with the medium-term development plan of the District Assembly as a condition of the company’s support. Ultimately, our investment will help the capacity of the district assembly to deliver on its development mandate. The community development activities that will be considered are as presented in Table 23.

Table 23: Priority Areas for Hujak Mining CSR Activities

Priority sector	Proposed Activities for the Community
Education and skills development	Scholarships and bursaries for tertiary and technical education. Support for technical education will target youth interested in developing their skills in craft and artisanship e.g., dump truck operators, auto-mechanics, welders, carpenters, hospitality stewards, Integrate skills training with the company's operational requirements
Health infrastructure and services	Construction of Wards at district hospitals Clinics/CHPs Compounds in local communities. Donation of medical equipment Short-term staff development training
Local economic development	Sponsorship for agriculture-based initiatives e.g. National Farmers Day Training and technical support for women in agro-processing initiatives Crop-based livelihoods activities linked with livelihood restoration programs for Project-affected persons Local Employment Strategy – 70 percent recruitment of unskilled labor from affected communities
Water and Sanitation	Provision of boreholes and small-town water systems Provision of household latrines
Others	Cash donations for community-driven projects

10.3 Occupational Health and Safety

Access to the mining area will be strictly limited to authorized operators and mine personnel to mitigate operational hazards. All equipment operators and supervisory staff will be equipped with two-way radio communication systems to facilitate continuous contact without requiring personnel to exit equipment cabins.

Personnel entering the pit area will be mandated to wear full Personal Protective Equipment (PPE), including safety goggles, steel-toe boots, high-visibility vests, and hard hats. Where feasible, mine surveying activities will be scheduled during operational shutdown periods to minimize interaction with active heavy machinery.

To control dust emissions on haul roads, regular water spraying will be implemented, supplemented by surfacing with sorted oversize material from the processing plant. This dust

suppression strategy will both reduce airborne particulate matter and enhance road stability during periods of heavy precipitation.

These integrated measures are designed to maintain operational safety while optimizing production efficiency under varying site conditions.

11.0 SIGNATURE

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Professional Profile

The author is a Senior Resource Geologist with over 18 years of industry experience (since 2005) in both hard rock and alluvial deposit evaluation, working with mining companies across Ghana and internationally. The author maintains professional accreditation as a member in good standing with the:

- Ghana Institute of Geoscientists (GhIG)
- *Australian Institute of Mining and Metallurgy (AusIMM)*

Alluvial Gold Mining Expertise

The author has provided technical evaluation and consultancy services to multiple alluvial gold mining operations, including:

- Kibi Goldfields
- Swed Mining
- Koantwi Mining Ltd

Hard Rock Exploration & Mining Consultancy

As a Principal Consultant, the author has led exploration programs, mine development, and feasibility studies for several hard rock mining companies, with key engagements at:

- Konongo Gold Mines (Owere Mines) – Project design, implementation, and evaluation
- African Gold Group Limited – (Ghana & Mali)
- Newmont Ghana Gold Limited
- Star Goldfields Limited
- Ghana Bauxite Limited – Resource assessment and development planning
- Maripoma Mining Limited – Exploration strategy and feasibility support
- JH Resources Ltd – Geological modeling and resource estimation

The author has a demonstrated track record in successful project execution, from early-stage exploration through to feasibility-stage deliverables.

12.0 APPENDICES

Appendix 1: Selected Logging Data Sheet

Pit-ID	Overburden Thickness (m)	Gravel Thickness (m)	Visible Specks	Assay (Grade) (g/m ³)
KP1	2.8	2.5	-	0.29
KP2	2	2.8	2	0.27
KP3	1.8	1.9	-	0.95
KP4	1.8	1.2	5	0.25
KP5	1.26	2.8	-	0.21
KP6	1.8	1.9	12	0.80
KP7	1.8	1.7	-	0.86
KP8	2	2.3	1	0.57
KP9	2	2.2	2	0.67
KP10	1.6	2.8	6	0.49
KP11	2.7	2.3	-	0.32
KP12	1.76	4.25	-	0.76
KP13	2.2	1.1	1	0.19
KP14	2.2	2.5	-	0.95
KP15	2.68	4.5	3	0.76
KP16	2.08	4.5	-	0.19
KP17	0.19	2.4	1	0.19
KP18	1.05	1.5	-	0.06
KP19	1	1	-	0.32
KP20	1	2.7	1	0.29
KP21	1	1.2	1	0.45
KP22	2	1.1	-	0.52
KP23	1.6	0.5	1	0.19
KP24	1.6	1.2	1	0.59
KP25	1.6	1.3	1	0.61
KP26	1	1.3	1	0.57
KP27	1.6	1.3	1	0.41
KP28	1	3.5	-	0.65
KP29	0.1	1.2	-	0.40
KP30	1	0.5	-	0.50
KP31	1.6	1.1	1	0.25
KP32	0.1	0.9	1	0.95
KP33	1.3	1.3	1	0.19
KP34	0.7	1	1	0.38
KP35	1.75	1.2	-	0.40
KP36	1.98	1.3	1	0.19

Pit-ID	Overburden Thickness (m)	Gravel Thickness (m)	Visible Specks	Assay (Grade) (g/m ³)
KP37	1.6	0.8	1	0.40
KP38	1.6	1	1	0.57
KP39	1.6	3.2	1	0.21
KP40	0.68	2.7	1	0.19
KP41	3.1	0.5	3	0.73
KP42	2	0.8	2	0.76
KP43	1.8	1.2	1	0.83
KP44	1	1.5	-	0.57
KP45	0.8	0.8	1	0.48
KP46	0.6	1	-	0.25
KP47	0.6	0.8	-	0.61
KP48	0.6	0.3	1	0.76
KP49	0.5	0.8	1	0.95
KP50	2	0.4		0.78
KP51	1.4	1.3		0.48
KP52	2.1	1.2		0.63
KP53	2.1	1.5		0.65
KP54	0.5	1.5		0.32
KP55	0.4	1.6		0.62
KP56	1	4.25		0.57
KP57	1	1	-	0.19
KP58	1	0.3	3	0.57
KP59	1	0.6	2	0.48
KP60	2	1.1	-	0.19
KP61	1.2	1	-	0.19
KP62	1.8	0.3	-	0.19
KP63	0.85	1	-	0.19
KP64	2.6	0.7	-	0.29
KP65	1.6	0.5	-	0.24
KP66	1.3	1.2	-	0.42
KP67	1.6	0.6	-	0.19
KP68	1.5	0.9		0.19
KP69	1.2	1.4	3	0.58
KP70	1	0.7	-	0.19
KP71	0.9	0.7	2	0.45
KP72	1.3	0.9	1	0.19
KP73	2.3	0.5	-	0.66
KP74	2.3	0.8	-	0.19
KP75	2.2	0.5	5	0.45

Pit-ID	Overburden Thickness (m)	Gravel Thickness (m)	Visible Specks	Assay (Grade) (g/m ³)
KP76	2.2	1	-	0.21
KP77	2.1	0.4	-	0.21
KP78	1.2	1	-	0.19
KP79	2.1	1	-	0.21
KP80	1	0.8	-	0.19
KP81	1	0.5	-	0.70
KP82	1	0.6	-	0.19
KP83	1	0.8	-	0.10
KP84	1.9	0.7	1	0.27
KP85	1.8	0.6	-	0.38
KP86	1.8	2.8	-	0.19
KP87	1.8	0.7	-	0.19
KP88	1.7	2.4	-	0.38
KP89	1.5	0.7	-	0.57
KP90	1.5	1	1	0.19
KP91	1.3	1	-	0.19
KP92	1.2	1.4	-	0.36
KP93	1.2	0.2	-	0.19
KP94	1.2	0.9	-	0.19
KP95	1.1	0.5	-	0.19
KP96	1.1	0.1	-	0.40
KP97	1	0.5	-	0.19
KP98	0.9	0.3	3	0.76
KP99	0.9	0.9	-	0.19
KP100	0.7	1.1	3	0.95
KP101	0.6	0.6	-	0.57
KP102	0.6	4.2	-	0.46
KP103	0.5	0.9	-	0.48
KP104	0.5	1	-	0.38
KP105	0.1	0.7	1	0.19
KP106	2	2.3	1	0.19
KP107	0.3	2.3	-	0.19
KP108	1	2.5	1	0.76
KP109	1	1	-	0.14
KP110	1	3	1	0.29
KP111	1	1.2	1	0.25
KP112	0.8	1.1	-	0.46
KP113	1.6	4.5	1	0.95
KP114	0.45	1.2	1	0.59

Pit-ID	Overburden Thickness (m)	Gravel Thickness (m)	Visible Specks	Assay (Grade) (g/m ³)
KP115	1.1	4.2	1	0.84
KP116	0.5	1.3	1	0.75
KP117	1.6	2.8	1	0.84
KP118	1.2	1	-	0.61
KP119	0.1	2.2	-	0.40
KP120	0.8	0.85	-	0.87
KP121	1.6	1.1	1	0.25
KP122	0.1	0.9	1	0.19
KP123	1.3	1.3	1	0.52
KP124	0.7	1	1	0.38
KP125	1.75	2.9	-	0.65
KP126	0.5	1.3	1	0.35
KP127	0.5	3.92	1	0.86
KP128	0.6	1	1	0.26
KP129	0.5	1.1	1	0.62
KP130	0.68	1.5	1	0.59
KP131	1.6	0.7	1	0.80
KP132	0.6	0.6	1	0.51
KP133	0.8	2.7	1	0.26
KP134	1.5	1.2	1	0.19
KP135	1.6	3.5	1	0.58
KP136	1.6	1.2	1	0.65
KP137	1.2	3.8	1	0.56
KP138	1.6	2.3	1	0.82
KP139	1.5	0.6	-	0.61
KP140	0.5	3.9	-	0.75
KP141	1.5	1.6	-	0.51
KP142	0.9	3.95	-	0.52

13.0 Organizational Chart Brief

